Local Differential Privacy: Refined Mechanism Design and Utility Analysis

Ye Zheng

Advisor: Dr. Yidan Hu

Committee: Dr. Sumita Mishra, Dr. Haibo Yang, Dr. Weijie Zhao

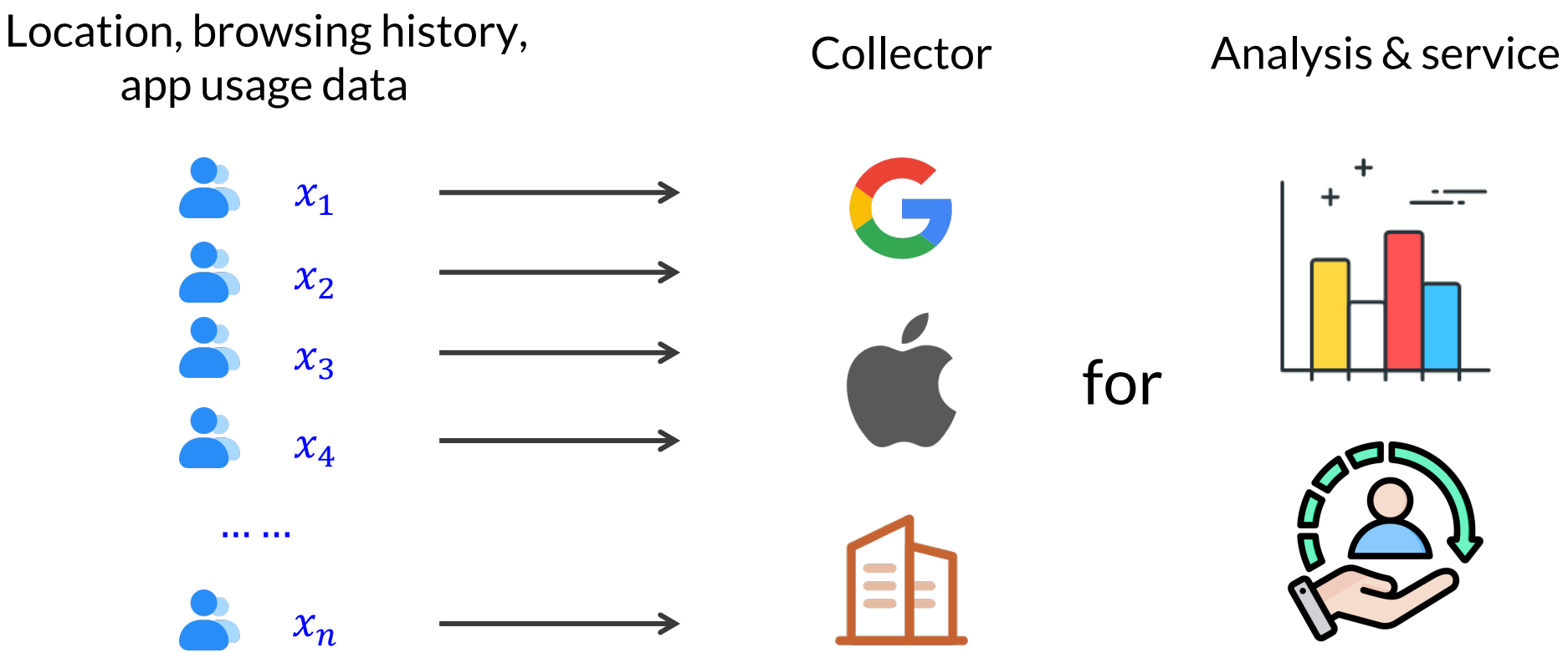
RIT | Rochester Institute of Technology

PDF & slides 👉 https://zhengyeah.com

Data Collection

RIT

- Users' personal data are collected by companies for analysis or services



Users' Data Privacy

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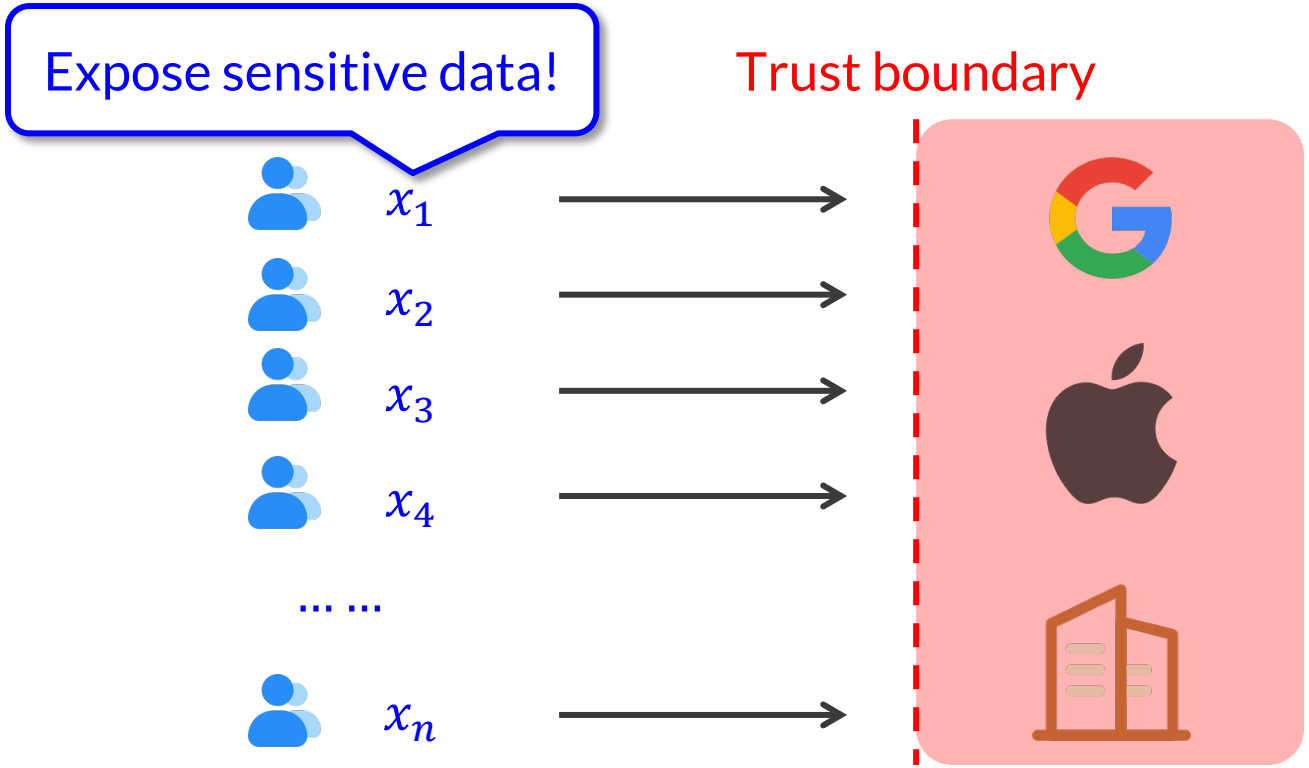
  - these companies may not be trusted to collect users' sensitive data



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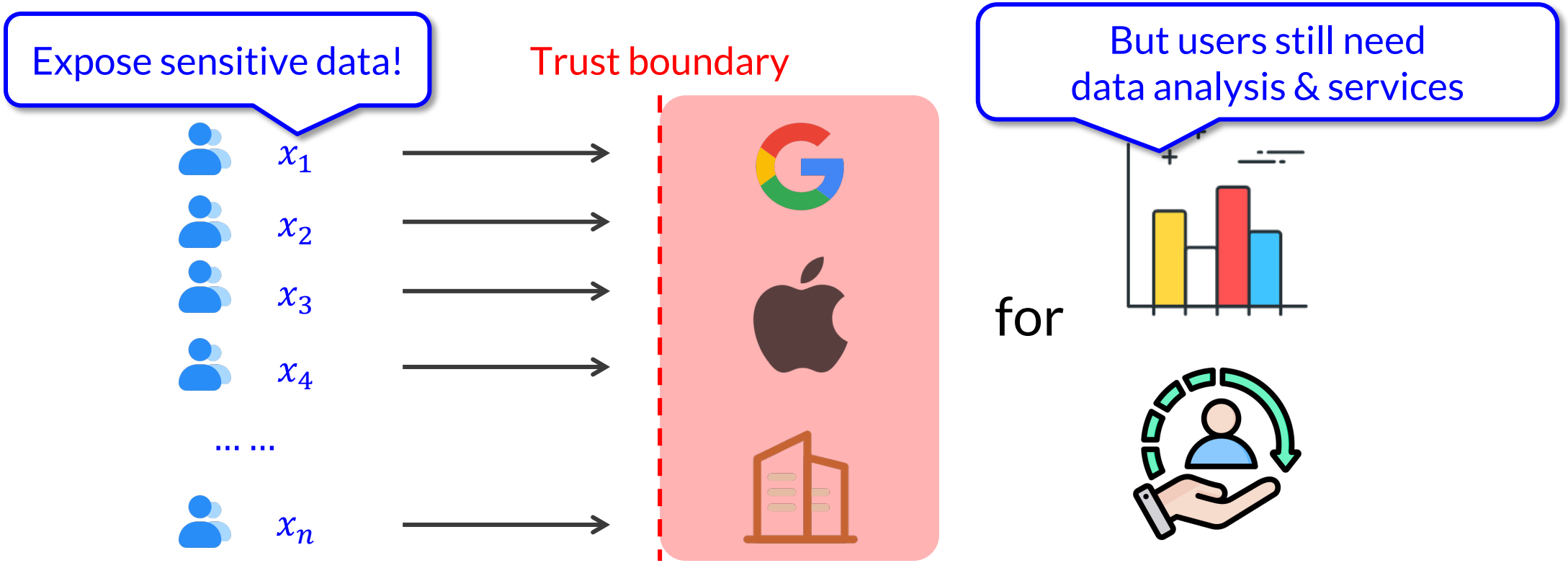

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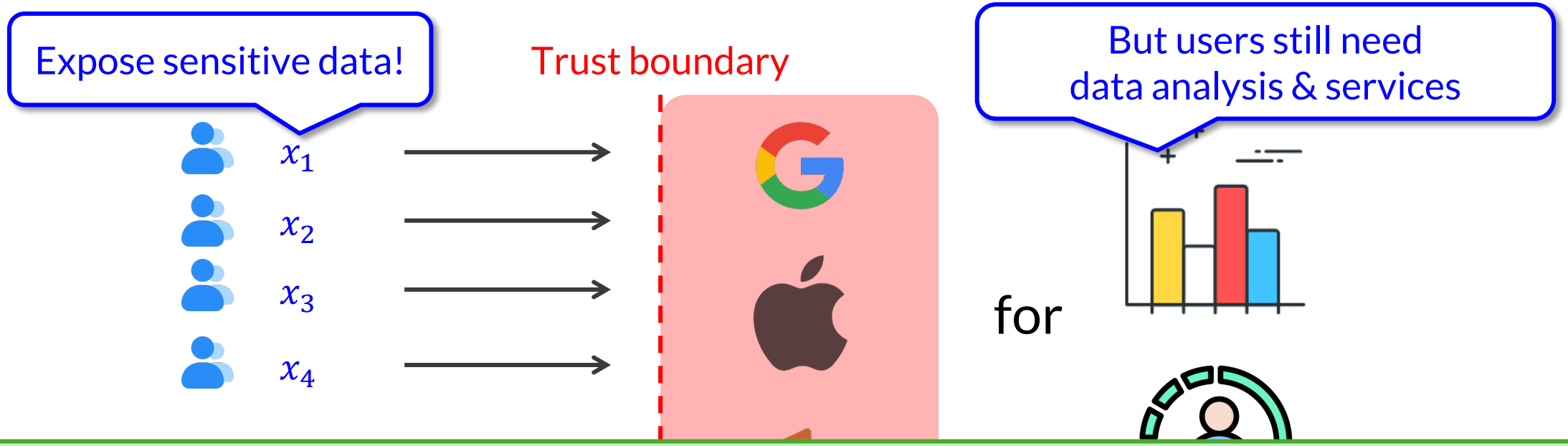
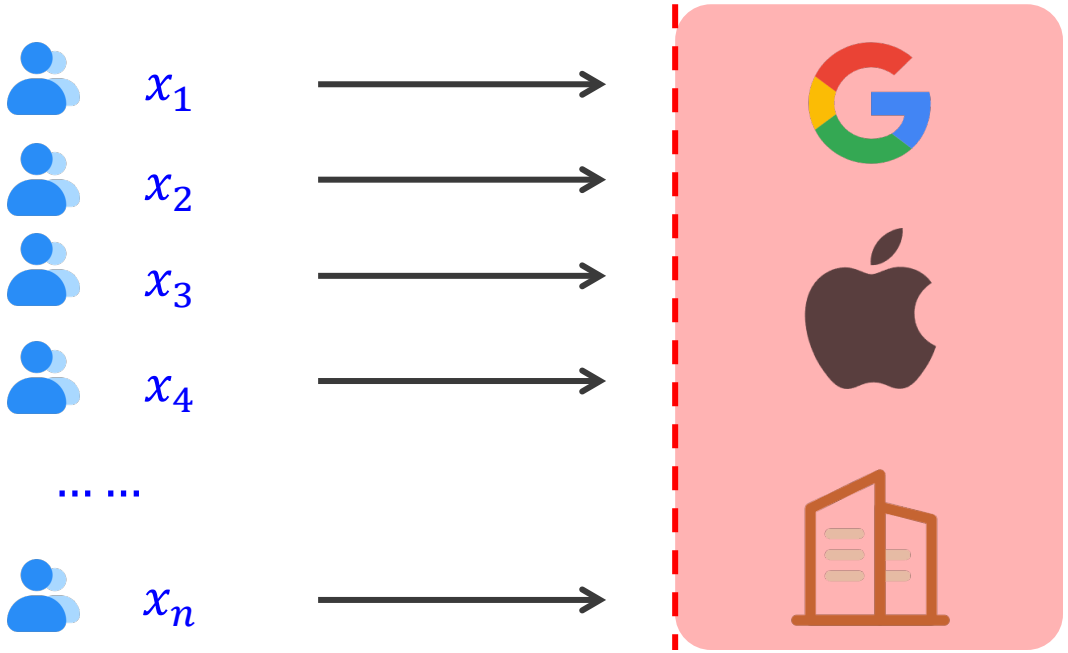


Q: How can we provide data analysis & services **while** protecting users' data privacy?

Privacy-Preserving Computation

RIT

- Users' personal data are collected by companies for analysis or services
  - these companies may be untrusted to collect users' sensitive data
  - how to compute $f(x_1, x_2, \dots, x_n)$ without revealing $x_1, x_2, \dots, x_n$?



for $f(x_1, x_2, \dots, x_n)$

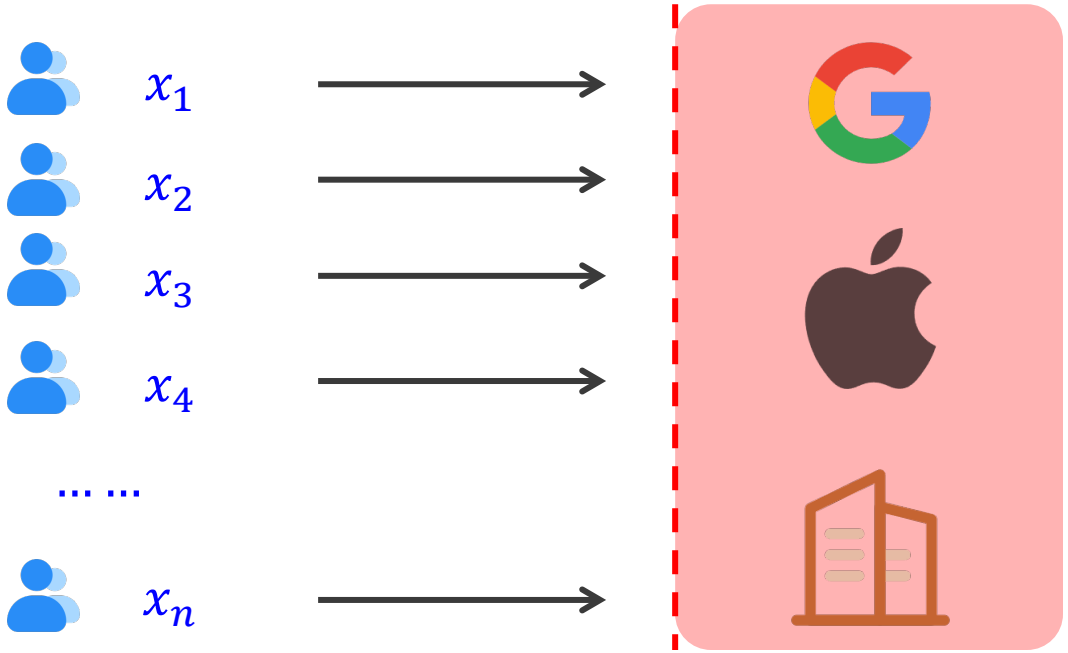
Privacy-Preserving Computation - **Techniques**

- Homomorphic encryption (HE), multi-party computation (MPC), local differential privacy (LDP), etc

Privacy-Preserving Computation - **HE**

RIT

- Homomorphic encryption (HE):
  - "homomorphic": preserving structure
  - design algorithms $\{Enc, Dec, Eval\} \rightarrow Dec\Big(Eval\big(f, Enc(x_1), Enc(x_2), \dots, Enc(x_n)\big)\Big) = f(x_1, x_2, \dots, x_n)$



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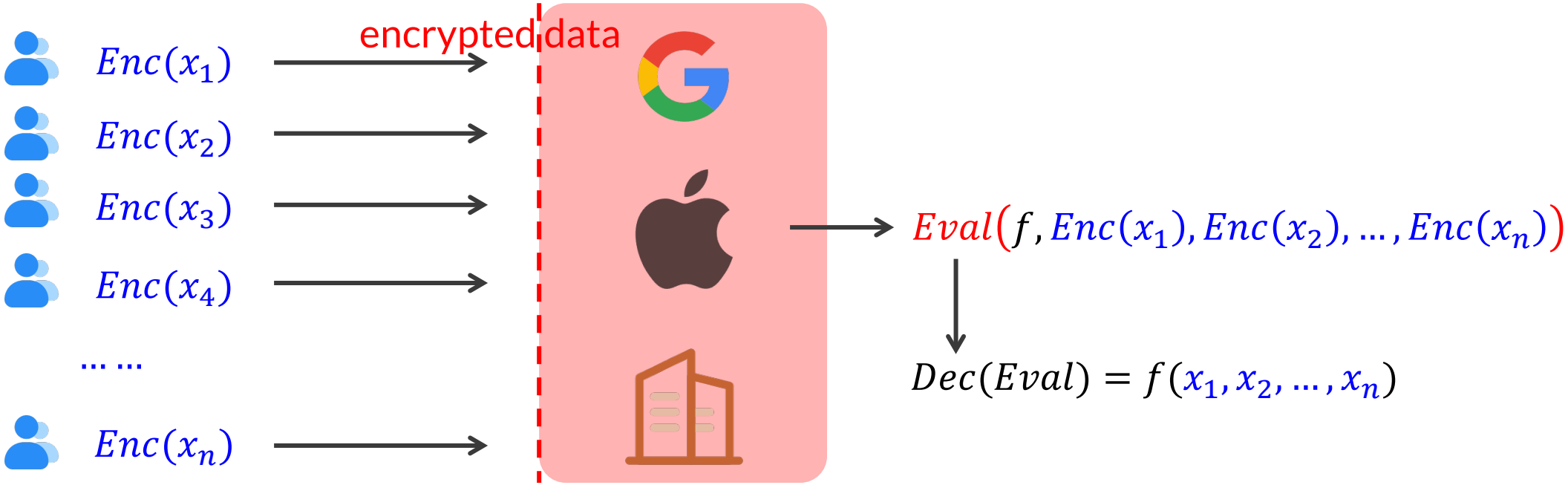

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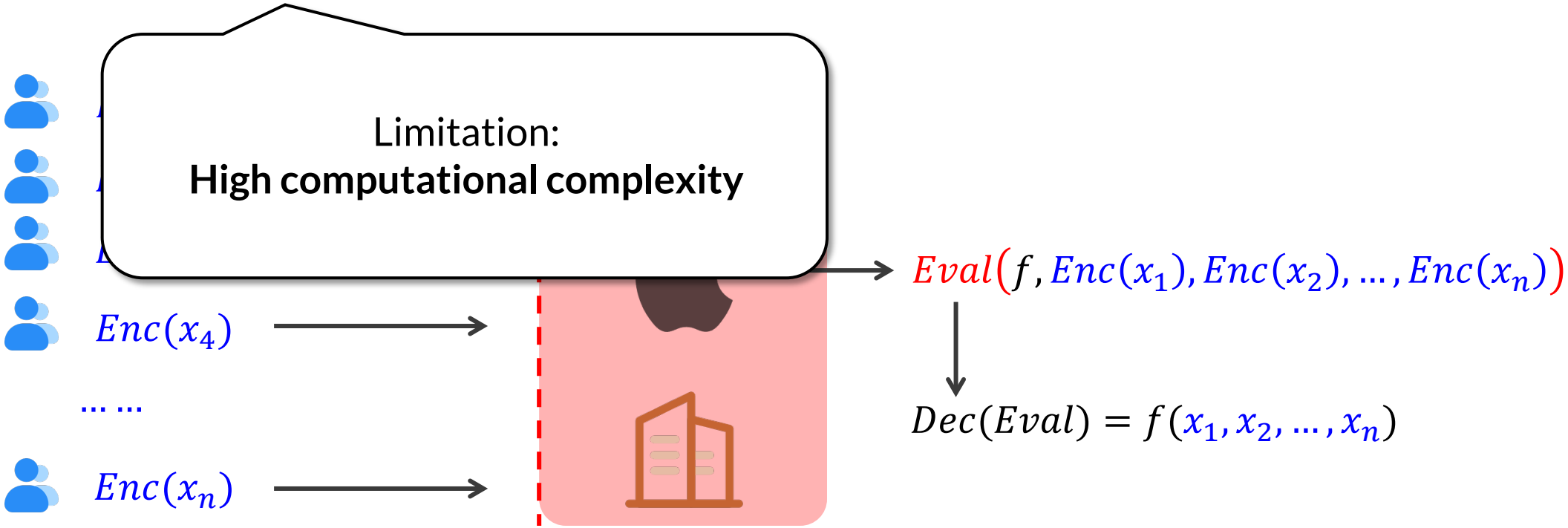

Privacy-Preserving Computation - **MPC**

RIT

- Multi-party computation (MPC):
  - no central party
  - jointly compute f without revealing x_i
- Example: $f(x_1, x_2) = x_1 + x_2$

x_1

for $f(x_1, x_2)$

x_2

… …

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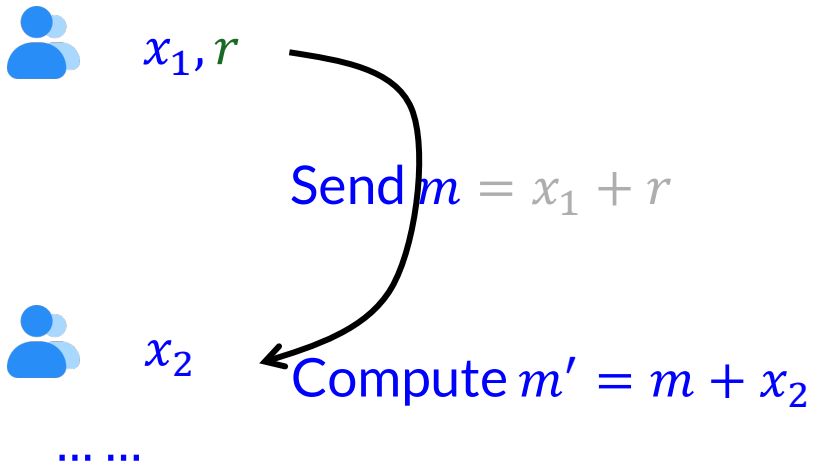

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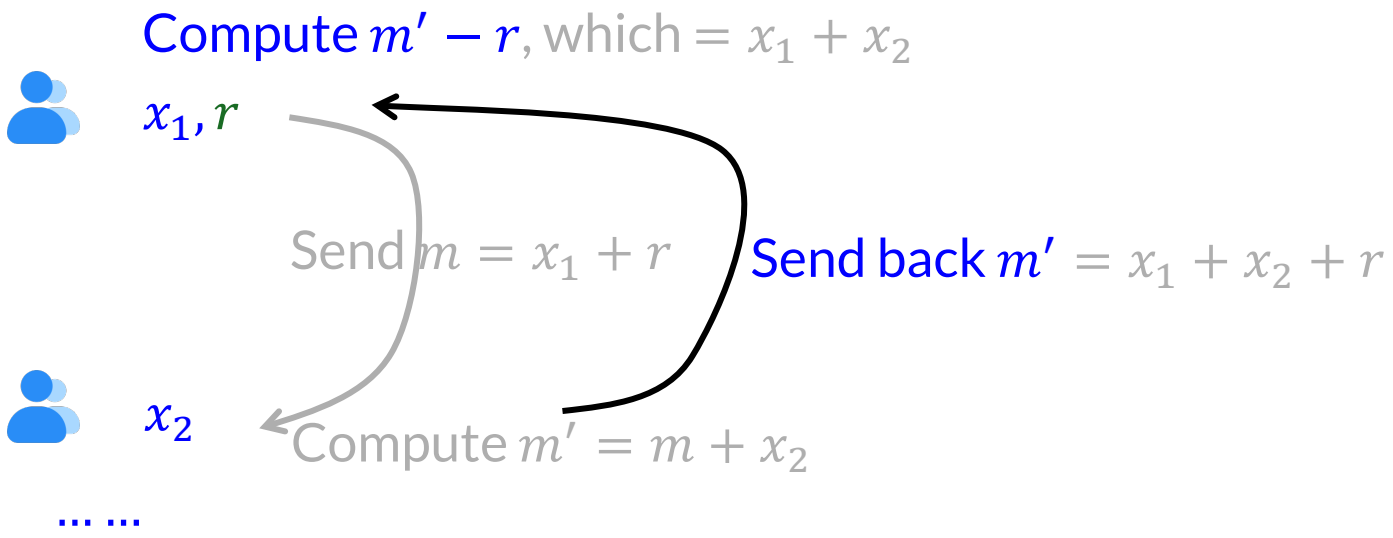

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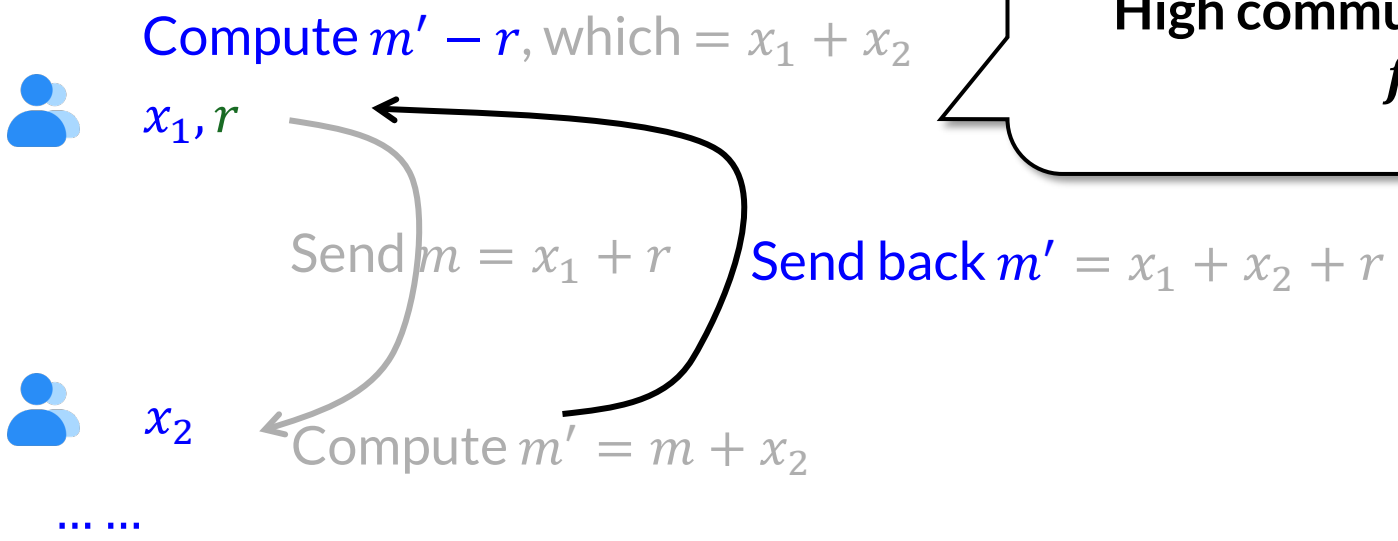

Limitations:
High communication complexity
f-specific

Privacy-Preserving Computation - **LDP**

RIT

- Local differential privacy (LDP):
  - hard to differentiate the sensitive data from other data
  - each user locally perturbs x_i to $\tilde{x}_i$ $\rightarrow$ $f'(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n) \approx f(x_1, x_2, \dots, x_n)$

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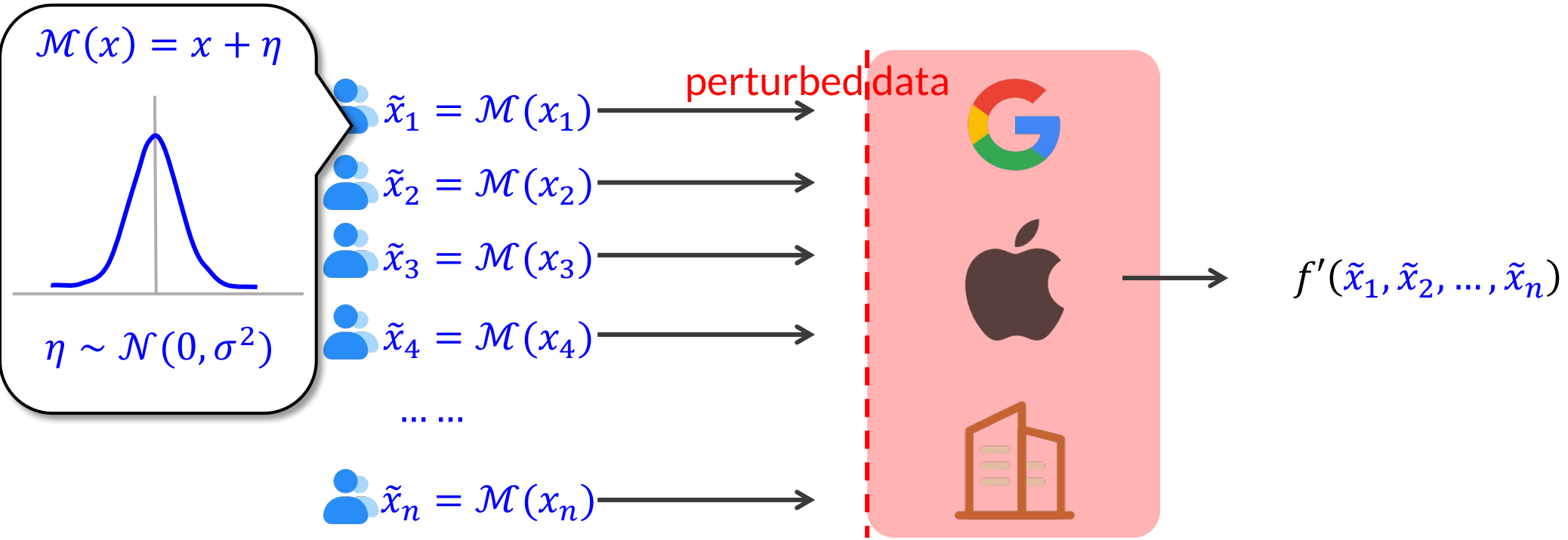

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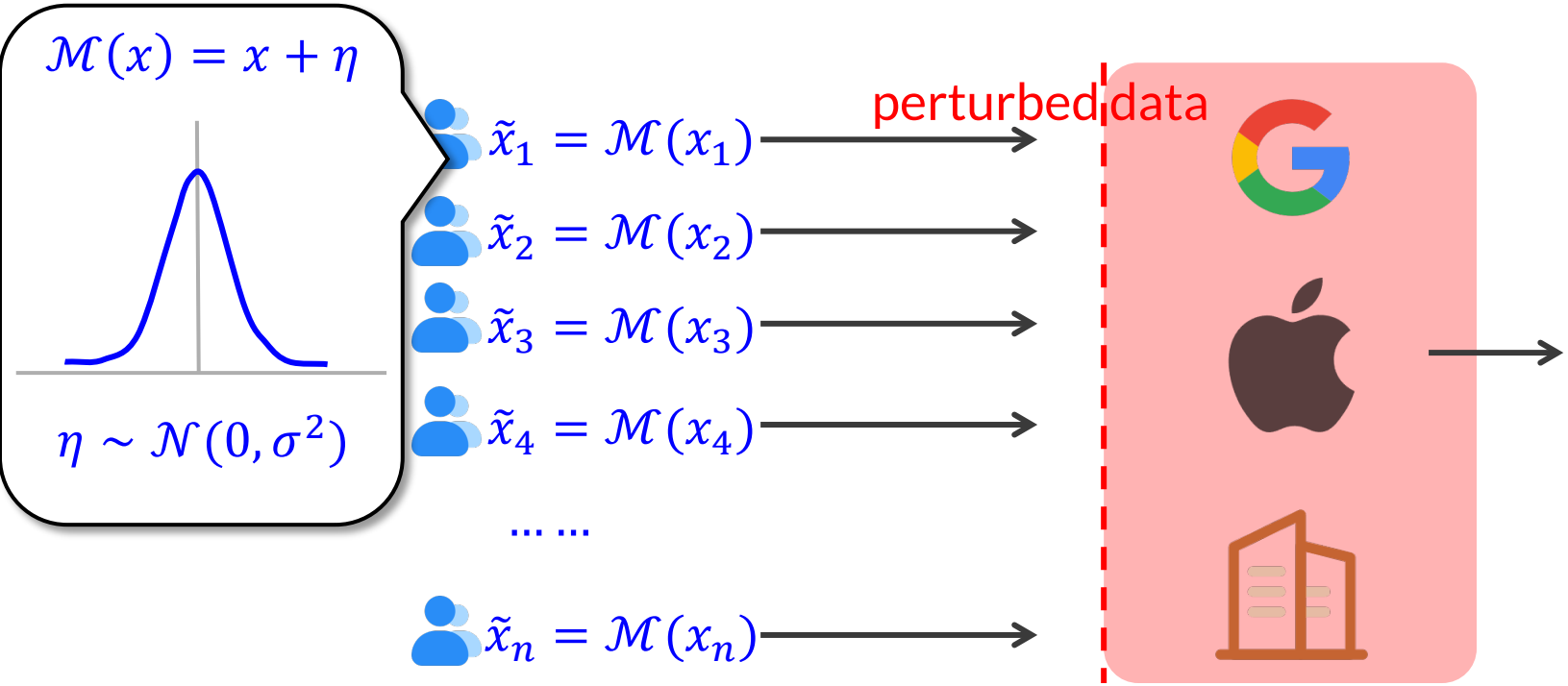


$f'(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)$

$\mathrm{E}[f'(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)] = f(x_1, x_2, \dots, x_n)$

Computation with noise

Privacy-Preserving Computation - **LDP**

RIT

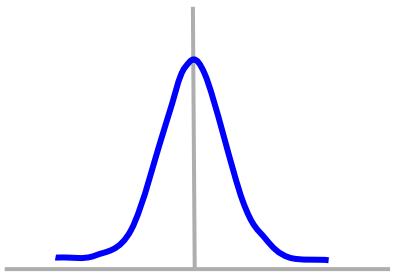
- Local differential privacy (L
  - hard to differentiate the s
  - each user locally perturb

$, x_n)$

Advantages:
Negligible computational complexity
No communication between users

But approximated f

$\mathcal{M}(x) = x + \eta$



$\eta \sim \mathcal{N}(0, \sigma^2)$

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perturbed data

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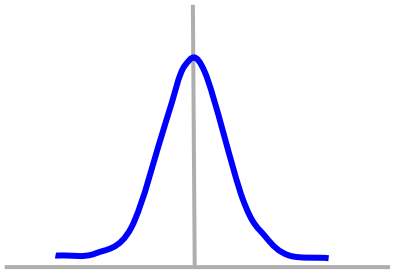
LDP – Applications

RIT

Chrome uses LDP to collect homepage settings, extension usage, etc



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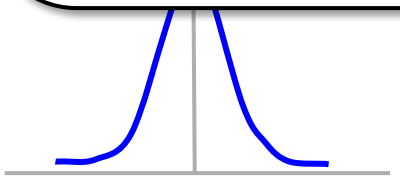
Emoji usage, new keyboard words, Safari URL statistics, health analytics



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LDP – Formal Privacy

RIT

- After applying $\mathcal{M}$, the confidence of distinguishing sensitive x_1 and x_2 from observation $\tilde{x}$:

$$\forall x_1, x_2 \in \mathcal{D}, \forall y \in \widetilde{\mathcal{D}} \qquad \max \frac{\Pr[\mathcal{M}(x_1) = \tilde{x}]}{\Pr[\mathcal{M}(x_2) = \tilde{x}]} \leq e^{\varepsilon}$$

Distinguishability

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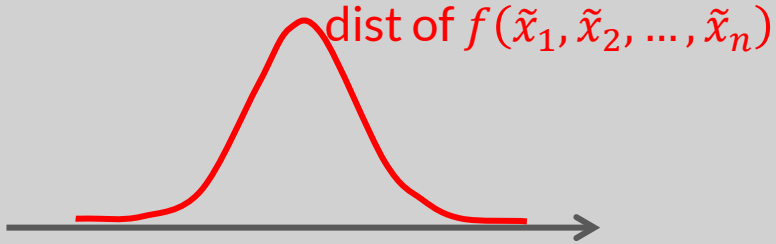
- The collector's / adversary's view: hard to infer the sensitive data

Privacy quantified by ε

$x_1 \rightarrow \mathcal{M} \rightarrow \tilde{x}$

Provable defense against data inference attacks

Data utility by approximation error



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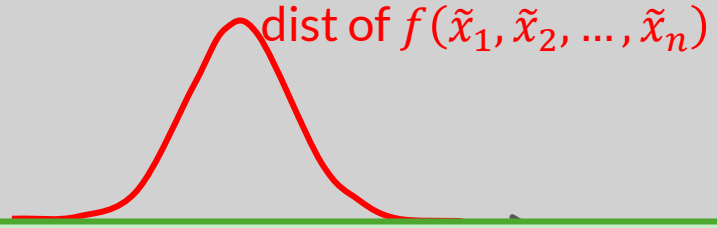

Fundamental direction: **Design** of $\mathcal{M}$ to **optimize** the privacy—utility tradeoff

LDP – Formal Privacy

RIT

Utility analysis of $f \circ \mathcal{M}$

$$f(x_1, x_2, \dots, x_n) := \sum_{i=1}^{n} x_i \quad \text{or} \quad f(x_1, x_2, \dots, x_n) := \{x_1, x_2, \dots, x_n\} \rightarrow \text{Variance, MSE}$$

$$f(x_1, x_2, \dots, x_n) := h: \mathbb{R}^n \rightarrow \{1,2, \dots, K\} \text{ is a classifier} \rightarrow$$

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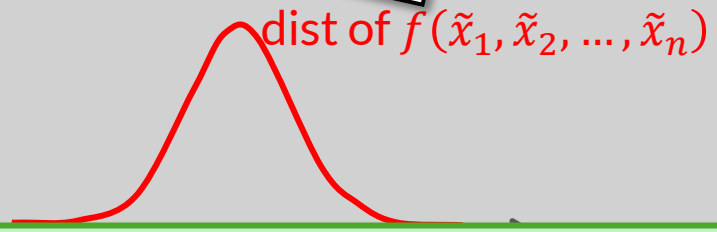

Fundamental direction: **Utility analysis** of complex task f

This Dissertation: LDP Theory

RIT

- Advancing LDP's mechanism design and utility analysis

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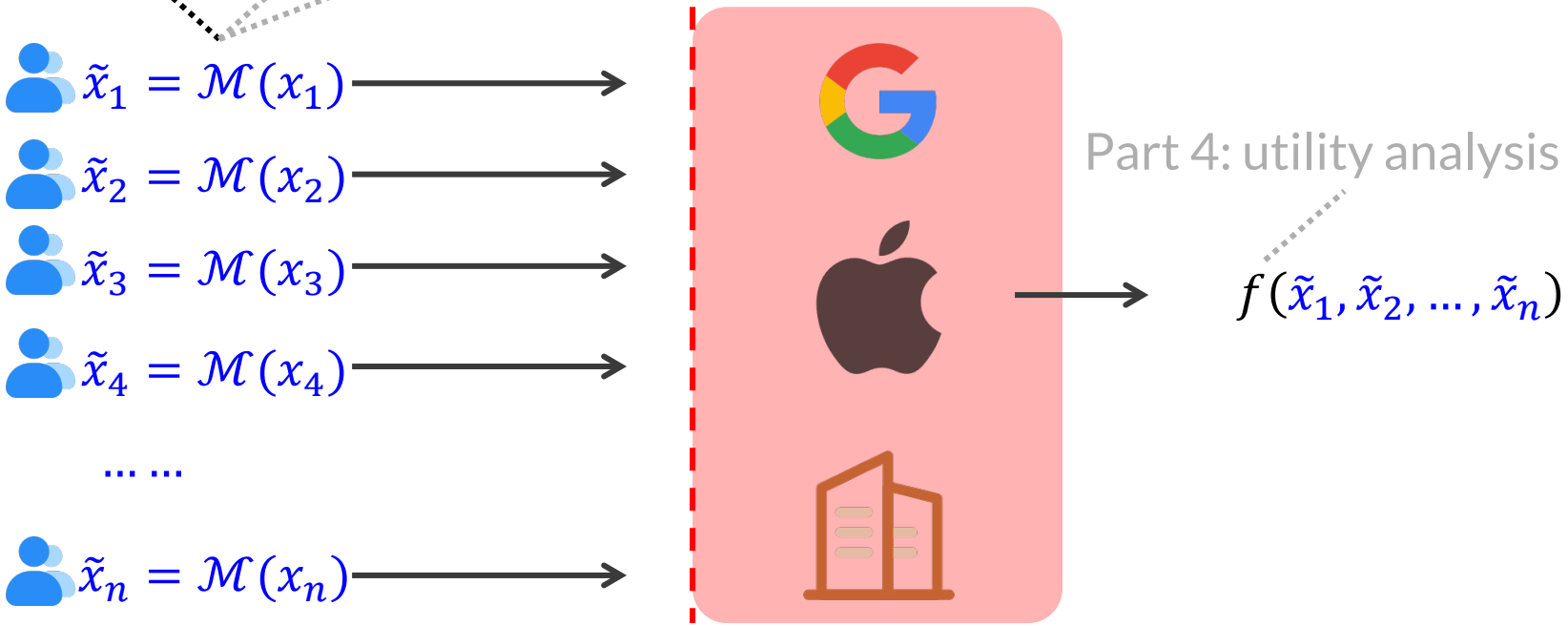
Part 1: correlated $\mathcal{M}$

Part 2: optimal piecewise-based $\mathcal{M}$

Part 3: $\mathcal{M}$ for trajectories in continuous space



Part 4: utility analysis for classifier $\circ\ \mathcal{M}$

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binary x → numerical x

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Mechanism-level
↓
Task-level

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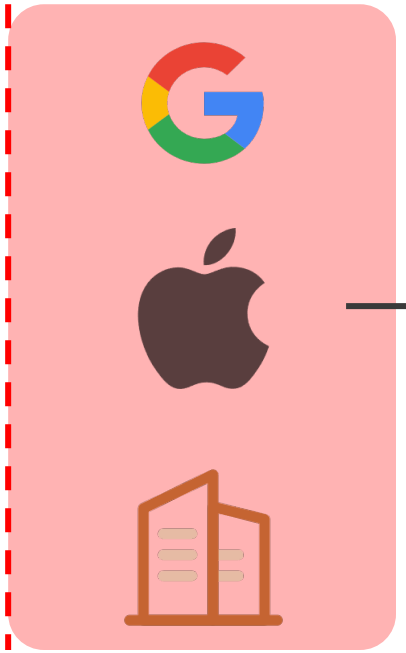

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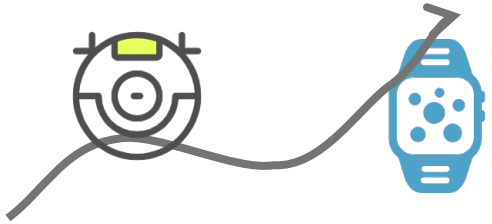
Societal Impacts

RIT

- **New LDP building blocks**
  - correlated LDP mechanisms
  - optimal piecewise-based mechanisms

Sensor networks & Federated learning, etc

- **Universal trajectory collection mechanisms**
  - applicable to both continuous / discrete space



Smart home & wearable devices' trajectories, etc

- **Analytical view of classifier utility under LDP-perturbed inputs**
  - choosing best ε when using classifiers

Local Differential Privacy: Refined Mechanism Design and Utility Analysis

Mechanism design

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- Part 2: Optimal piecewise-based $\mathcal{M}$
- Part 3: $\mathcal{M}$ for trajectories in continuous space

Utility analysis

- Part 4: Utility analysis for classifier $\circ$ $\mathcal{M}$

More complex data types & tasks

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Utility analysis

- Part 4: Utility analysis for $\text{classifier} \circ \mathcal{M}$

More complex data types & tasks

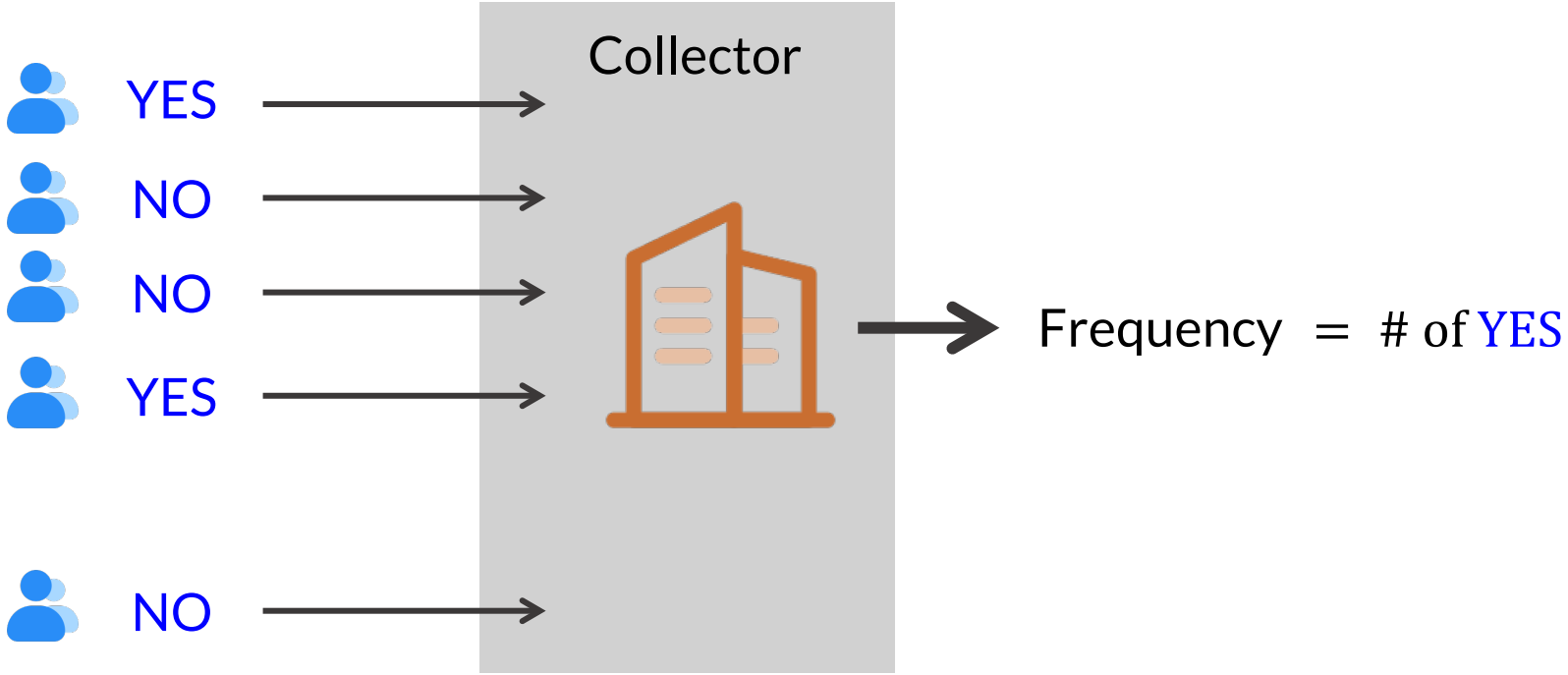
Frequency Estimation of YES/NO

RIT

- Social science: How to know how many people engage in tax evasion?

  - ask one person if they had evaded tax

  - the person answers YES or NO



Formal Privacy

RIT

- People have privacy concerns on sensitive/embarrassing question

  - i.e. don't want to answer → how to collect answers in a privacy-preserving way?

- A privacy mechanism $\mathcal{M}$ satisfies LDP if

For any truth x_1, x_2,
and randomized answer y:

$$\max \frac{\Pr[\mathcal{M}(x_1) = y]}{\Pr[\mathcal{M}(x_2) = y]} \leq e^{\varepsilon}$$

Distinguishability of x_1 (YES) and x_2 (NO)
from y (randomized answer)

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Distinguishability of x_1 (YES) and x_2 (NO)
from y (randomized answer)

  - **quantifiable hardness** to distinguish x_1 (YES) and x_2 (NO) from the randomized answer y

  - defense against inference from data collectors or adversaries

Randomized Response for Privacy

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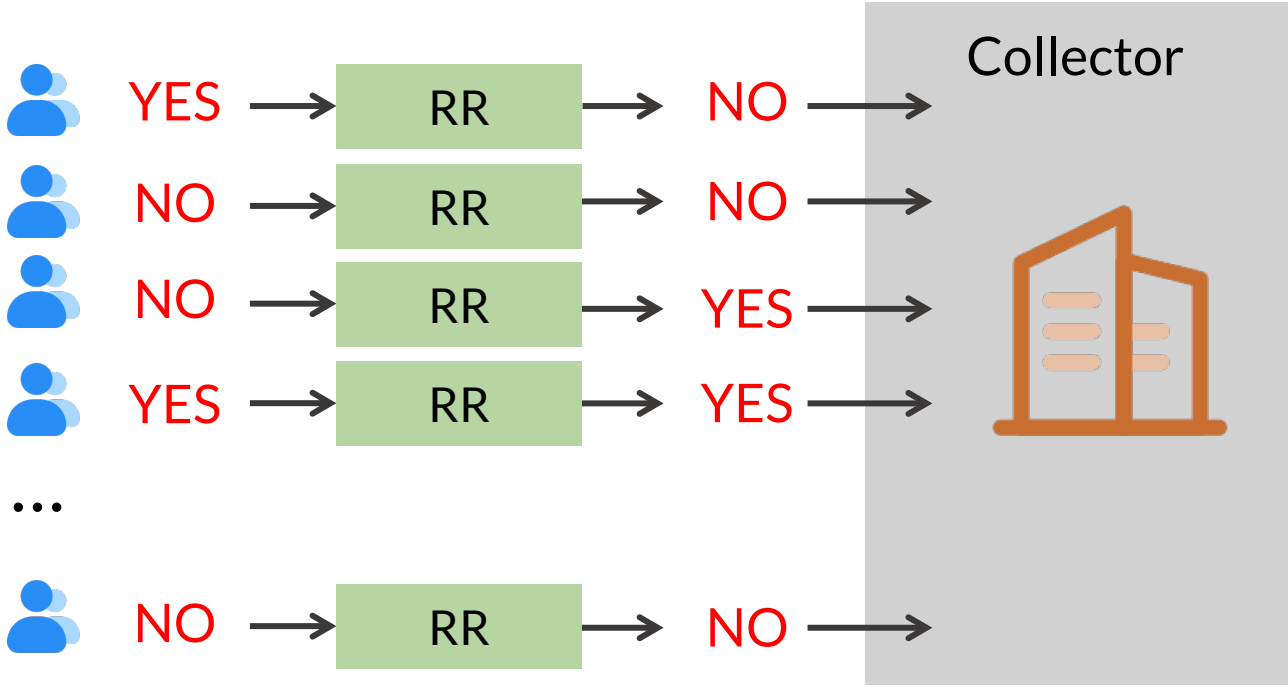
- People have privacy concerns on sensitive/embarrassing question

  - i.e. don't want to answer → how to collect answers in a privacy-preserving way?

- Randomized Response: Randomize the truth before answering the collector



Randomized Response for Privacy

$$\max \frac{\Pr[\mathbf{RR}(x_1) = y]}{\Pr[\mathbf{RR}(x_2) = y]} \le e^{\ln \frac{p}{1-p}}$$

- People have privacy concerns on sensitive/embarrassing questi
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Private

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answer truth with probability p

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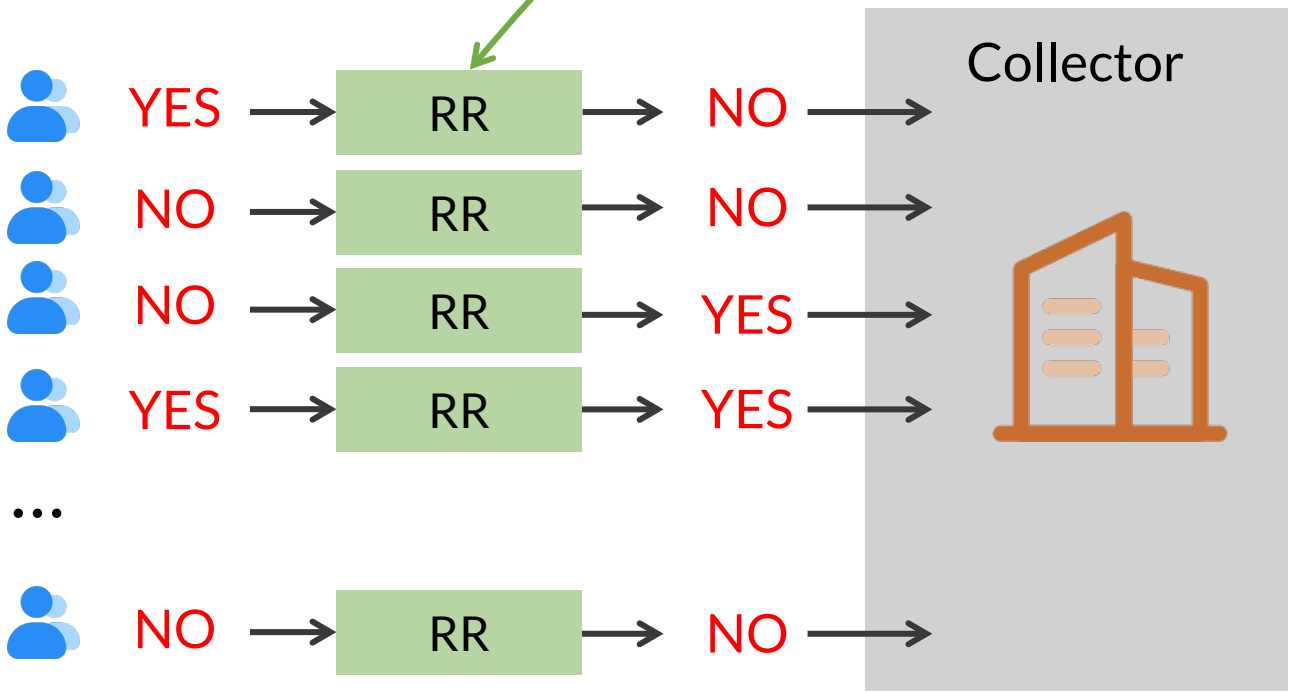

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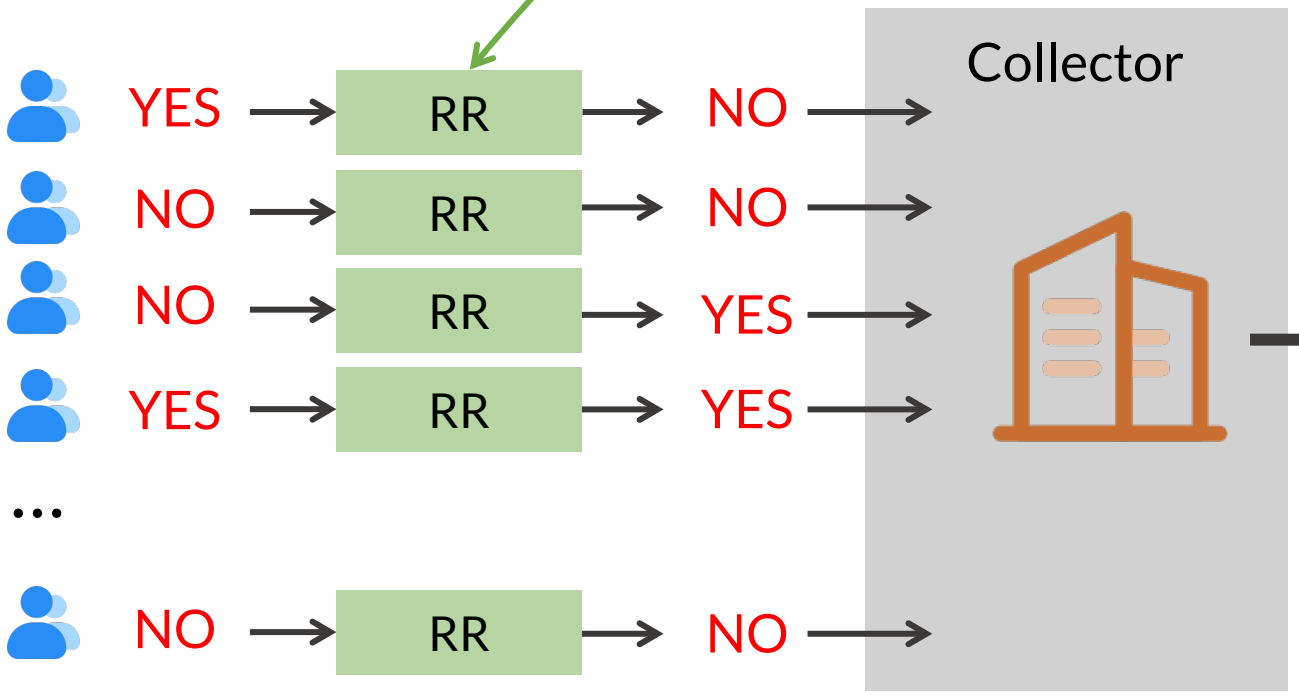

estimated frequency

$$= \frac{\#\text{ of YES} - \#\,\text{👥}\times q}{p-q}$$

Unbiased:
expectation = truth

Utility: RR's Variance

RIT

- Randomization reduces data utility

$$\mathrm{Var}\left[\frac{\#\text{ of YES} - \#\,\text{👤} \times q}{p-q}\right] = \frac{\mathrm{Var}[\#\text{ of YES}]}{(p-q)^2} = \frac{npq}{(p-q)^2}$$

- classical result: summation of variance from all n **independent** randomization

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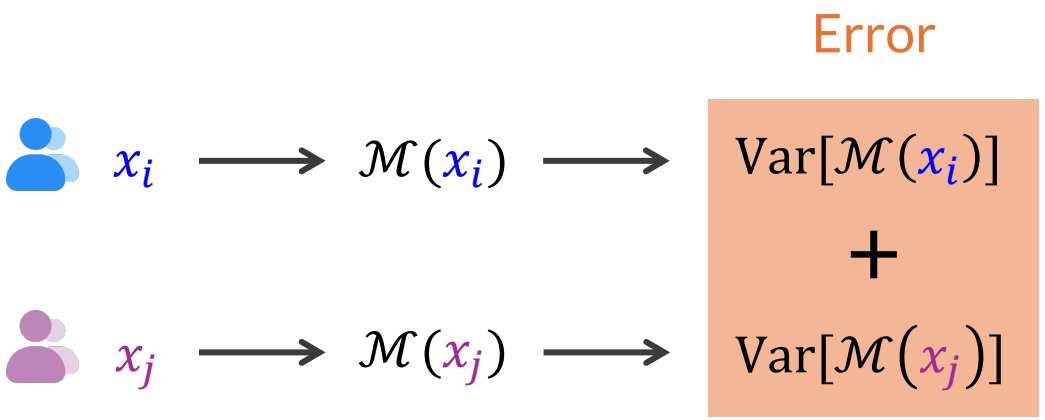
- Q: Can we do better?
  - **yes, by correlated (joint) randomization**

Joint/Correlated RR (JRR)

RIT

- Existing LDP mechanisms: Each user perturbs their data independently



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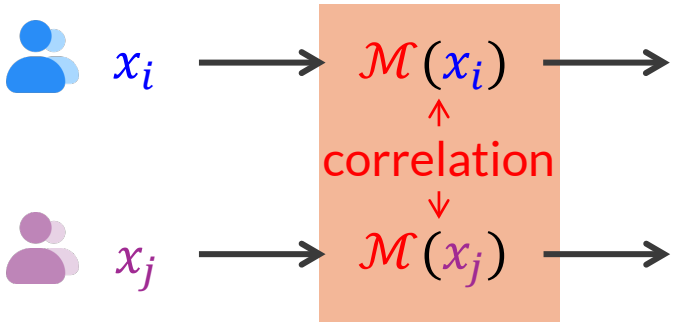

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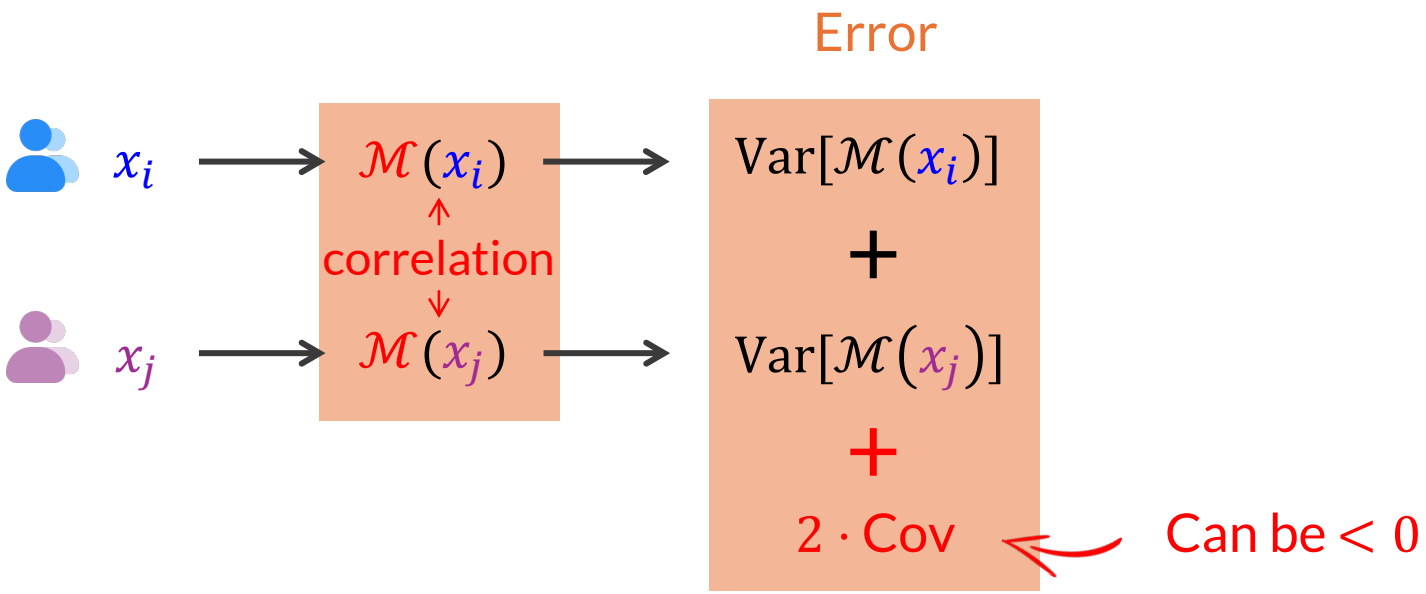

Joint/Correlated RR (JRR)

RIT

- JRR: Better data utility by joint/correlated randomization
- **Example:** 2-person ($x_1 = \text{YES}$ and $x_2 = \text{YES}$) with $p = 0.8$ ($\text{P}[T = 1] = 0.8$)

RR: Joint distribution

	$T_1 = 1$	$T_1 = 0$
$T_2 = 1$	0.64 ($= p^2$)	0.16 ($= pq$)
$T_2 = 0$	0.16 ($= pq$)	0.04 ($= q^2$)

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Truthfulness of x_2

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$\mathbf{Cov}[y_1, y_2] = -0.04$

$\text{P}[T_1 = 0 \cap T_2 =$ $0 \neq \text{P}[T_1 = 0] \cdot \text{P}[T_2 = 0]$=0.04

NOT independent T_1 and T_2

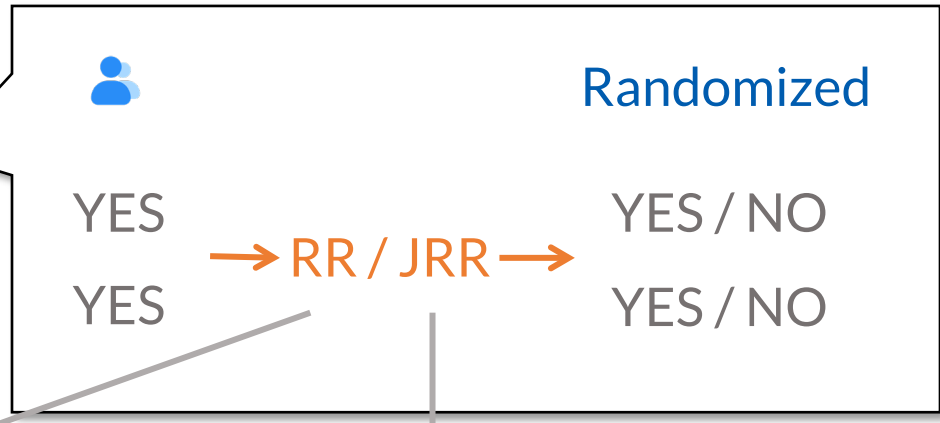
Joint probability $\neq$ Π of marginal probabilities

Utility: JRR's Variance

RIT

- Variance: (# 👥 = 2, $p = 0.8$)

$$\mathrm{Var}[\hat{n}_{\mathrm{YES}}] = \frac{\mathrm{Var}[\#\text{ of YES}]}{(0.8-0.2)^2}$$



RR

$$\mathrm{Var}[\#\text{ of YES}] = 2 \times pq = \mathbf{0.32}$$

JRR

$$\mathrm{Var}[\#\text{ of YES}] = 0.32 + 2 \cdot \mathbf{Cov}[\boldsymbol{y_1}, \boldsymbol{y_2}]$$

$$= 0.32 + 2 \cdot (-0.04)$$

$$= \mathbf{0.24}$$

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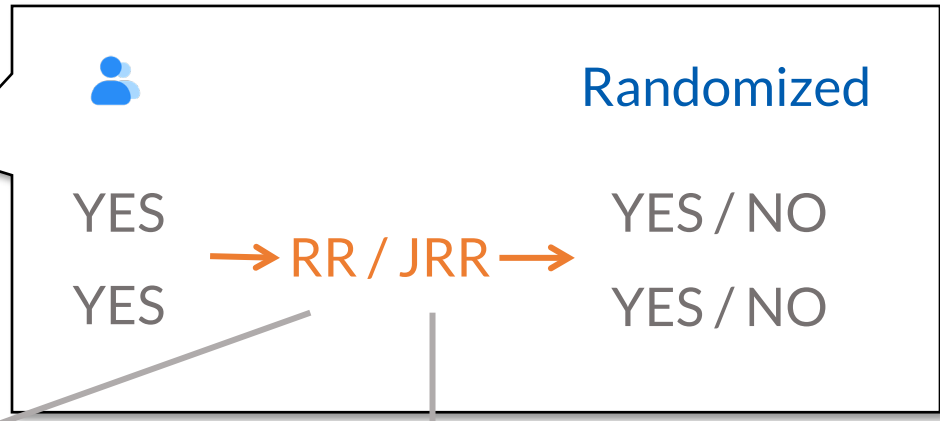

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$$= 0.32 + 2 \cdot (-0.04)$$
$$= \mathbf{0.24}$$

Better data utility

JRR's General Form

RIT

- Correlated randomization with 2 persons x_{2i-1} and x_{2i}

JRR: Joint distribution

	$T_{2i-1} = 1$	$T_{2i-1} = 0$
$T_{2i} = 1$	$p^2 + \rho pq$	$(1 - \rho)pq$
$T_{2i} = 0$	$(1 - \rho)pq$	$q^2 + \rho pq$

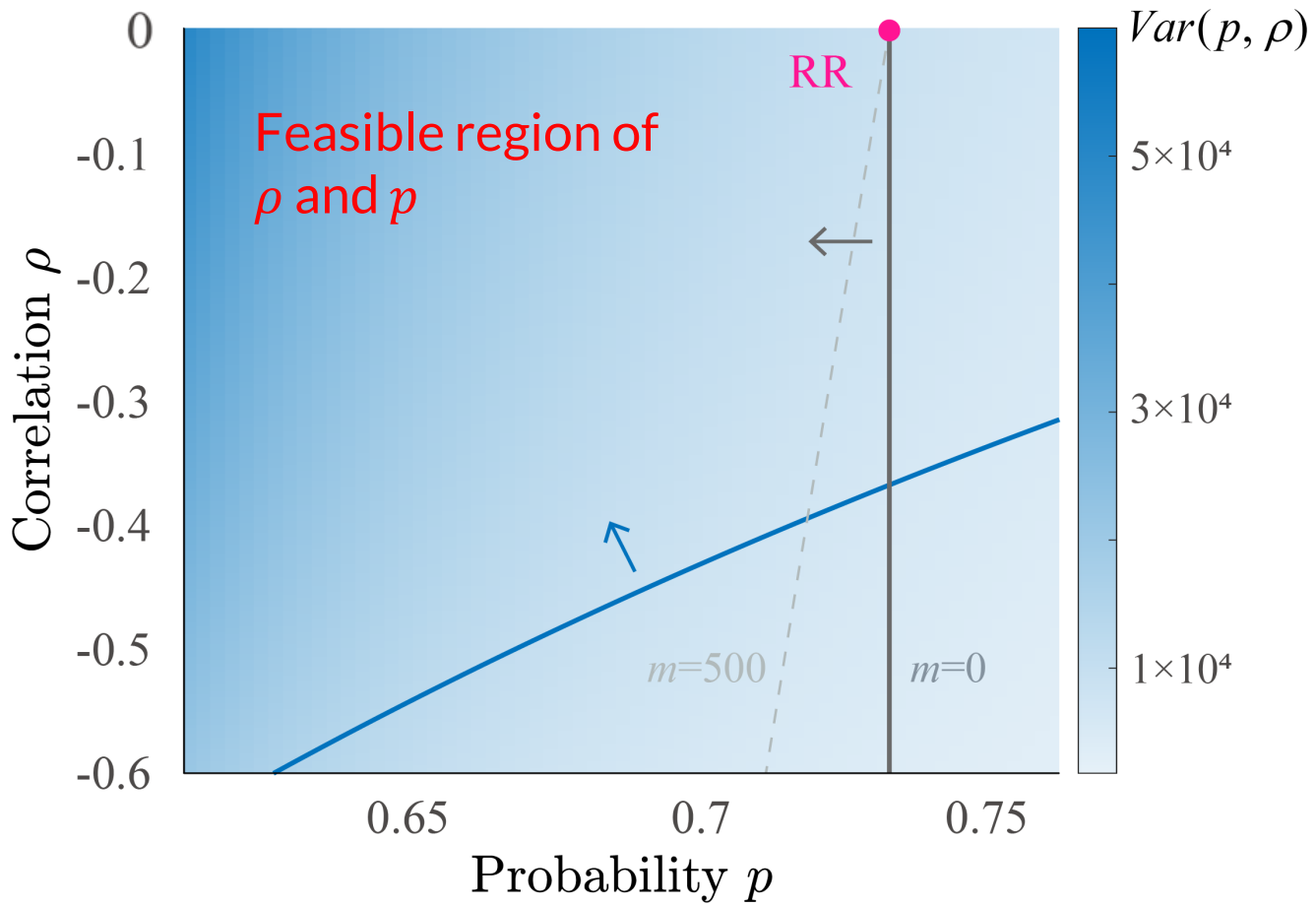
$\rho \in [-1,1]$: correlation coefficient

- RR is a special case of JRR with $\rho = 0$ (no correlation)

JRR – Variance Heatmap

RIT

- Effect of ρ and p (when $\varepsilon = 1, n = 10^4, n_{\text{Yes}} = 200$, and malicious users $m = 0$ & 500)



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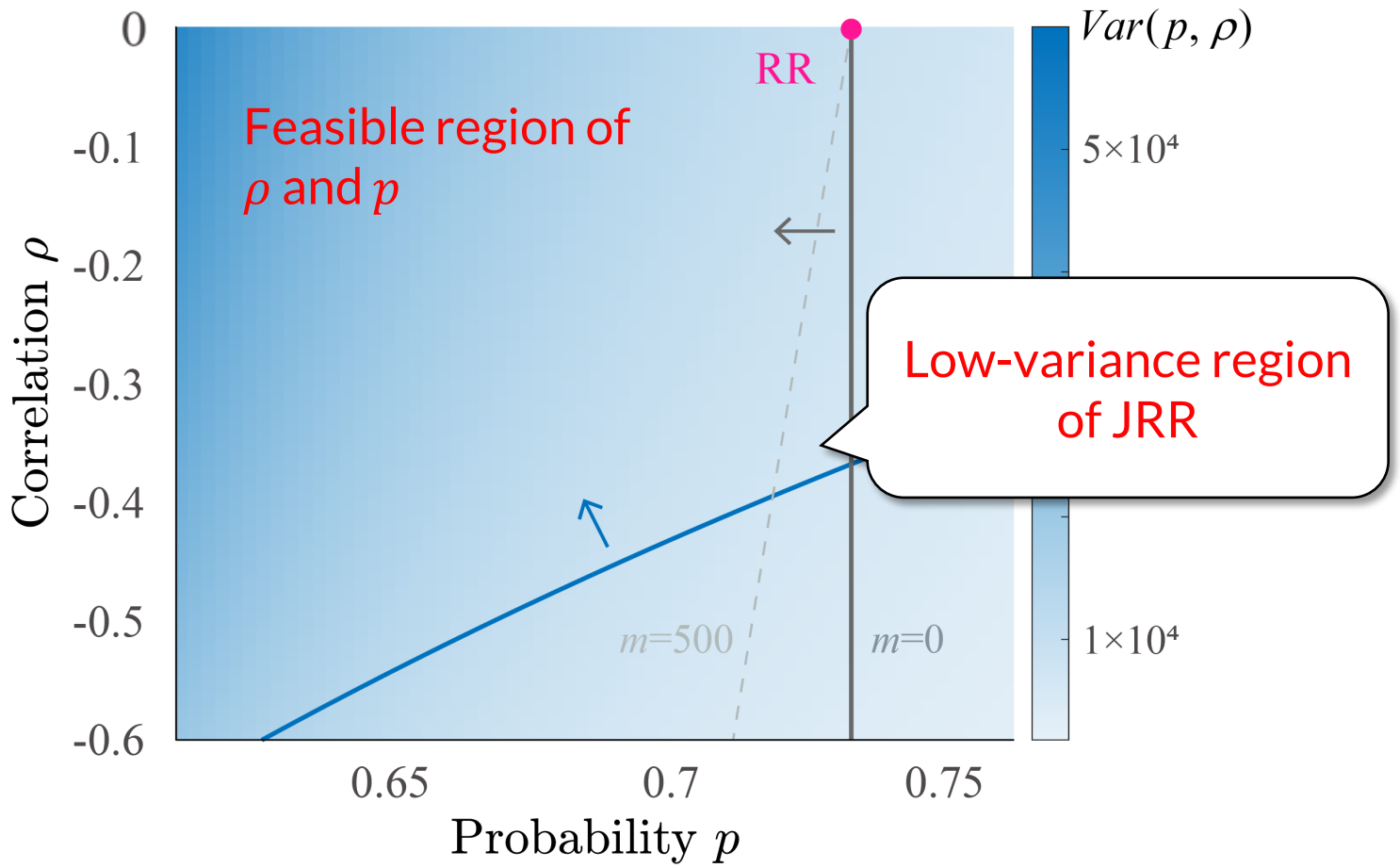
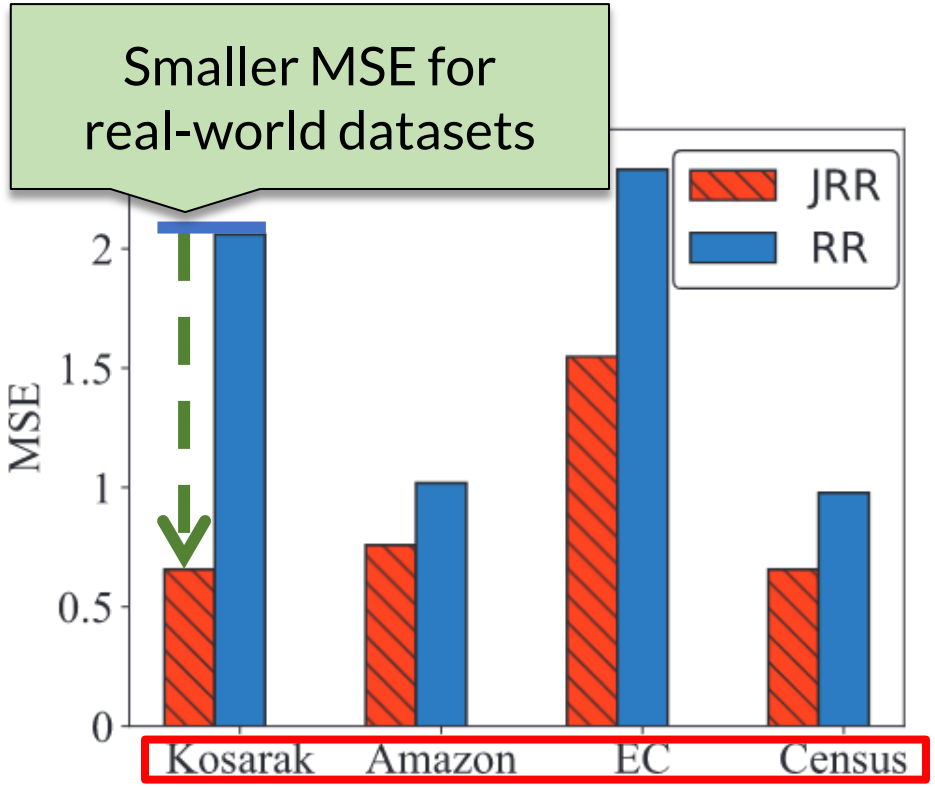


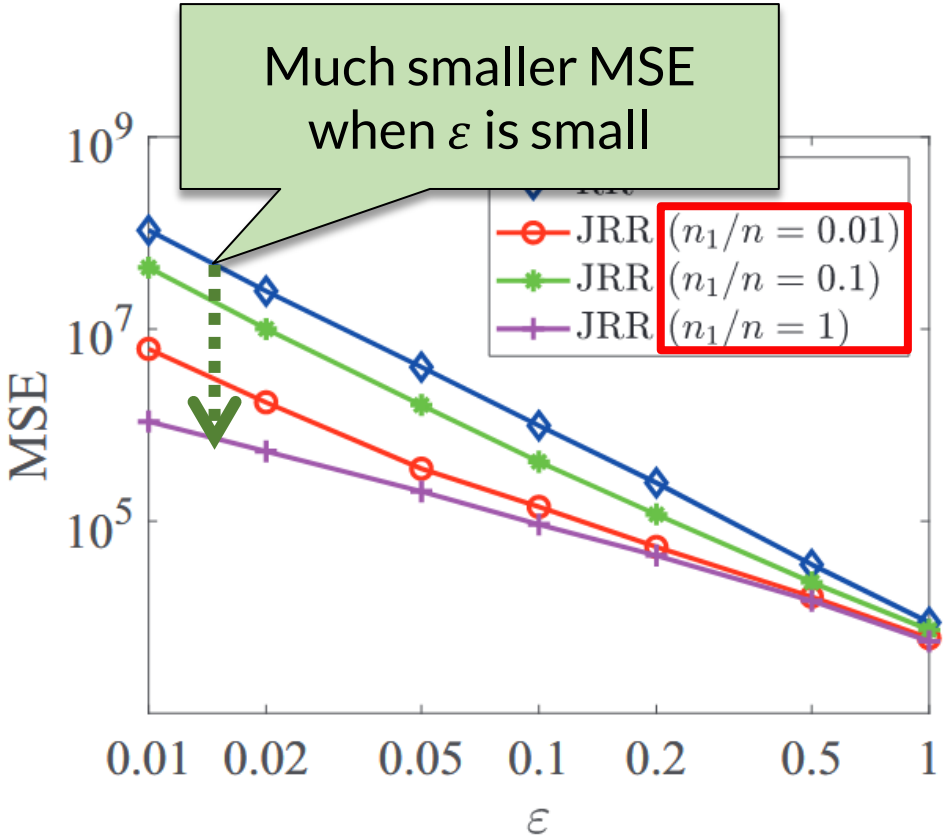
Evaluations

RIT

- Comparison with RR under the same privacy level - JRR: $\varepsilon(n, m, \rho, p)$, RR: $\varepsilon(p)$



Real-world datasets ($\varepsilon = 0.1$)



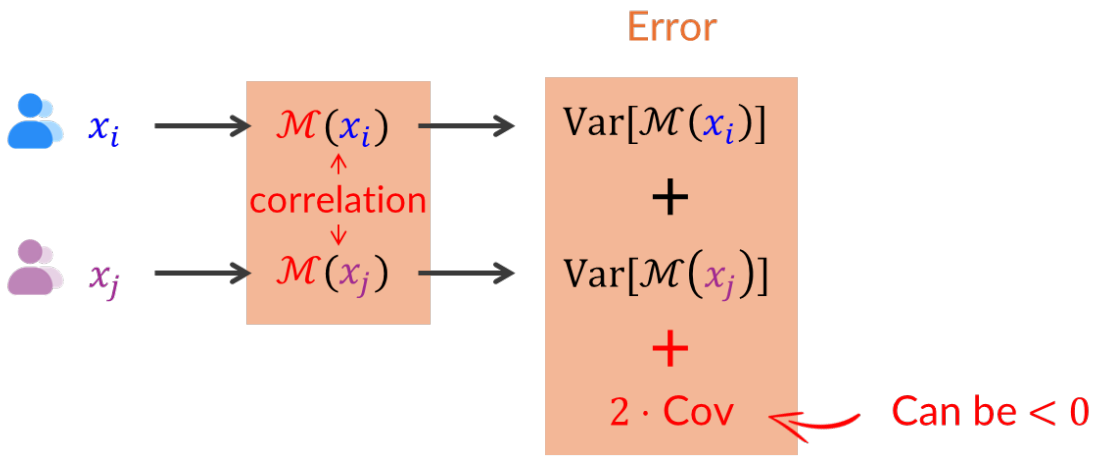
Synthetic datasets ($n = 10^4$)

Summary & Takeaway

RIT

- **RQ: Can the classical RR be better?**

- Contributions:*
  - **introducing correlation into RR**
  - extensible to other LDP mechanisms
  - RR as a special case of $\rho = 0$
  - analytical privacy & data utility analysis



* Locally Differentially Private Frequency Estimation via Joint Randomized Response, PETS'25

Local Differential Privacy: Refined Mechanism Design and Utility Analysis

Mechanism design

- Part 1: Correlated $\mathcal{M}$
- Part 2: Optimal piecewise-based $\mathcal{M}$
- Part 3: $\mathcal{M}$ for trajectories in continuous space

binary x $\rightarrow$ numerical x

Utility analysis

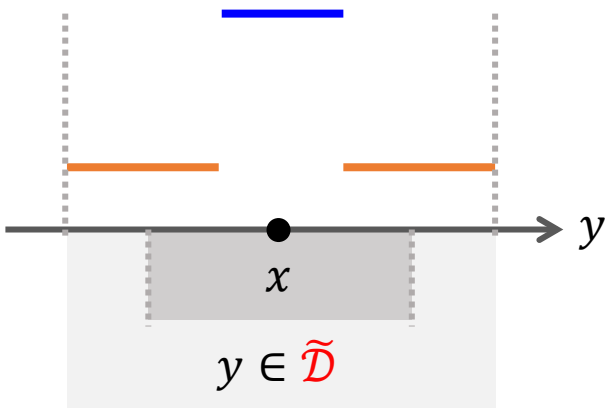
- Part 4: Utility analysis for $\mathrm{classifier} \circ \mathcal{M}$

3-Piecewise Mechanism

RIT

- 3-piecewise distributions on bounded numerical domain $\mathcal{D} \to \widetilde{\mathcal{D}}$

  - given input x, sample output y from a distribution



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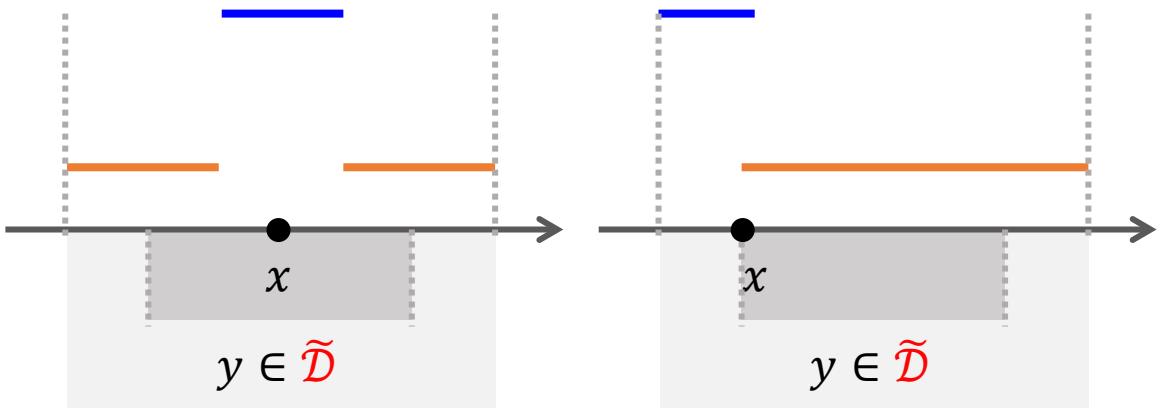

Sampling probability depends on ε

Sampling interval depends on x and ε

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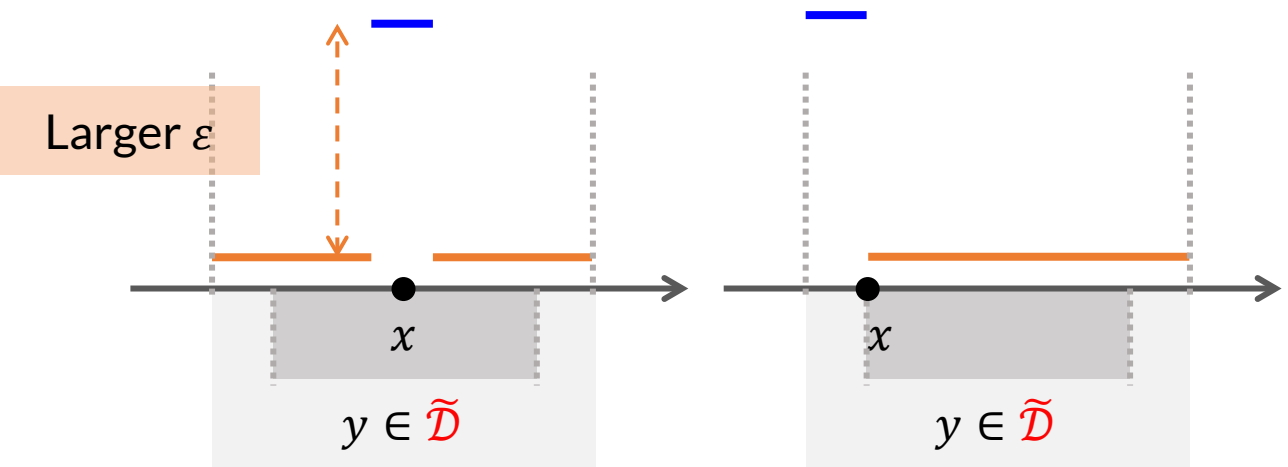

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  - different errors, but **without optimality,** which is important for building blocks

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NOT enough to study optimality of piecewise-based mechanism

- only 3 pieces, two probabilities

- 3-piecewise distributions on boun

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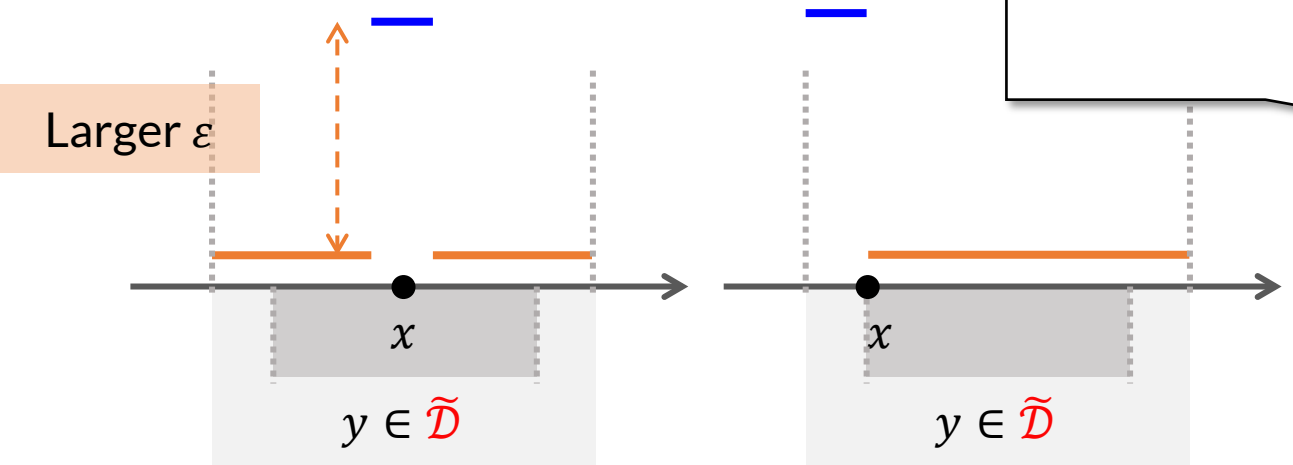

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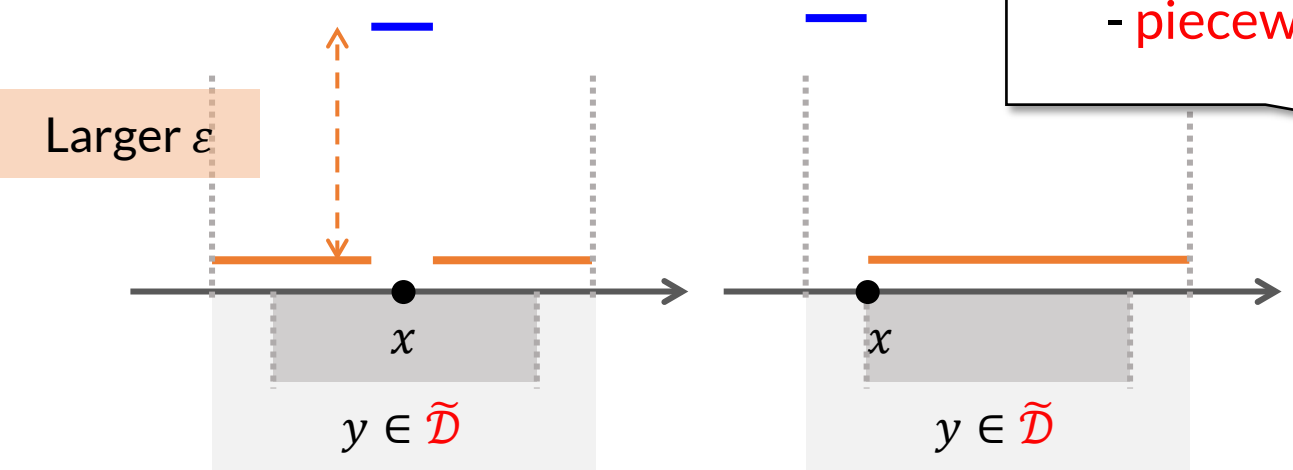

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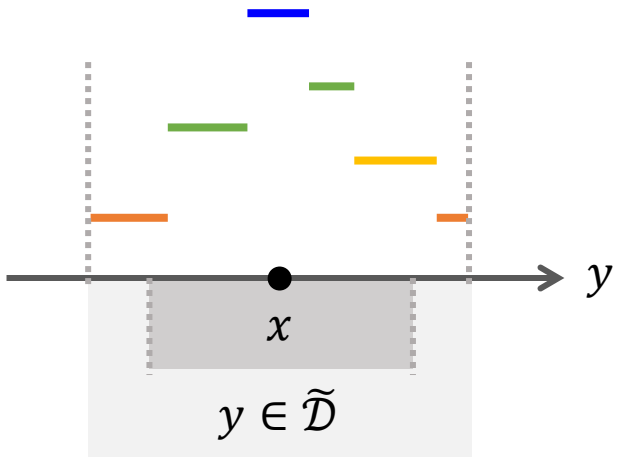
Generalized Piecewise-based Mechanism

- **Most generalized version:** m-piecewise distribution



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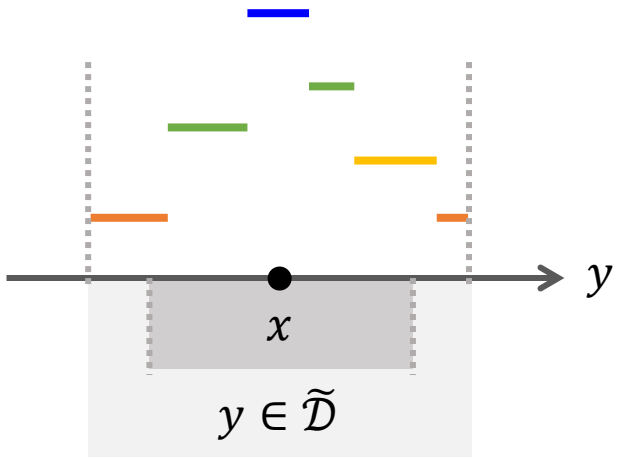

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$\frac{p_{i,\varepsilon}}{p_{j,\varepsilon}} \leq e^{\varepsilon}$ (LDP constraint)

- Error (data utility): $\mathcal{L}(y, x)$

$\uparrow$

$\mathcal{L}(y, x) := |y - x|^p$

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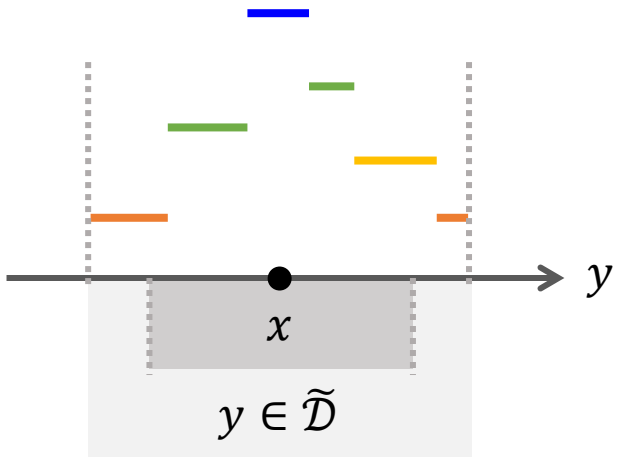

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$$\frac{p_{i,\varepsilon}}{p_{j,\varepsilon}} \leq e^{\varepsilon} \text{ (LDP constraint)}$$

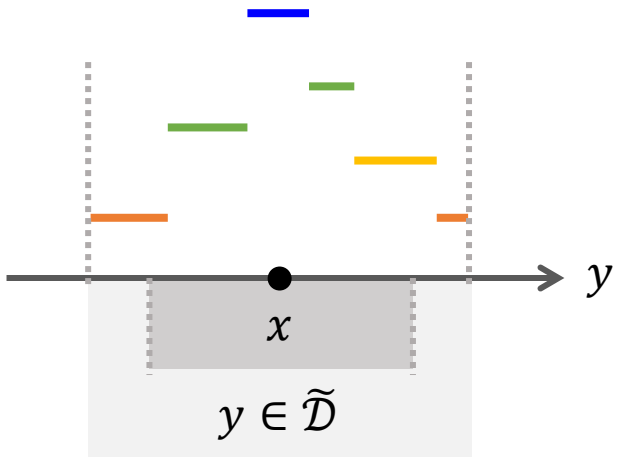
- Expected error:

$$\int_{\widetilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

Optimal Piecewise-based Mechanism

RIT

- **Most generalized version:** m-piecewise distribution



$$pdf[\mathcal{M}(x) = y] = \begin{cases} p_{1,\varepsilon} & \text{if } y \in [l_{1,x,\varepsilon}, r_{1,x,\varepsilon}] \\ p_{2,\varepsilon} & \text{if } y \in [l_{2,x,\varepsilon}, r_{2,x,\varepsilon}] \\ & \dots \\ p_{m,\varepsilon} & \text{if } y \in [l_{m,x,\varepsilon}, r_{m,x,\varepsilon}] \end{cases}$$

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- Expected error:

$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \int_{\widetilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

Find $\mathcal{M}$ to minimize the error at x

Optimal Piecewise-based Mechanism

RIT

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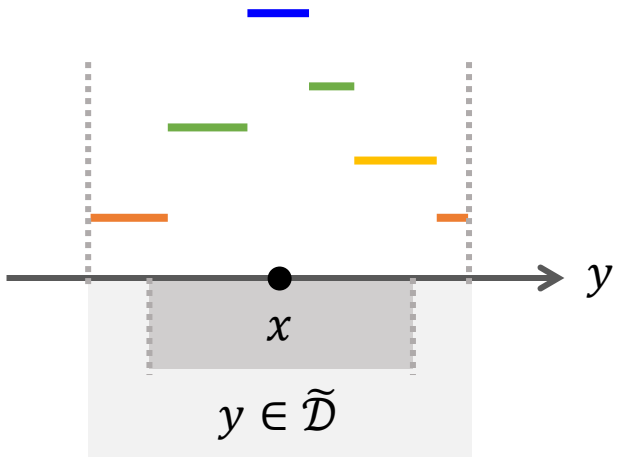

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- Expected error:

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Find $\mathcal{M}$ to minimize the worst-case error

Optimal Piecewise-based Mechanism

RIT

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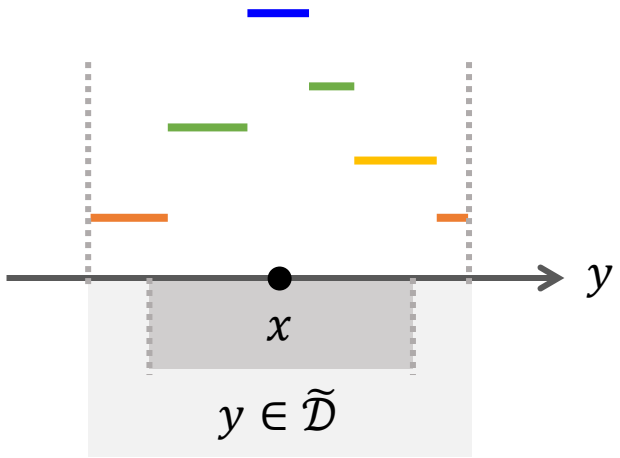

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Find $\mathcal{M}$ to minimize the worst-case error

Solved $\mathcal{M}$ is
the optimal piecewise-based mechanism

Mathematically ≡ to find the **optimal piecewise distribution under the LDP constraint**

Challenges & Proofs

RIT

- Challenges
  1. min-max problem & multiple variables
  2. optimal results only for a specific m

$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \max_{x \in \mathcal{D}} \int_{\widetilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

2. $i \in [m]$

1. variables p_i, l_i, r_i

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1

Inner max
has a closed form

Worst-case from
x = endpoints of $\mathcal{D}$

Reduced to:
minimization problem

Challenges & Proofs

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minimization problem

2

The optimal #piece
is m

If optimal m-piecewise =
$(m+1)$-piecewise

Optimal m:
Check $m = 1,2, \ldots$ until optimal

"Solvable"

Challenges & Proofs

RIT

- Challenges
  1. min-max problem & multiple variables
  2. optimal results only for a specific m

$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \max_{x \in \mathcal{D}} \int_{\widetilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

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1 Inner max has a closed form

Worst-case from

2 The optimal #piece is m

If optimal m-piecewise = $(m+1)$-piecewise

"Solvable"

Monte Carlo testing for 2

Hypothesis. For any domain $\mathcal{D} \to \mathcal{D}$, under error metrics $\mathcal{L}(y, x) := |y - x|$ and $\mathcal{L}(y, x) := (y - x)^2$, the optimal piecewise-based mechanism falls into 3-piecewise mechanism

minimization problem

different from existing instantiations

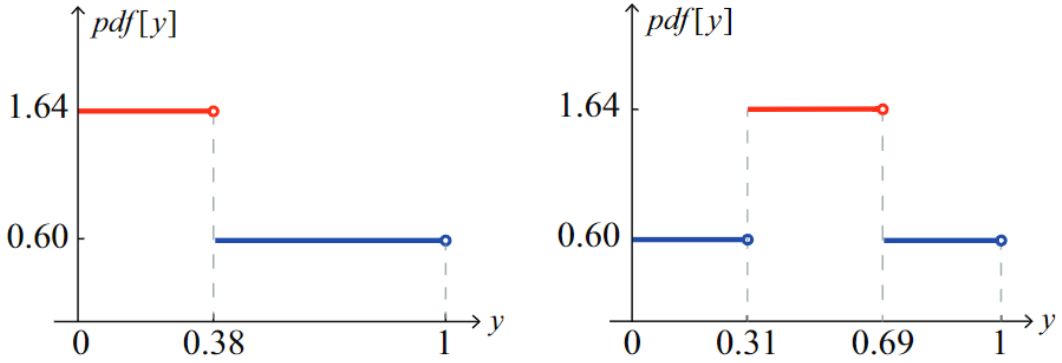
Optimal Closed-Form Mechanism

RIT

- Optimal $\mathcal{M}: [0,1) \to [0,1)$ under $\mathcal{L} := |y - x|$

$$pdf[\mathcal{M}(x) = y] = \begin{cases} \exp\left(\frac{\varepsilon}{2}\right) & \text{if } y \in [l_{x,\varepsilon}, r_{x,\varepsilon}) \\ \exp\left(-\frac{\varepsilon}{2}\right) & \text{if } y \in [0,1) \setminus [l_{x,\varepsilon}, r_{x,\varepsilon}) \end{cases} \longleftarrow [l_{x,\varepsilon}, r_{x,\varepsilon}) = \begin{cases} [0,2C) & \text{if } x \in [0,C) \\ x + [-C,C) & \text{if } x \in [C,1-C) \\ [1-2C,1) & \text{otherwise} \end{cases} \longleftarrow C = \frac{\exp(\varepsilon/2) - 1}{2(\exp(\varepsilon) - 1)}$$

- When $\varepsilon = 1$:



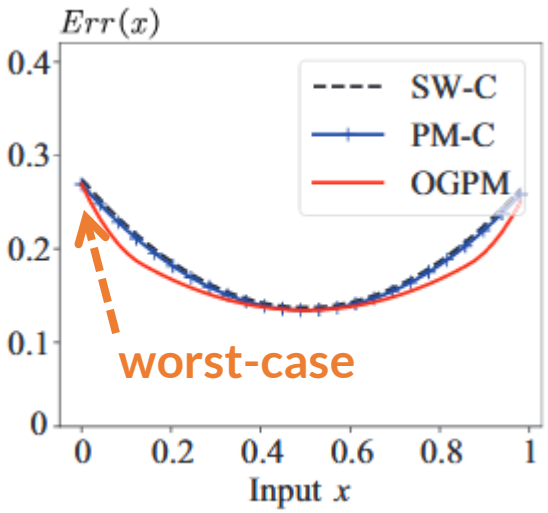
(a) $x = 0$. **(b)** $x = 0.5$.

Comparison of Expected Errors

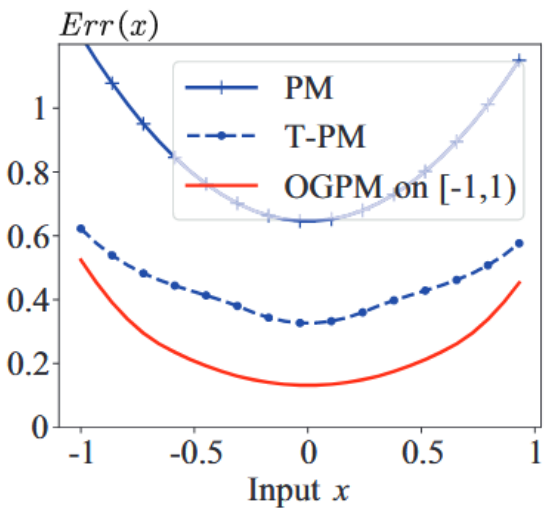
RIT

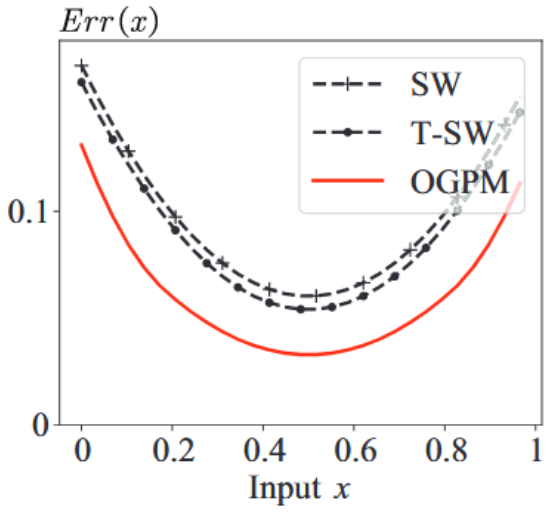
Whole-domain error (i.e. each point in $\mathcal{D}$) **($\varepsilon = 2$)**

Compressed PM, SW



Original and **truncated** PM, SW

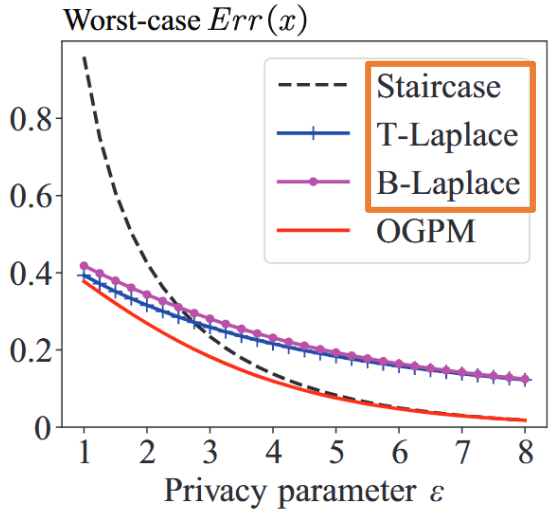




Lowest error

Worst-case error

Non-piecewise-based

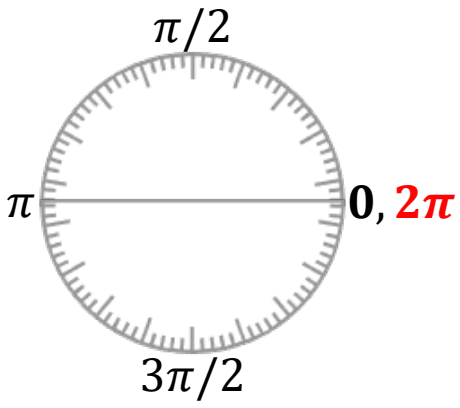


Lowest error

Circular Domain

RIT

- Different meaning of distance, e.g. $\text{distance}(0, 2\pi) = 0 \rightarrow$ requires different mechanisms



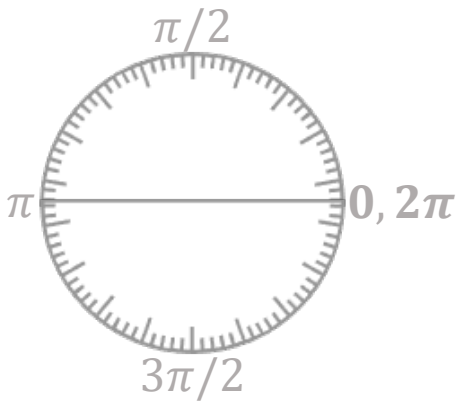
$\mathcal{L} \rightarrow \mathcal{L}_{\text{mod}}$

$$\mathcal{L}_{\text{mod}}(y, x) \coloneqq \min(\mathcal{L}(y, x), \mathcal{L}(y, 2\pi - x))$$

Circular Domain

RIT

- Different meaning of distance, e.g. $\text{distance}(0, 2\pi) = 0 \rightarrow$ requires different mechanisms



$$\mathcal{L} \rightarrow \mathcal{L}_{\text{mod}}$$

$$\mathcal{L}_{\text{mod}}(y, x) := \min(\mathcal{L}(y, x), \mathcal{L}(y, 2\pi - x))$$

$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \max_{x \in [0, 2\pi]} \int_{\tilde{\mathcal{D}}} \mathcal{L}_{\text{mod}}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

- **Linking** to problems in the classical domain

min-max under $\mathcal{L}_{\text{mod}}$	=	min under $\mathcal{L}$ at π		min under $\mathcal{L}_{\text{mod}}$ at x	=	min under $\mathcal{L}$ at π + Transform
		p_i				$l_{i,x}^{\text{mod}}, r_{i,x}^{\text{mod}}$

Optimal Closed-Form Mechanism

RIT

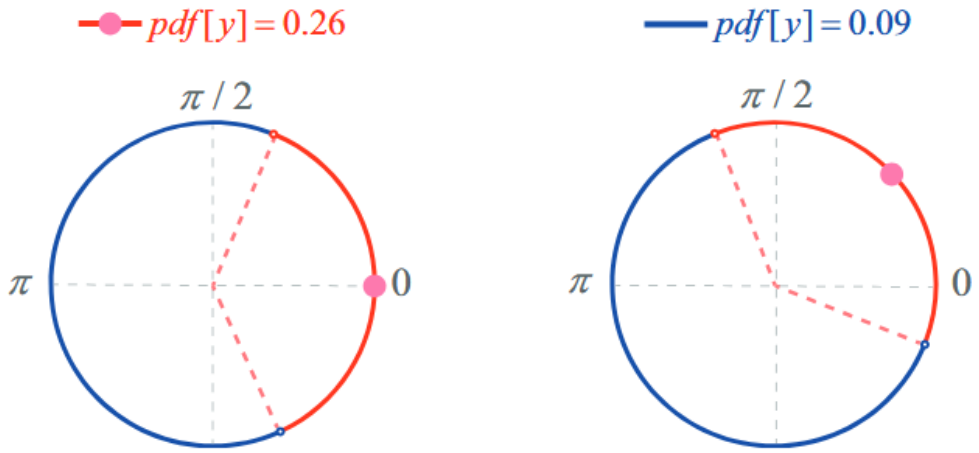
- Optimal $\mathcal{M}: [0,2\pi) \to [0,2\pi)$ under $\mathcal{L}_{\text{mod}}$

$$pdf[\mathcal{M}(x) = y] = \begin{cases} \frac{1}{2\pi}\exp\left(\frac{\varepsilon}{2}\right) & \text{if } y \in [l_{x,\varepsilon}^{\text{mod}}, r_{x,\varepsilon}^{\text{mod}}) \\ \frac{1}{2\pi}\exp\left(-\frac{\varepsilon}{2}\right) & \text{if } y \in [0,2\pi)\backslash[l_{x,\varepsilon}^{\text{mod}}, r_{x,\varepsilon}^{\text{mod}}) \end{cases}$$

$$[l_{x,\varepsilon}^{\text{mod}}, r_{x,\varepsilon}^{\text{mod}}) = [x - C, x + C) \mod 2\pi$$

$$C = \pi\frac{\exp(\varepsilon/2) - 1}{\exp(\varepsilon) - 1}$$

- When $\varepsilon = 1$:



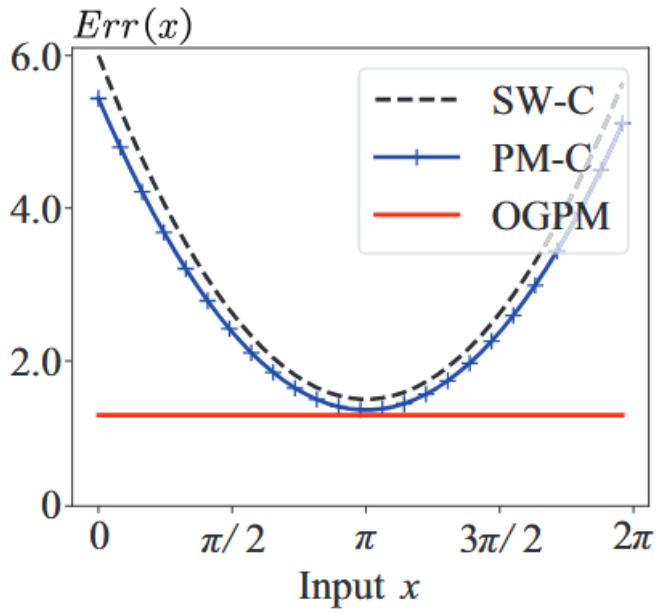
(a) $x = 0$. **(b)** $x = \pi/4$.

Comparison of Expected Errors

RIT

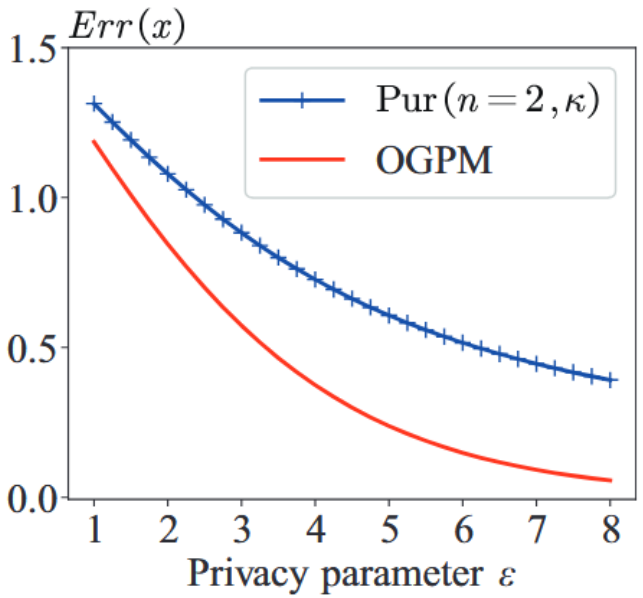
Whole-domain error ($\varepsilon = 2$)

PM, SW on the **flattened** domain



Worst-case error

Purkayastha mechanism [CCS'21]*



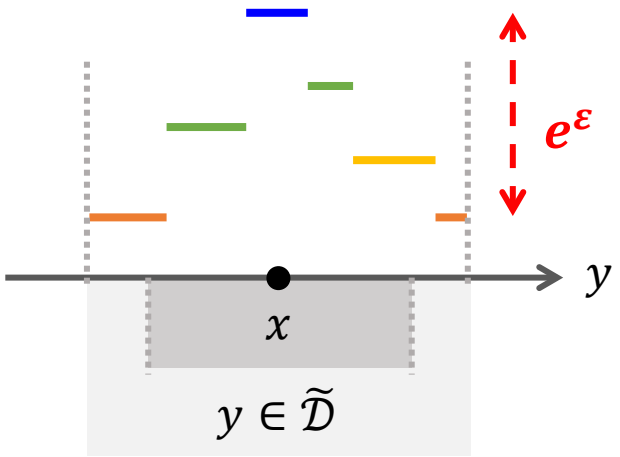
Lowest error

* Differential Privacy for Directional Data, CCS'21

Summary & Takeaway

RIT

- **RQ: What is the optimal piecewise-based mechanism?**



- Contributions:*

  - solving framework for the optimality

  - closed-form mechanisms for classical domains & circular domains

  - comparison with non-piecewise-based mechanisms

* Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy, PETS'25

Local Differential Privacy: Refined Mechanism Design and Utility Analysis

Mechanism design

- Part 1: Correlated $\mathcal{M}$
- Part 2: Optimal piecewise-based $\mathcal{M}$
- Part 3: $\mathcal{M}$ for trajectories in continuous space

1D x $\rightarrow$ 2D x

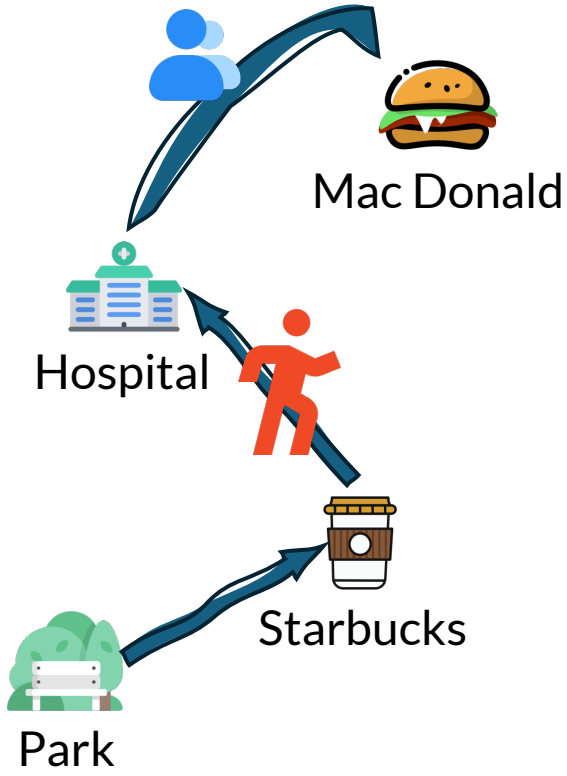
Utility analysis

- Part 4: Utility analysis for $\text{classifier} \circ \mathcal{M}$

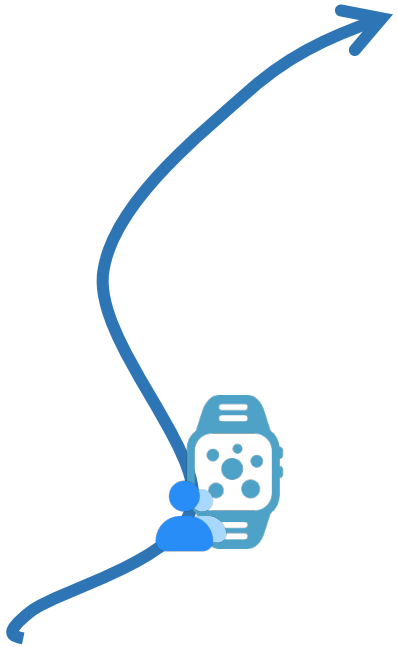
Trajectory Collection

RIT

- Sensitive trajectories: daily routine, wearable-sensor trajectories



Discrete locations w/ labels



Continuous locations w/o labels

Trajectory Collection

RIT

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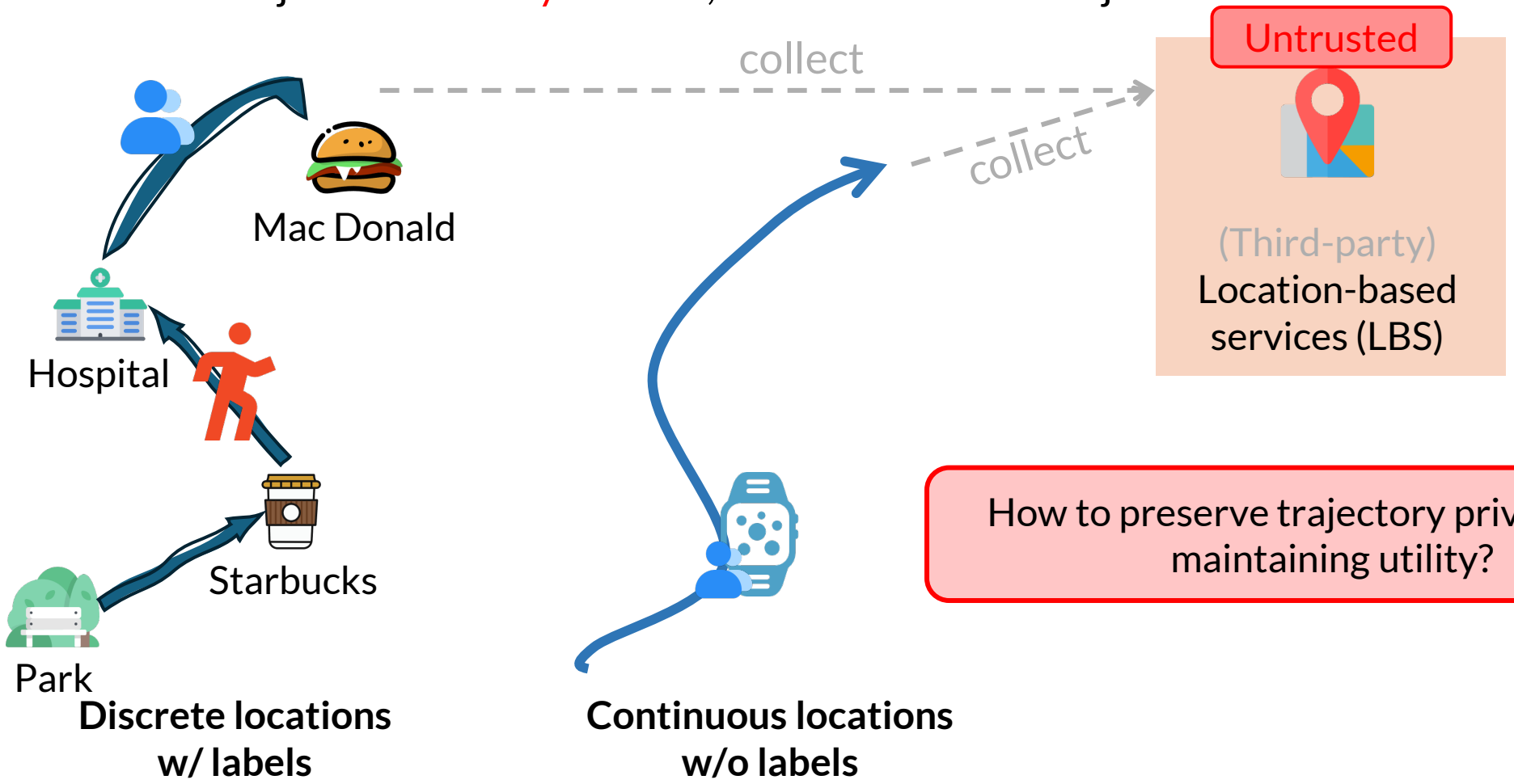

How to preserve trajectory privacy while maintaining utility?

LDP-fy a Trajectory

RIT

- LDP-fy: perturb a trajectory with LDP guarantee (provable privacy)
  - cannot distinguish location τ_1 from τ_2 with confidence quantified by e^{ε}

$$\forall \tau_1, \tau_2, \tau_* \in \mathcal{S}: \quad \frac{\Pr[\mathcal{M}(\tau_1) = \tau_*]}{\Pr[\mathcal{M}(\tau_2) = \tau_*]} \leq e^{\varepsilon}$$

location space

distinguishability of τ_1 from τ_2 when observing τ_*

LDP-fy a Trajectory

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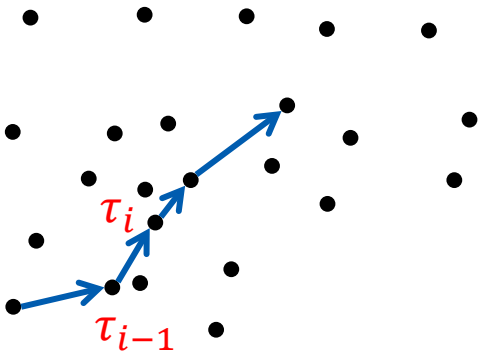
location space

distinguishability of τ_1 from τ_2 when observing τ_*

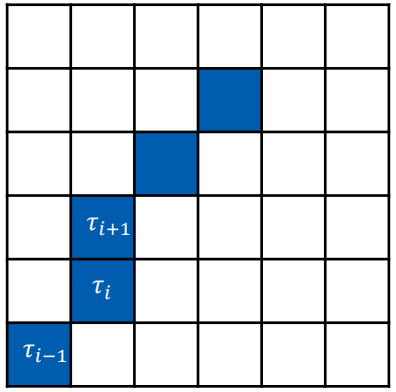
- Target: LDP mechanism $\mathcal{M}$ (for the location space)
  - provable privacy for users' trajectories, in continuous spaces and discrete spaces
  - as high trajectory utility as possible

LDP-fy a Trajectory in A **Discrete Space**

- Discrete location spaces: Point of interests or discretized cells



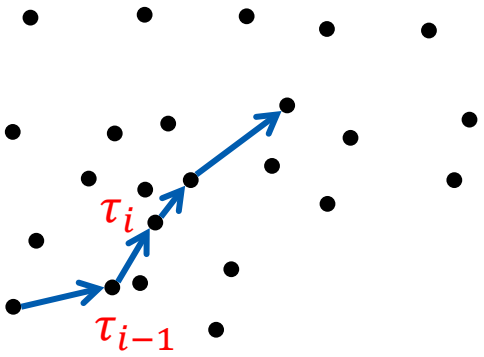
$\mathcal{S} = \{p_1, \dots, p_n\}$



$\mathcal{S} = \{c_1, \dots, c_n\}$

LDP-fy a Trajectory in A **Discrete Space**

- Discrete location spaces: Point of interests or discretized cells



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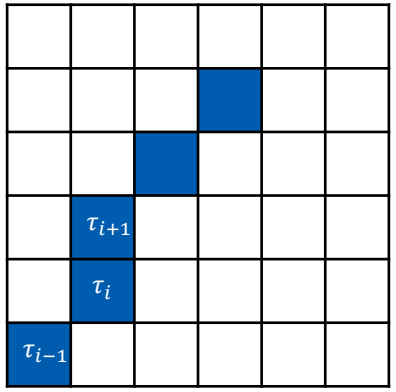


$\mathcal{S} = \{c_1, \dots, c_n\}$

- Exponential mechanism:

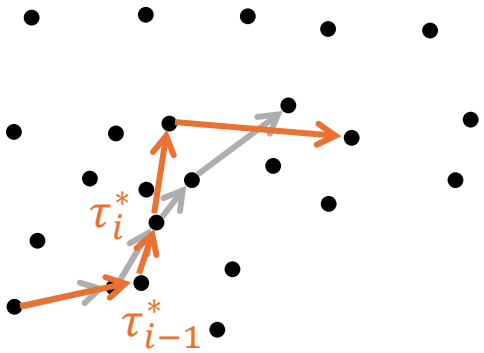
$$\Pr[\mathcal{M}_{\exp}(\tau) = \tau^*] = \frac{\exp\big(\varepsilon d(\tau, \tau^*)\big)}{\sum_{\tau' \in S} \exp\big(\varepsilon d(\tau, \tau')\big)}$$

← Pairwise distance

← Sum of distance

LDP-fy a Trajectory in A **Discrete Space**

- Discrete location spaces: Point of interests or discretized cells



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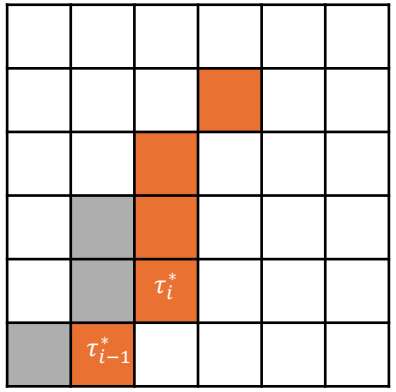

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LDP-fy a Trajectory in A **Discrete Space**

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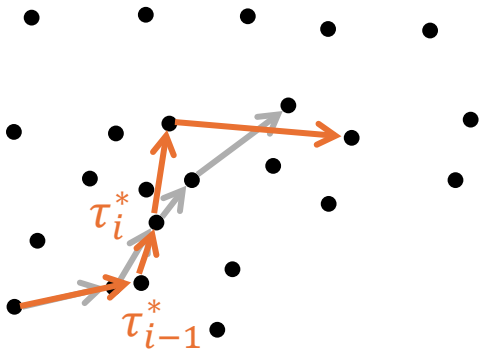

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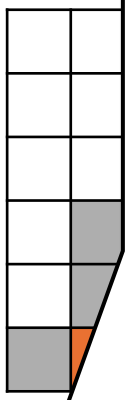


$\mathcal{S}$

Limitations

1. Efficiency

- each sample costs $\mathcal{O}(n)$

2. Trajectory Utility

- decreases as n increases

3. Applicability

- cannot apply to continuous $\mathcal{S}$

- Exponential mechanism:

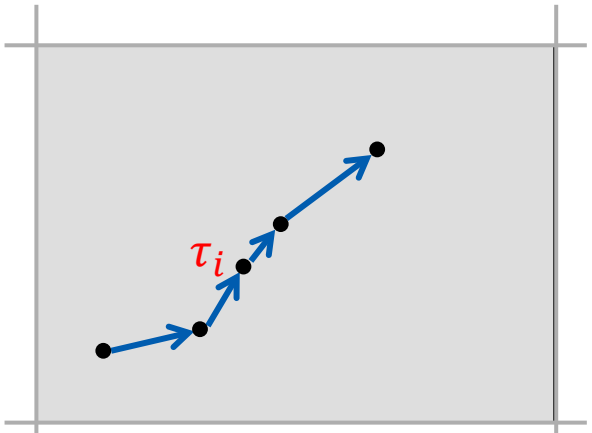
$$\Pr[\mathcal{M}_{\exp}(\tau) = \tau^*] = \frac{\exp(\varepsilon d(\tau, \tau^*))}{\sum_{\tau' \in S} \exp(\varepsilon d(\tau, \tau'))}$$

← Pairwise distance

← Sum of distance

Continuous Spaces: Better & Universal

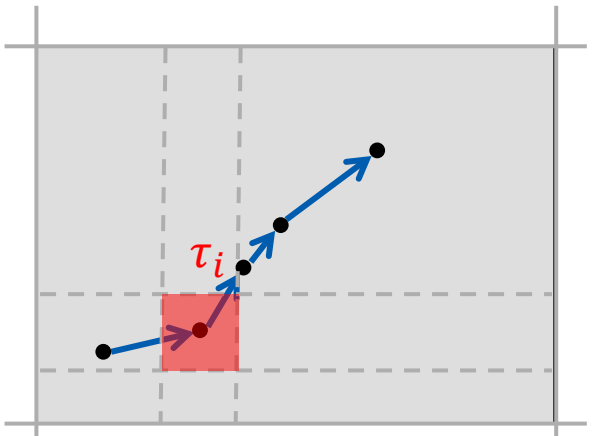
- Operate on a continuous space (covering the discrete space)



$\mathcal{S} = [a_{\text{sta}}, a_{\text{end}}] \times [b_{\text{sta}}, b_{\text{end}}]$

Continuous Spaces: Better & Universal

- Operate on a continuous space (covering the discrete space)



$\mathcal{S} = [a_{\text{sta}}, a_{\text{end}}] \times [b_{\text{sta}}, b_{\text{end}}]$

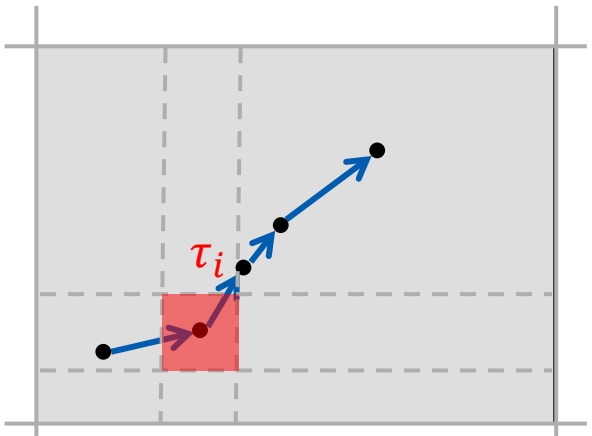
Simple sampling: (satisfying LDP for $\mathcal{S}$)

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_{\text{high}}$

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_{\text{low}}$

Continuous Spaces: Better & Universal

- Operate on a continuous space (covering the discrete space)



$\mathcal{S} = [a_{\text{sta}}, a_{\text{end}}] \times [b_{\text{sta}}, b_{\text{end}}]$

Simple sampling: (satisfying LDP for $\mathcal{S}$)

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_{\text{high}}$

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- Benefits:
  1. **Efficiency:** $\mathcal{O}(1)$ sampling complexity
  2. **Trajectory utility:** "n-independent"
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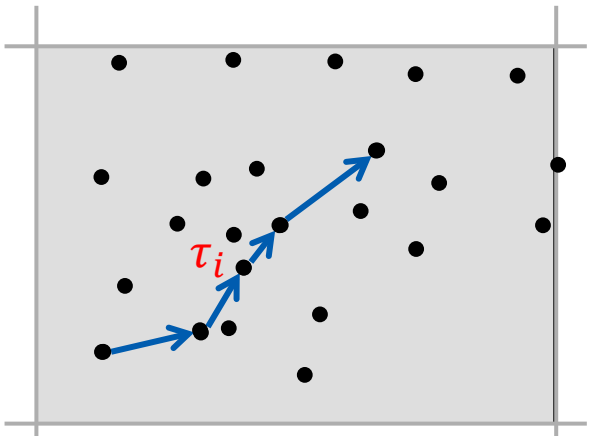

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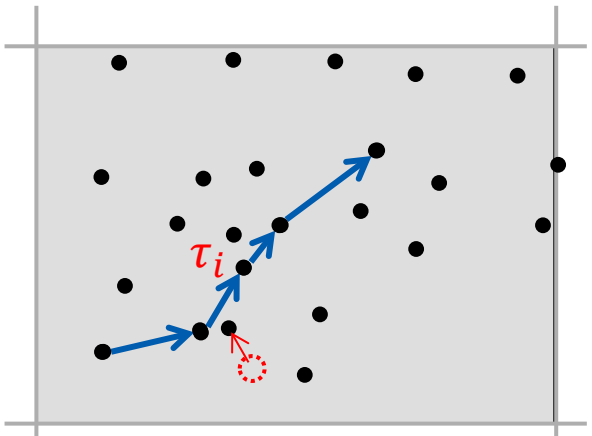

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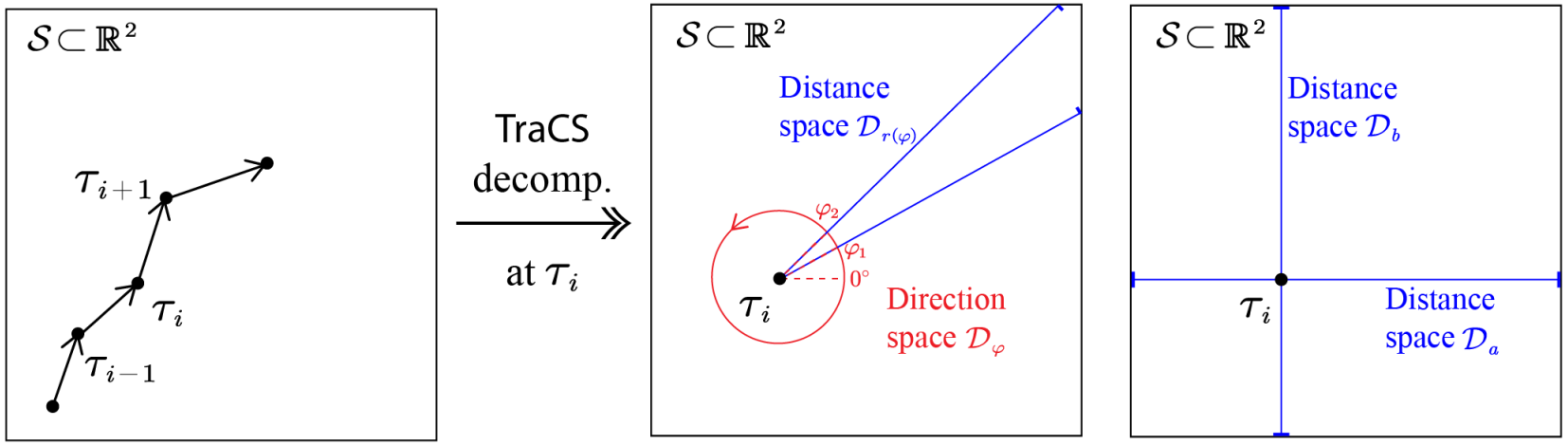
Rounding-to-the-nearest
(post-processing)

TraCS: Trajectory Collection in Continuous Spaces

- TraCS-D: direction-distance perturbation
- TraCS-C: coordinates perturbation

- **Key idea:** decomposes $\mathcal{S}$ into two subspaces



TraCS-D: $\mathcal{D}_\varphi \otimes \mathcal{D}_{r(\varphi)}$ TraCS-C: $\mathcal{D}_a \times \mathcal{D}_b$

TraCS: Trajectory Collection in Continuous Spaces

- TraCS-D: direction-distance perturbation
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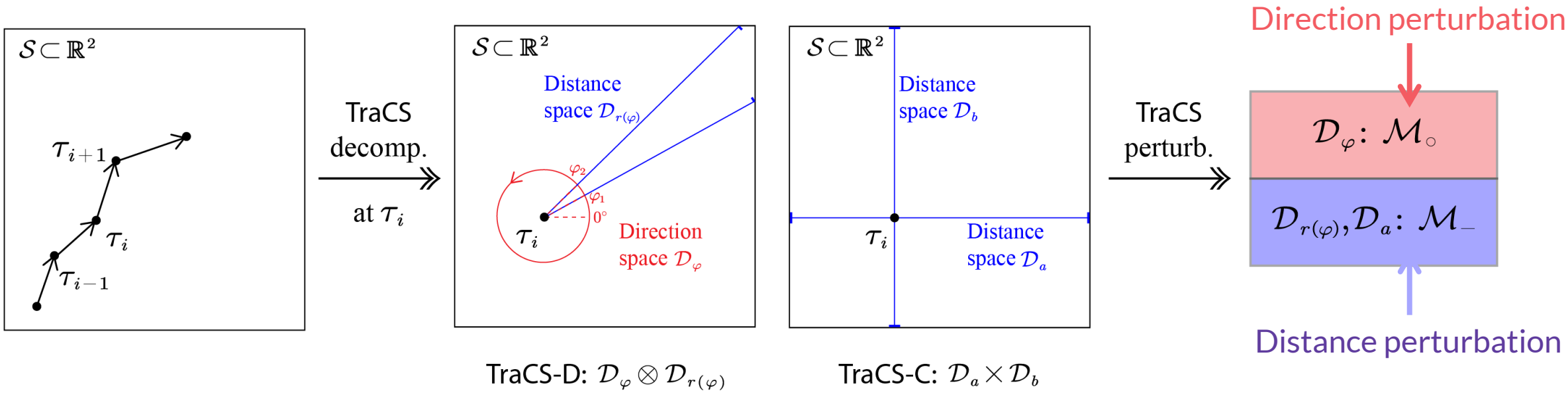
- **Key idea:** decomposes $\mathcal{S}$ into two subspaces → design $\mathcal{M}$ for each subspace



TraCS-D: $\mathcal{D}_\varphi \otimes \mathcal{D}_{r(\varphi)}$

TraCS-C: $\mathcal{D}_a \times \mathcal{D}_b$

LDP Mechanisms $\mathcal{M}_\circ$ and $\mathcal{M}_-$

RIT

- Leverage piecewise-based mechanisms $\mathcal{M}_\circ$ and $\mathcal{M}_-$

  $[0,2\pi) \to [0,2\pi)$ $\quad$ $[0,1) \to [0,1)$

  - other LDP mechanisms for bounded numerical domain also applicable

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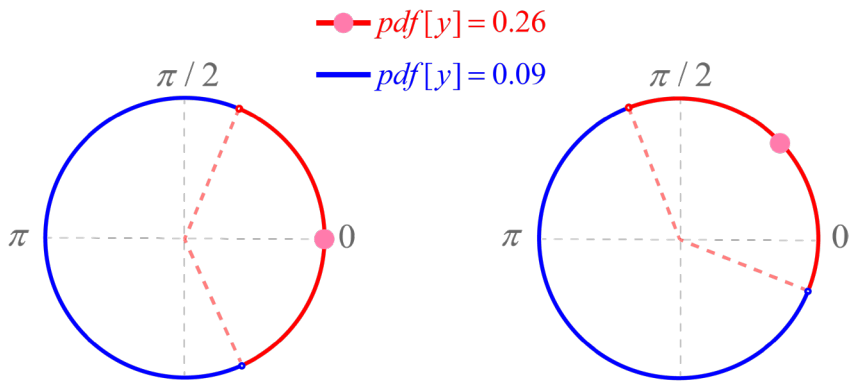
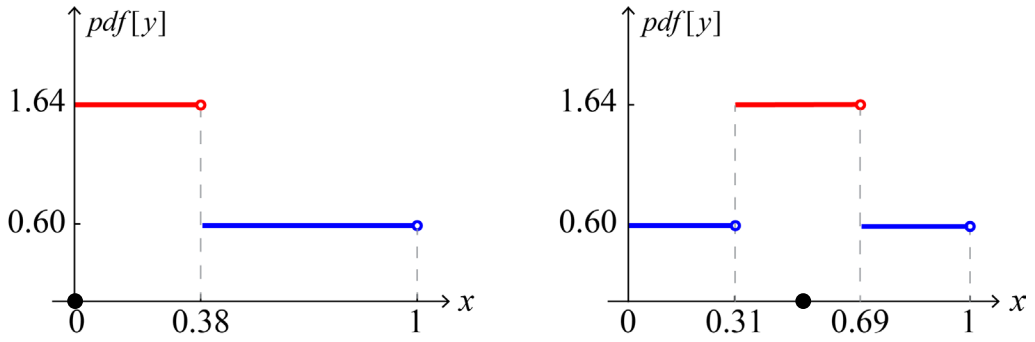
- Examples of $\mathcal{M}_\circ$ and $\mathcal{M}_-$*



(a) $x = 0$ (b) $x = \pi / 4$

$\mathcal{M}_\circ(x; \varepsilon = 1)$



(a) $x = 0$ (b) $x = 0.5$

$\mathcal{M}_-(x; \varepsilon = 1)$

* Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy, PETS'25

Evaluations

RIT

- Claims
  - continuous spaces: better utility than naïve baselines, e.g. planar Laplace + truncation
  - discrete spaces: better efficiency and utility than existing methods, ATP*, NGram**, L-SRR***

* Trajectory Data Collection with Local Differential Privacy, VLDB'23

** Real-World Trajectory Sharing with Local Differential Privacy, VLDB'21

*** L-SRR: Local Differential Privacy for Location-Based Services with Staircase Randomized Response, CCS'22

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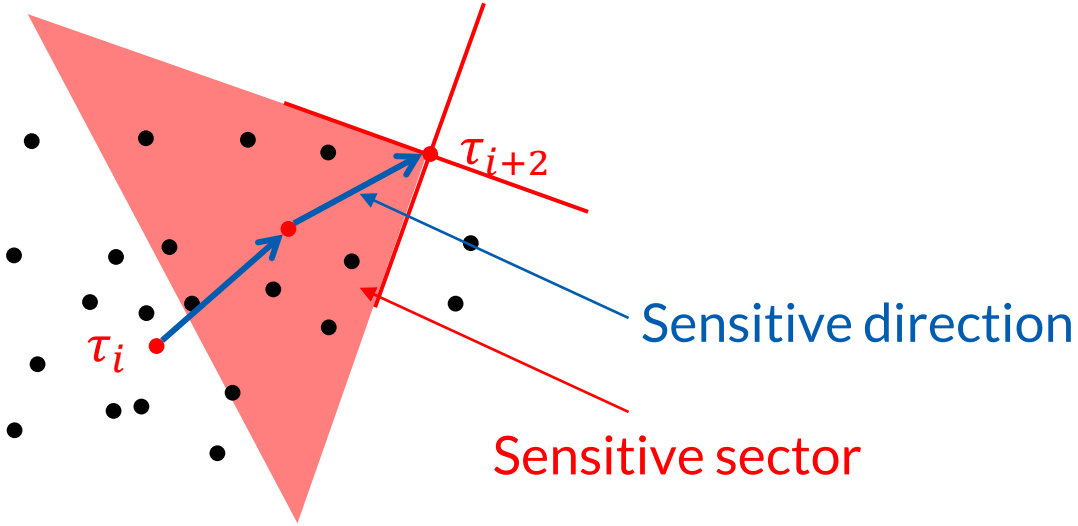
- **ATP:** direction perturbation

  1. divide direction sectors, e.g. $k = 4$



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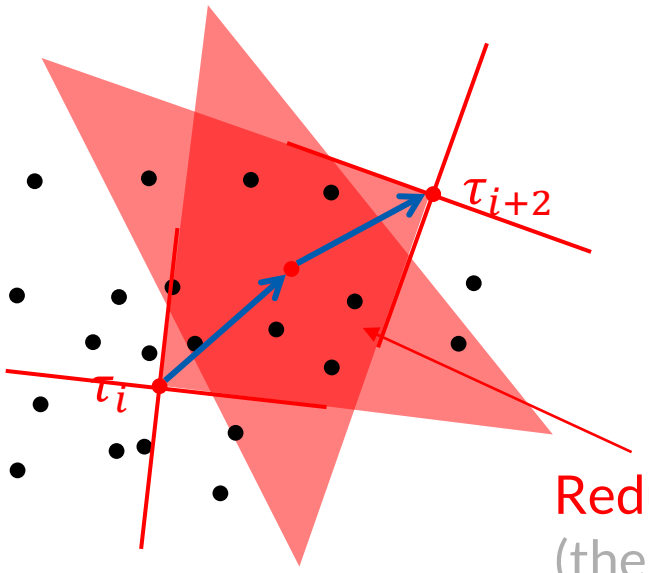
  1. divide direction sectors, e.g. $k = 4$
  2. perturb sector
  3. $\mathcal{M}_{\exp}(\tau)$ on reduced $\mathcal{S}$



Reduced location space $\mathcal{S}$
(then use EM)

* Trajectory Data Collection with Local Differential Privacy, VLDB'23

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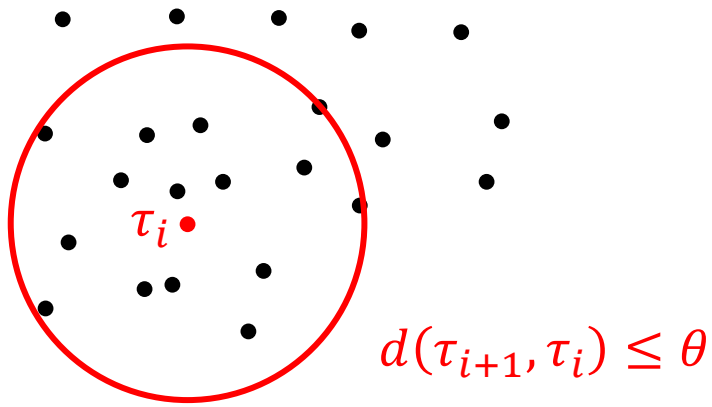
- **NGram:** reachability constraint from public knowledge

  1. distance reachability, i.e. the next location cannot be too far
  2. $\mathcal{M}_{\exp}(\tau)$ on reduced $\mathcal{S}$



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  - continuous spaces: better utility than naïve baselines, e.g. planar Laplace + truncation
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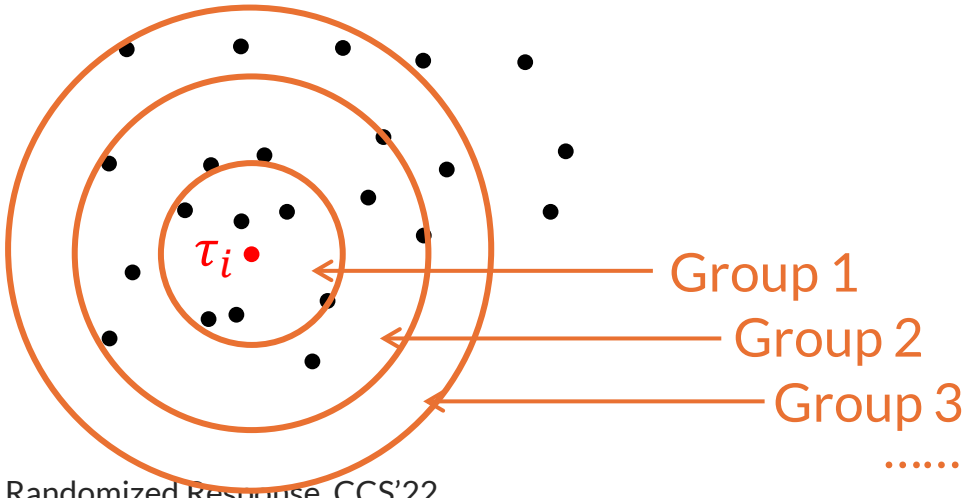
- **L-SRR:** Staircase Randomized Response

  1. group the locations to k groups
  2. $\mathcal{M}_{\mathrm{SRR}}(\tau)$ on k groups



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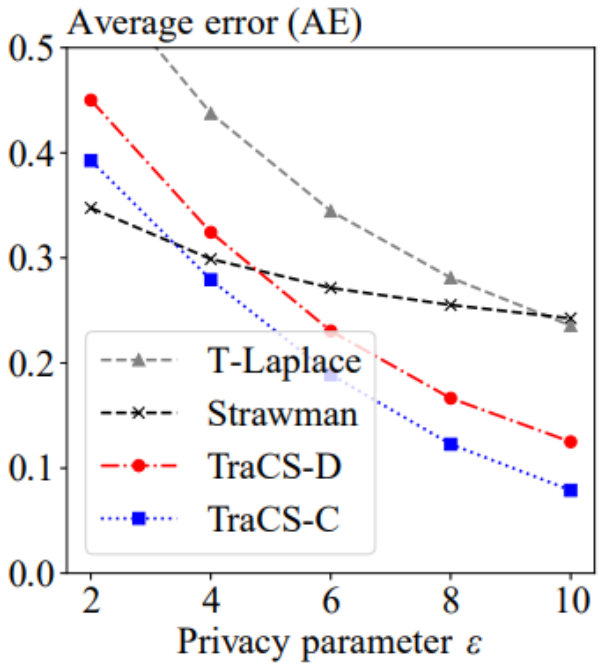
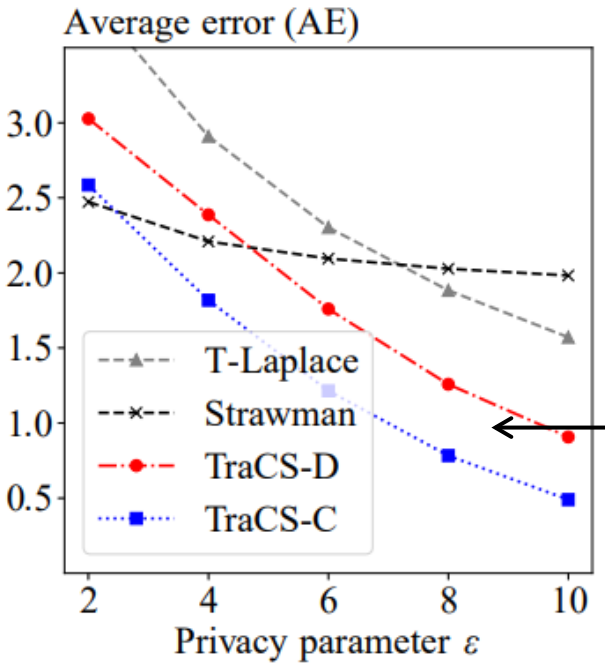
Evaluations – Continuous Spaces

RIT

- Trajectory utility metric: Average error (AE)
  - T-Laplace: Truncated Laplace; Strawman: k-RR + uniform sampling for direction perturbation (similar to SRR)



(a) $\mathcal{S} = [0, 1) \times [0, 1)$



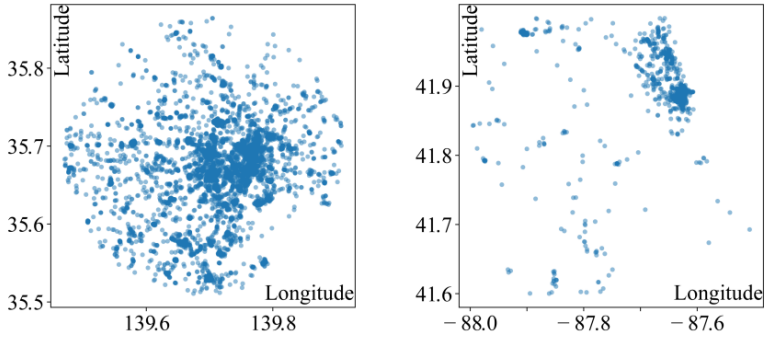
(b) $\mathcal{S} = [0, 2) \times [0, 10)$

Lower error when ε becomes larger

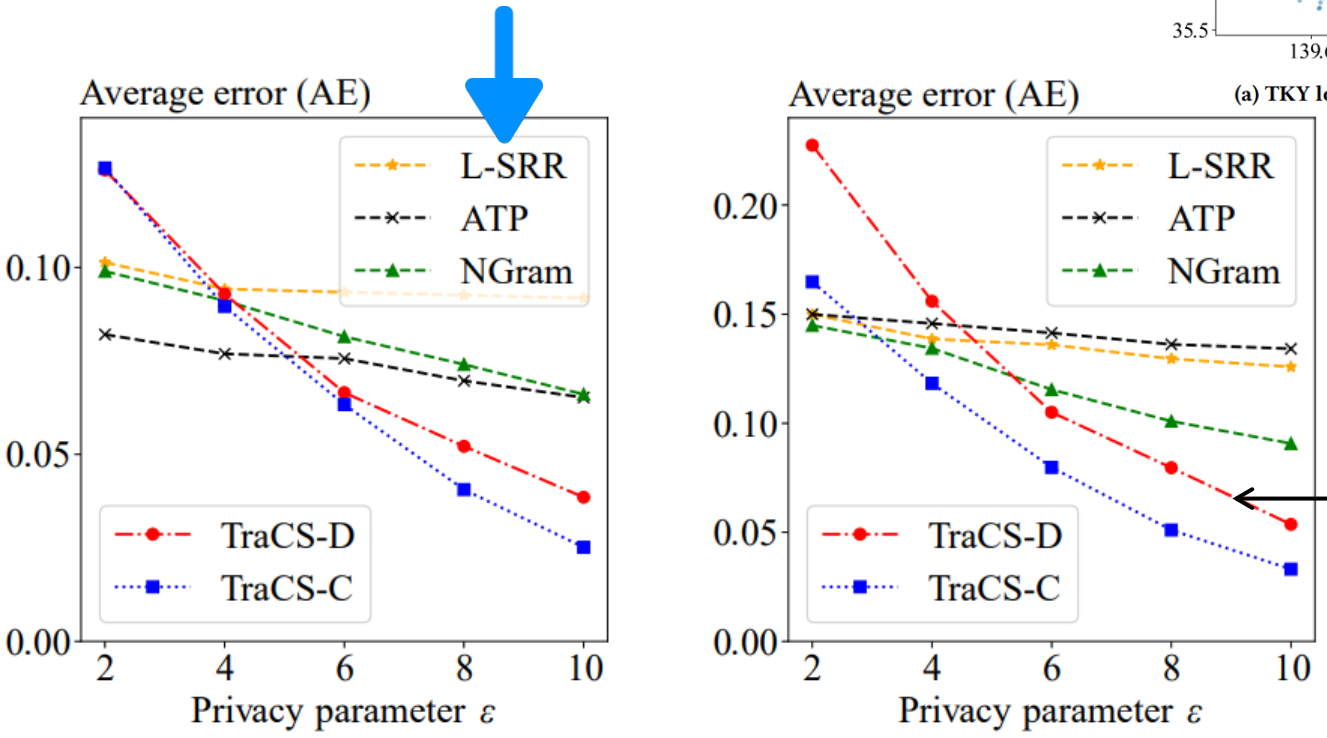
Evaluations – Discrete Spaces

RIT

- TKY and CHI datasets
  - TraCS for the rectangular area + rounding-to-the-nearest



(a) TKY location space (b) CHI location space



(a) TKY dataset (b) CHI dataset

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Evaluations – Discrete Spaces

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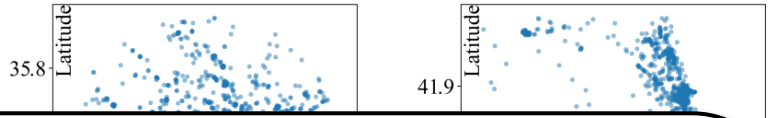
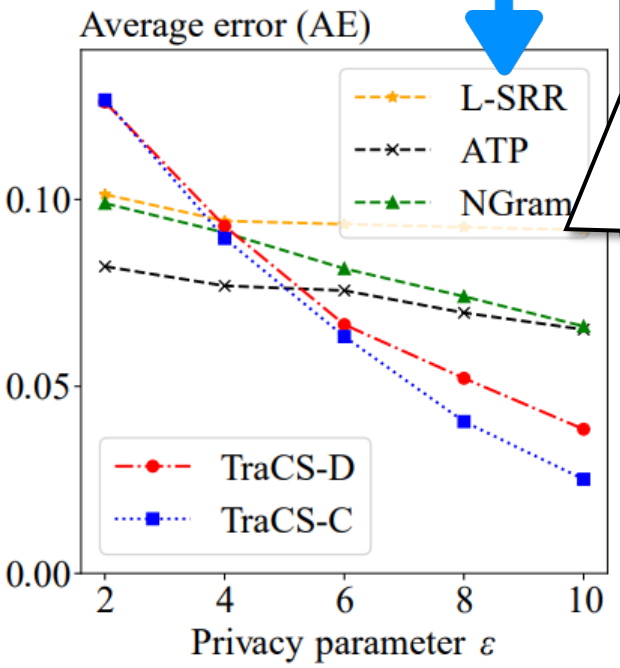


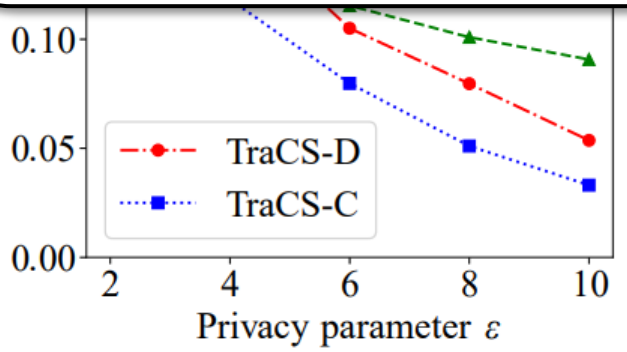
Table 3: Time cost comparison (in milliseconds).
averaged

	ATP	NGram	L-SRR	TraCS-D	TraCS-C
Total	145.7	100.9	6.2	0.06	0.05
Perturb	125.8	92.8	0.003	0.018	0.003

Negligible time cost



(a) TKY dataset



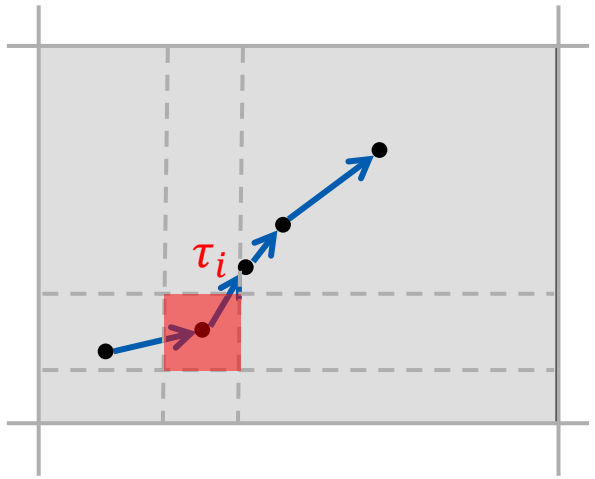
(b) CHI dataset

Summary & Takeaway

RIT

- **RQ: Trajectory collection under LDP**

- Our results:*
  - **operating in continuous spaces can do better**
  - better efficiency: $\mathcal{O}(1)$ sampling complexity
  - better trajectory utility, especially for larger ε
  - better applicability, for both continuous and discrete $\mathcal{S}$



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* TraCS: Trajectory Collection in Continuous Space under Local Differential Privacy, PETS'26

Local Differential Privacy: Refined Mechanism Design and Utility Analysis

Mechanism design
- Part 1: Correlated $\mathcal{M}$
- Part 2: Optimal piecewise-based $\mathcal{M}$
- Part 3: $\mathcal{M}$ for trajectories in continuous space

Utility analysis
- Part 4: Utility analysis for classifier $\circ$ $\mathcal{M}$

Mechanism-level utility
↓
Task-level utility

Classifier Utility under Noisy Inputs

RIT

- Input data for classifiers may be noisy

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RIT

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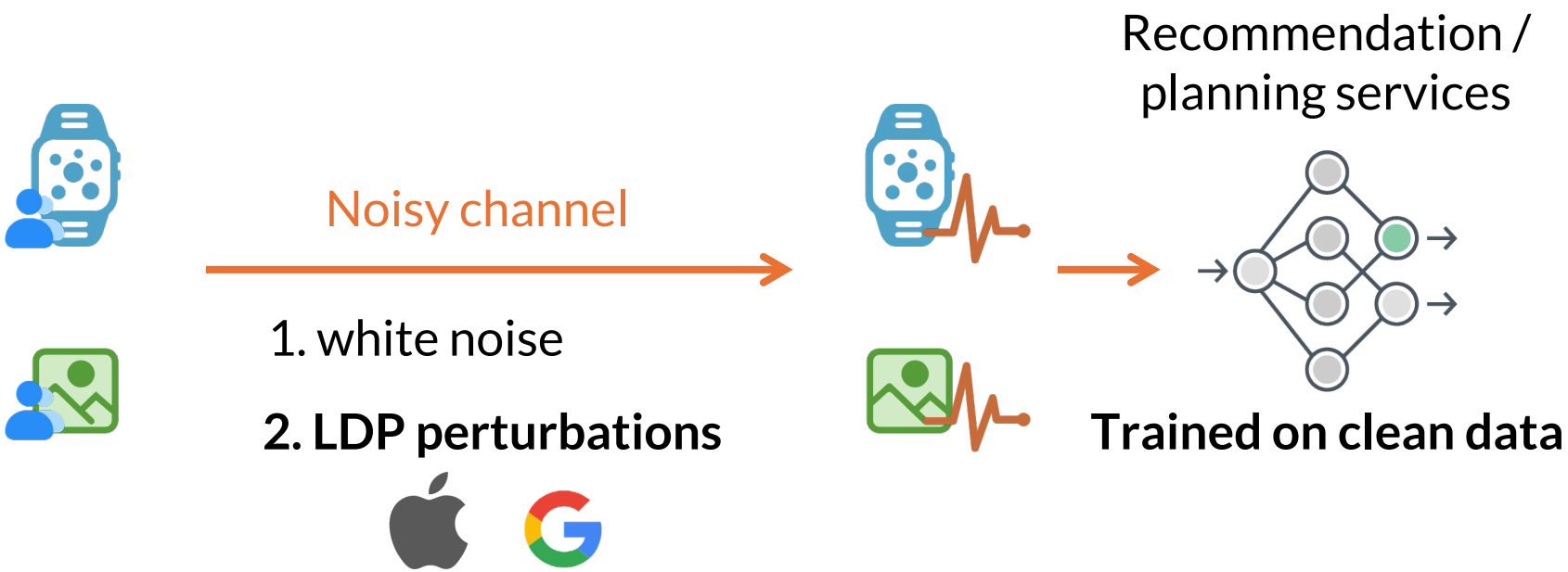

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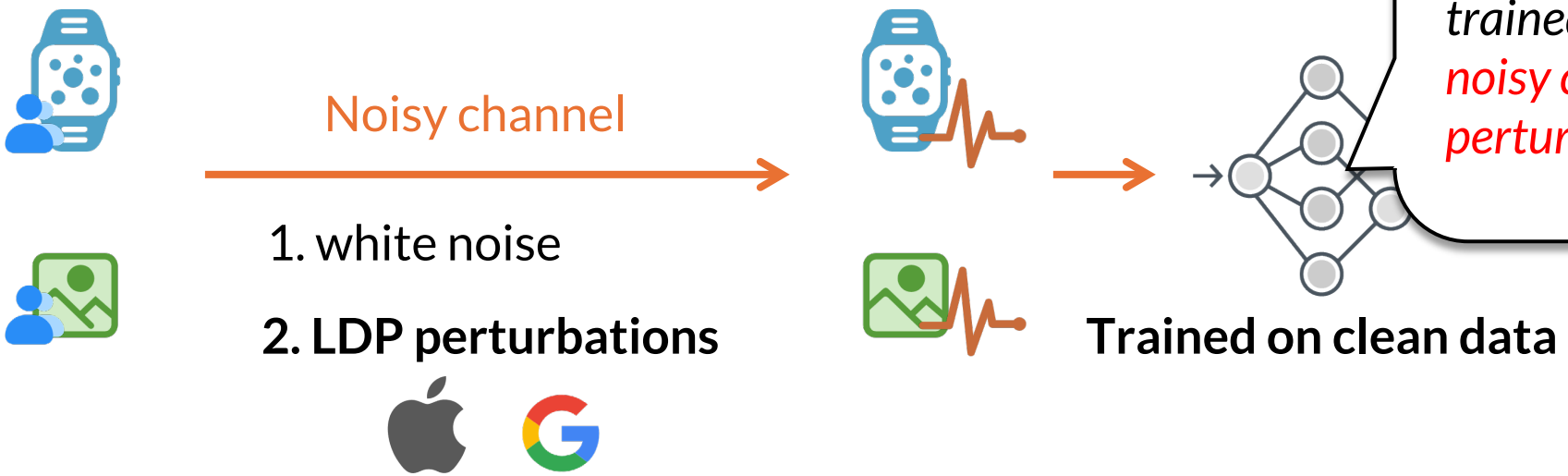

> *What's the utility of the trained classifier under a noisy channel, e.g. LDP perturbation?*

Classifier Utility under Noisy Inputs

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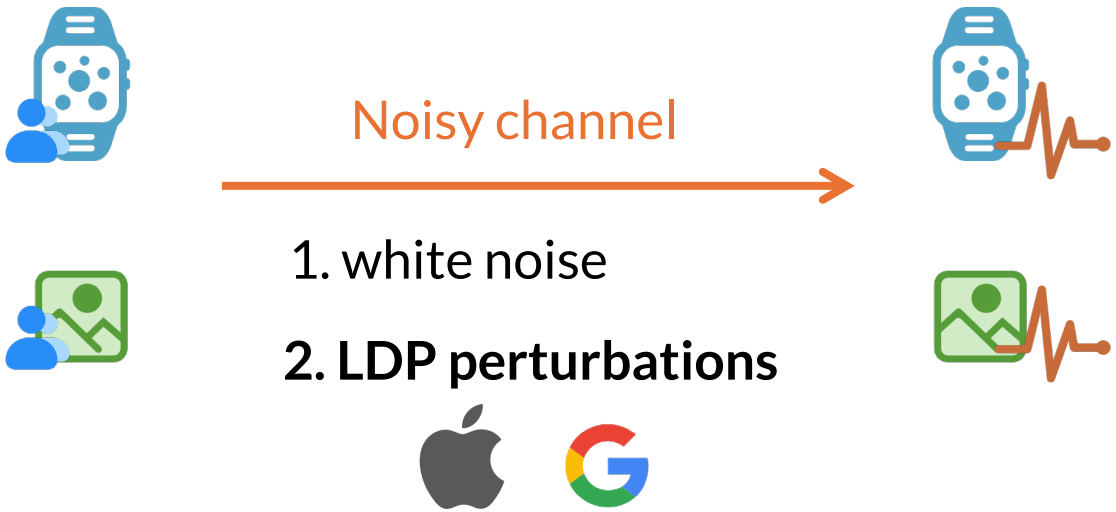


What's the utility of the trained classifier under a noisy channel, e.g. LDP perturbation?

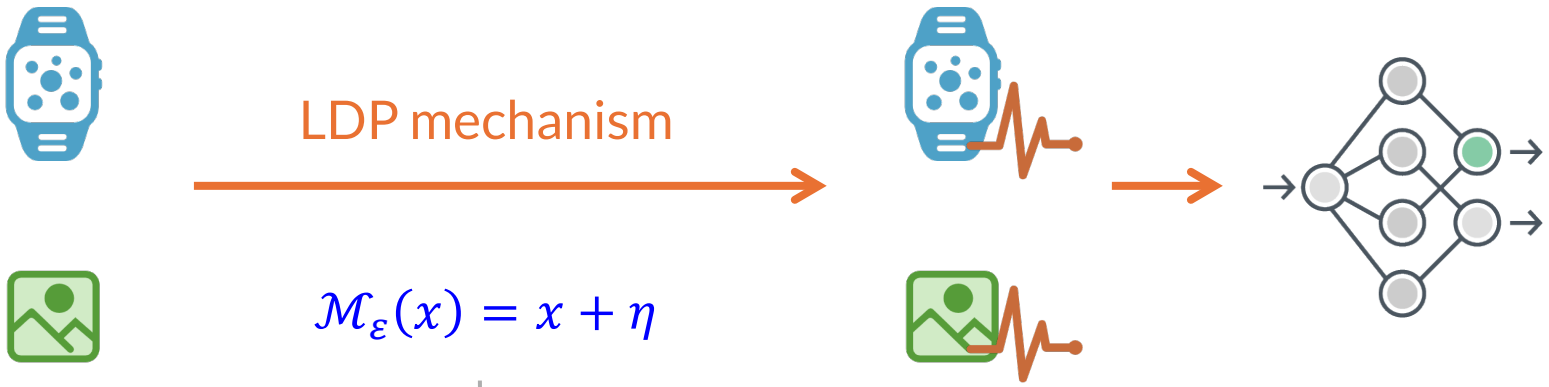
- Q: How can classifier designers/users know the classifier's accuracy under LDP-perturbed data?

For an LDP-friendly classifier

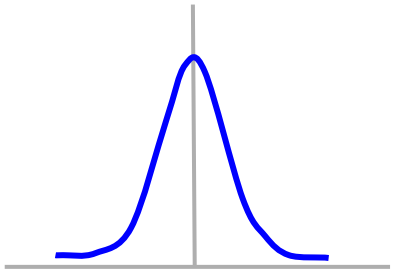
For a better privacy-utility balance

Empirical Classifier Utility under LDP-Data

- Empirical approach: Sample and then test
  - $\mathcal{M}$'s variance or MSE **doesn't help** – cannot provide a classifier accuracy



$$\mathcal{M}_\varepsilon(x) = x + \eta$$



$$\eta \sim \mathcal{N}(0, \sigma^2)$$

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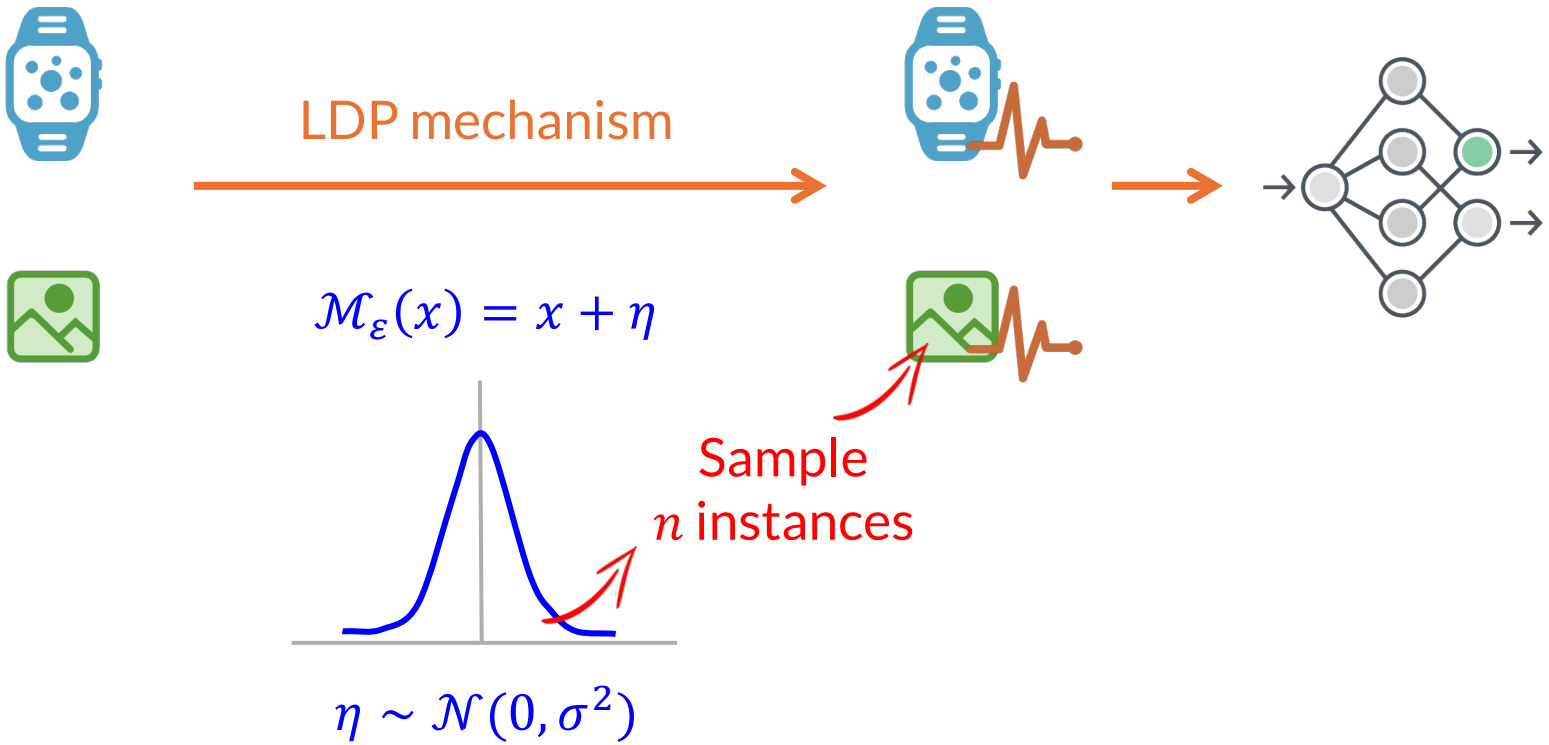

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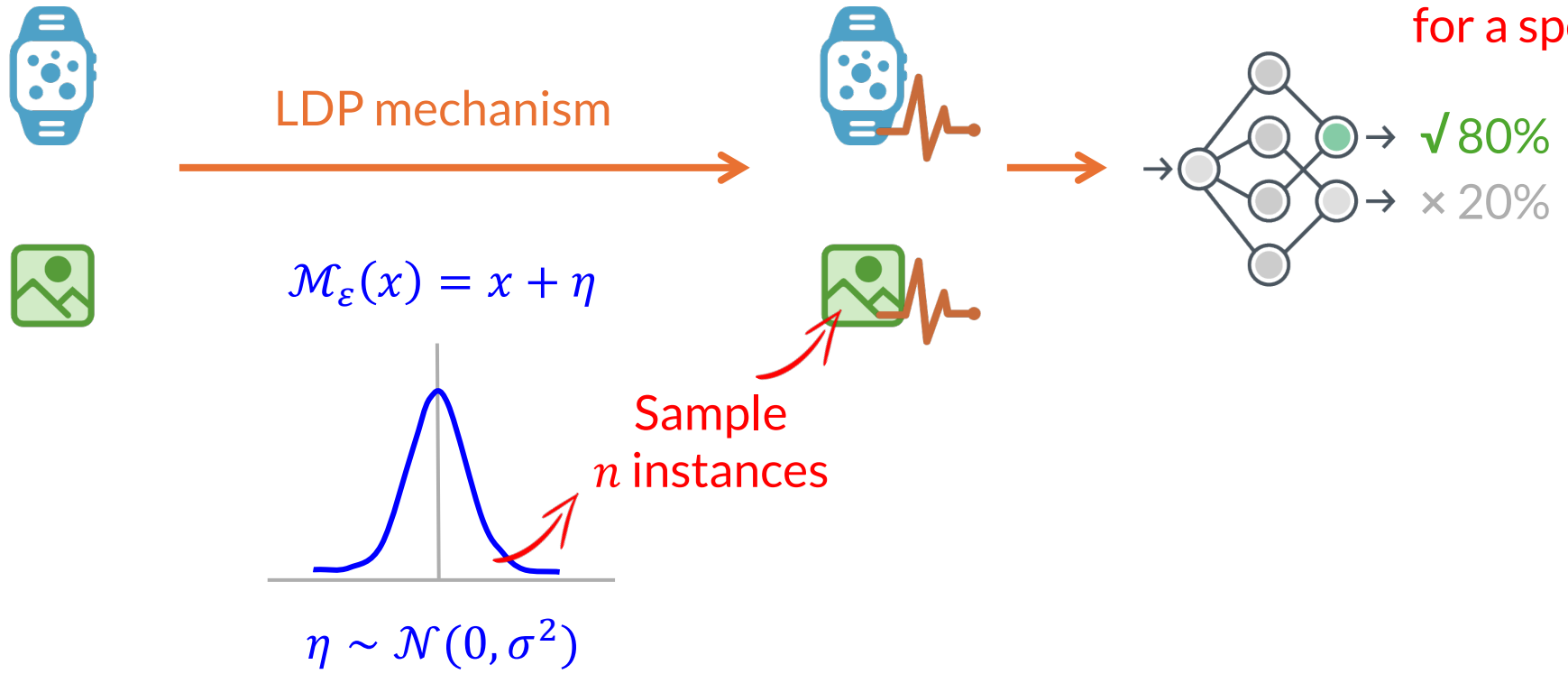

Empirical classifier utility
for a specific ε

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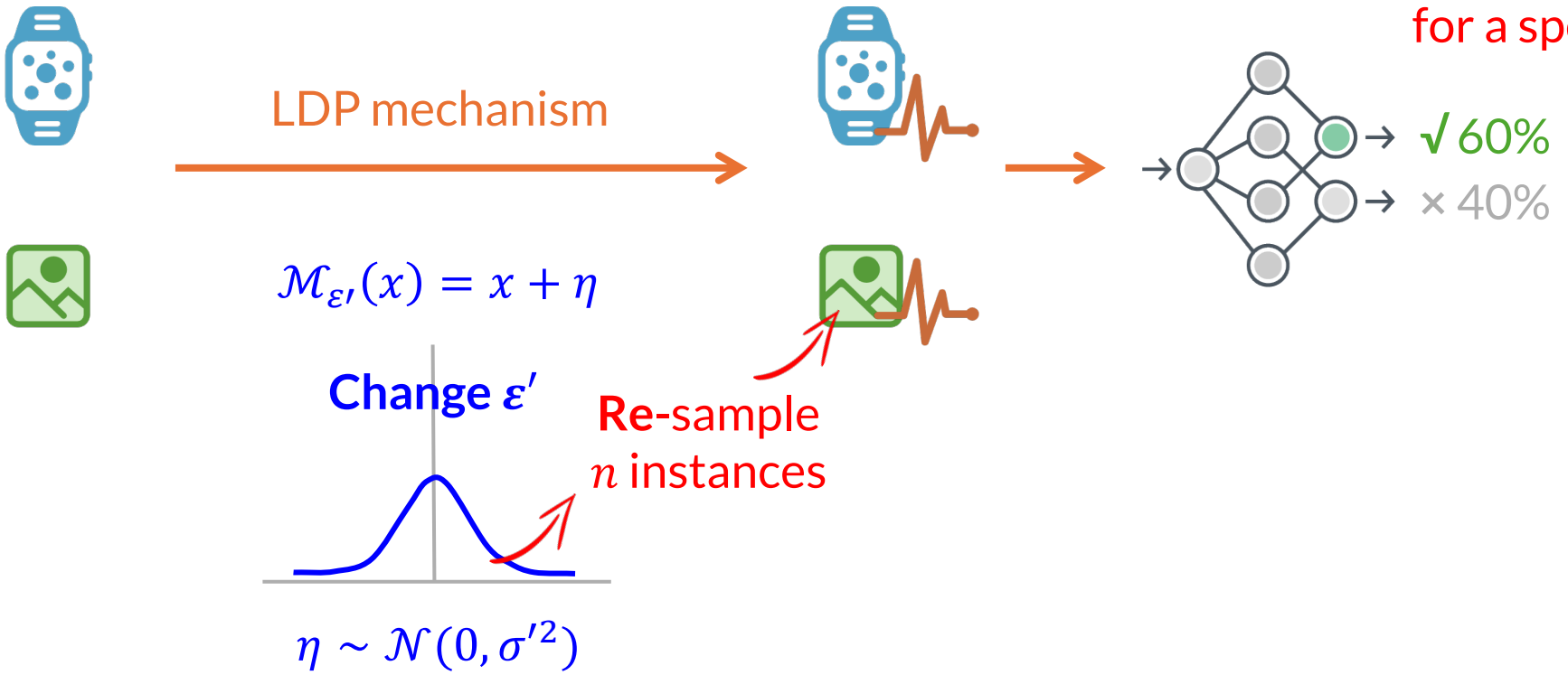

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√ 60%
× 40%

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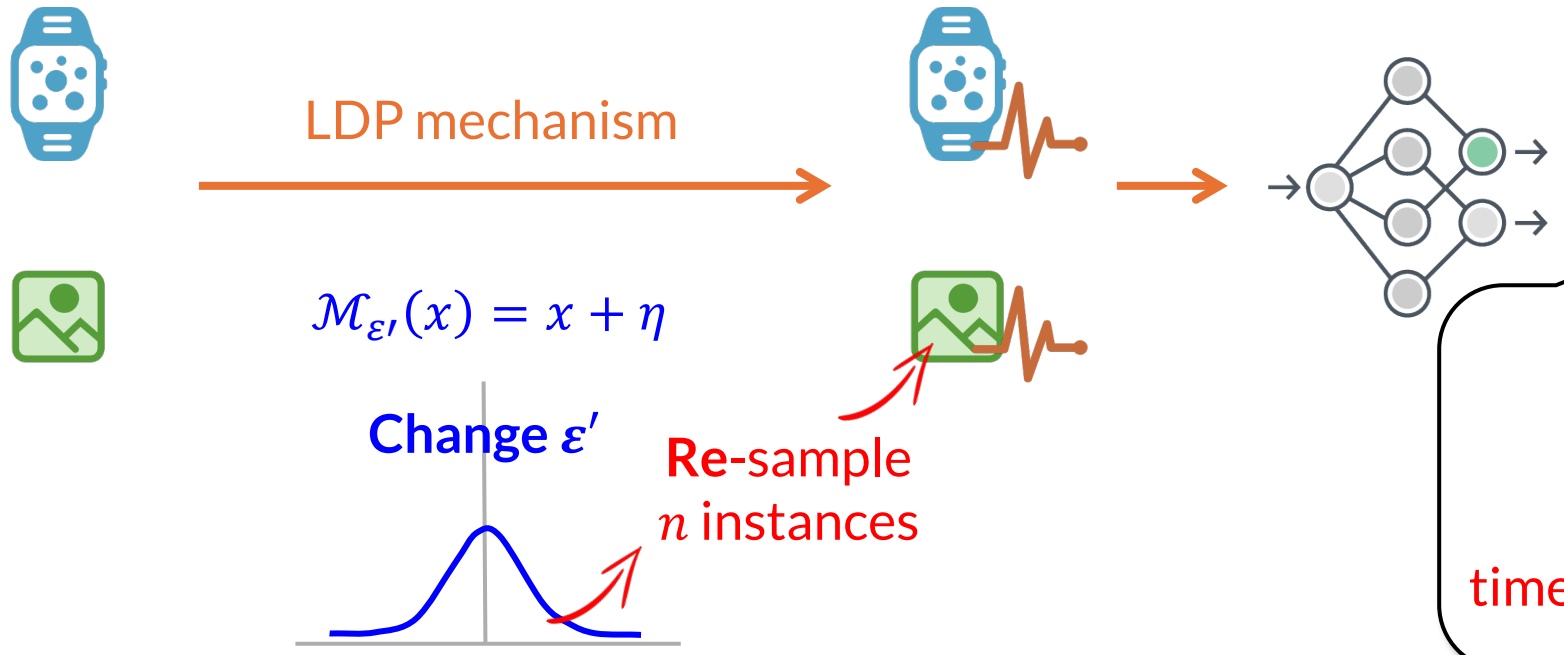

Empirical classifier utility
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Limitation

Empirical result
&
time-consuming resampling

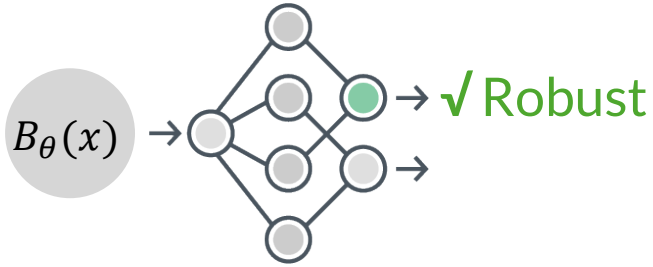
Lacks of an **analytical view** of classifier utility under LDP data

Empirical Utility → Analytical Utility

RIT

- Analytical approach: connecting LDP with robustness

$B_\theta(x)$ is robust



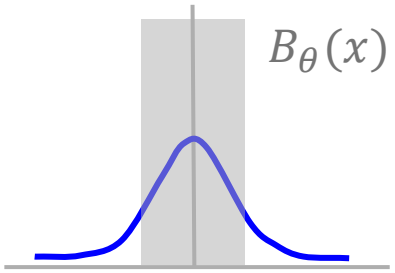
Robustness:
A θ-ball around x that doesn't change classification

Empirical Utility → Analytical Utility

RIT

- Analytical approach: connecting LDP with robustness

$\mathcal{M}_\varepsilon(x)$ concentrates in $B_\theta(x)$



Concentration:
Probability of falling into a region
$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)] \coloneqq p(\varepsilon, \theta)$

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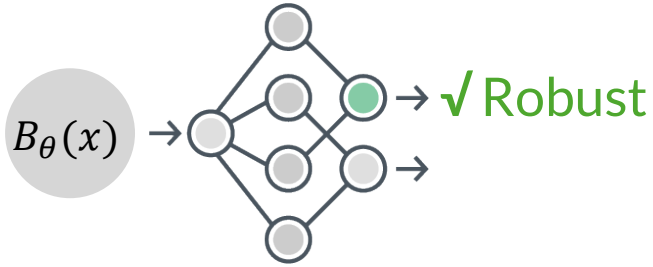

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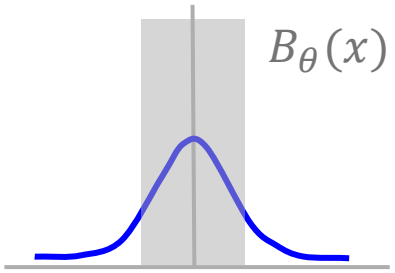

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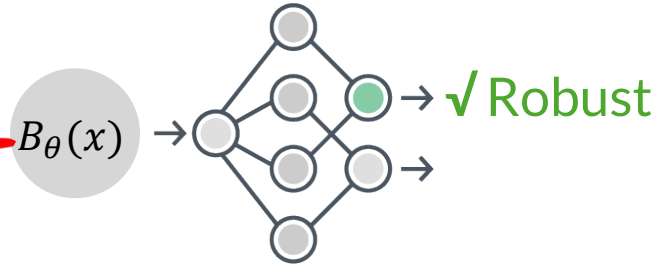

Concentration:
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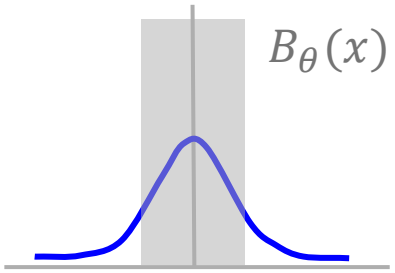
Robustness:
A θ-ball around x that doesn't change classification

Empirical Utility → Analytical Utility

RIT

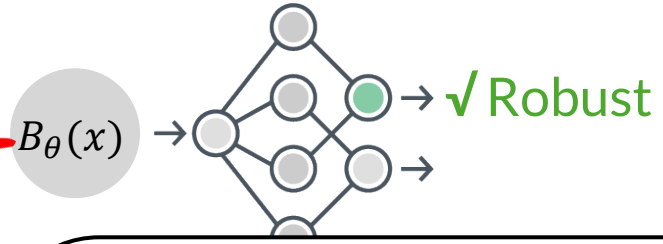
- Analytical approach: connecting LDP with robustness

$\mathcal{M}_\varepsilon(x)$ concentrates in $B_\theta(x)$



Concentration:
Probability of falling into a region
$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)] \coloneqq p(\varepsilon, \theta)$

$B_\theta(x)$ is robust



Analytical classifier utility

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_\varepsilon(x)$.

Empirical Utility → Analytical Utility

RIT

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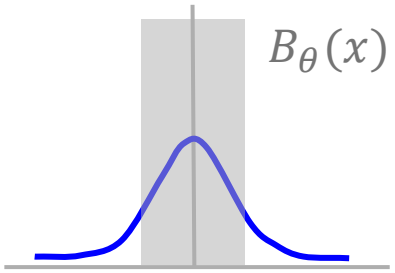
$\mathcal{M}_\varepsilon(x)$ concentrates in $B_\theta(x)$



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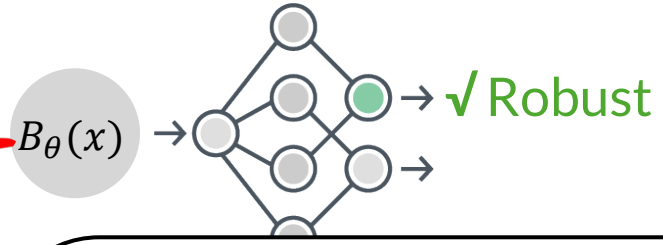


Analytical classifier utility

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_\varepsilon(x)$.

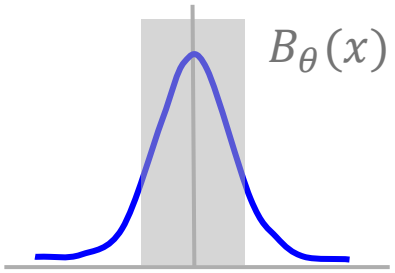
- Counterpart to the empirical "80%" accuracy

Empirical Utility → Analytical Utility

RIT

- Analytical approach: connecting LDP with robustness

$\mathcal{M}_{\varepsilon'}(x)$ concentrates in $B_\theta(x)$



Concentration:
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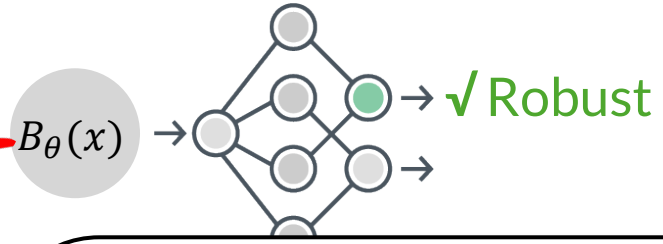
$\Pr[\mathcal{M}_{\varepsilon'}(x) \in B_\theta(x)] \coloneqq p(\varepsilon', \theta)$

$B_\theta(x)$ is robust



Analytical classifier utility

With probability at least $p(\varepsilon', \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_{\varepsilon'}(x)$.

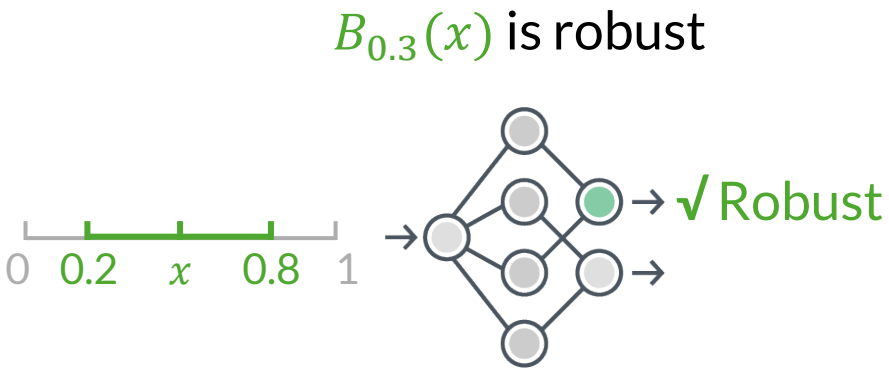
Analytical & systematic view for any ε without resampling

One-Dimension Example

RIT

- Classifier h: $[0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\mathrm{Lap}}(x)$

Classifier:
(robustness analysis)



One-Dimension Example

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- Classifier h: $[0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

$B_{0.3}(x)$ is robust

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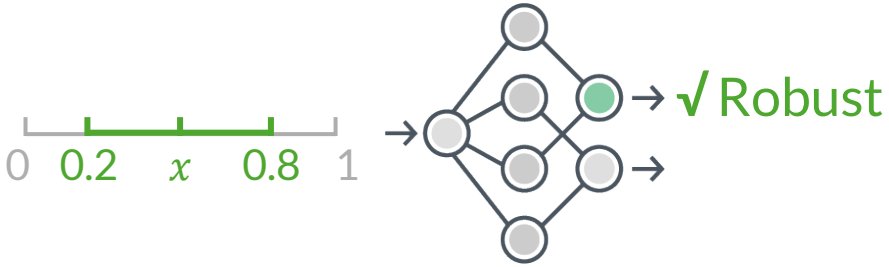
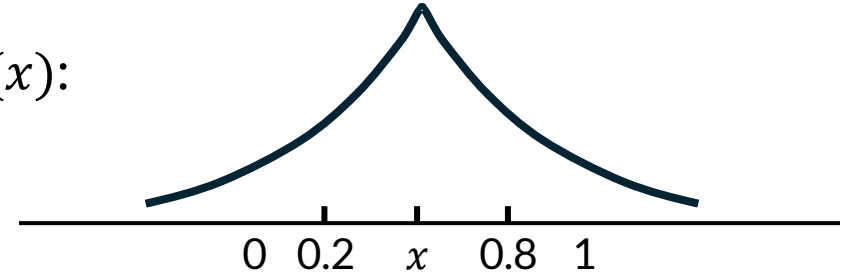


LDP mechanism $\mathcal{M}_{\text{Lap}}(x)$:



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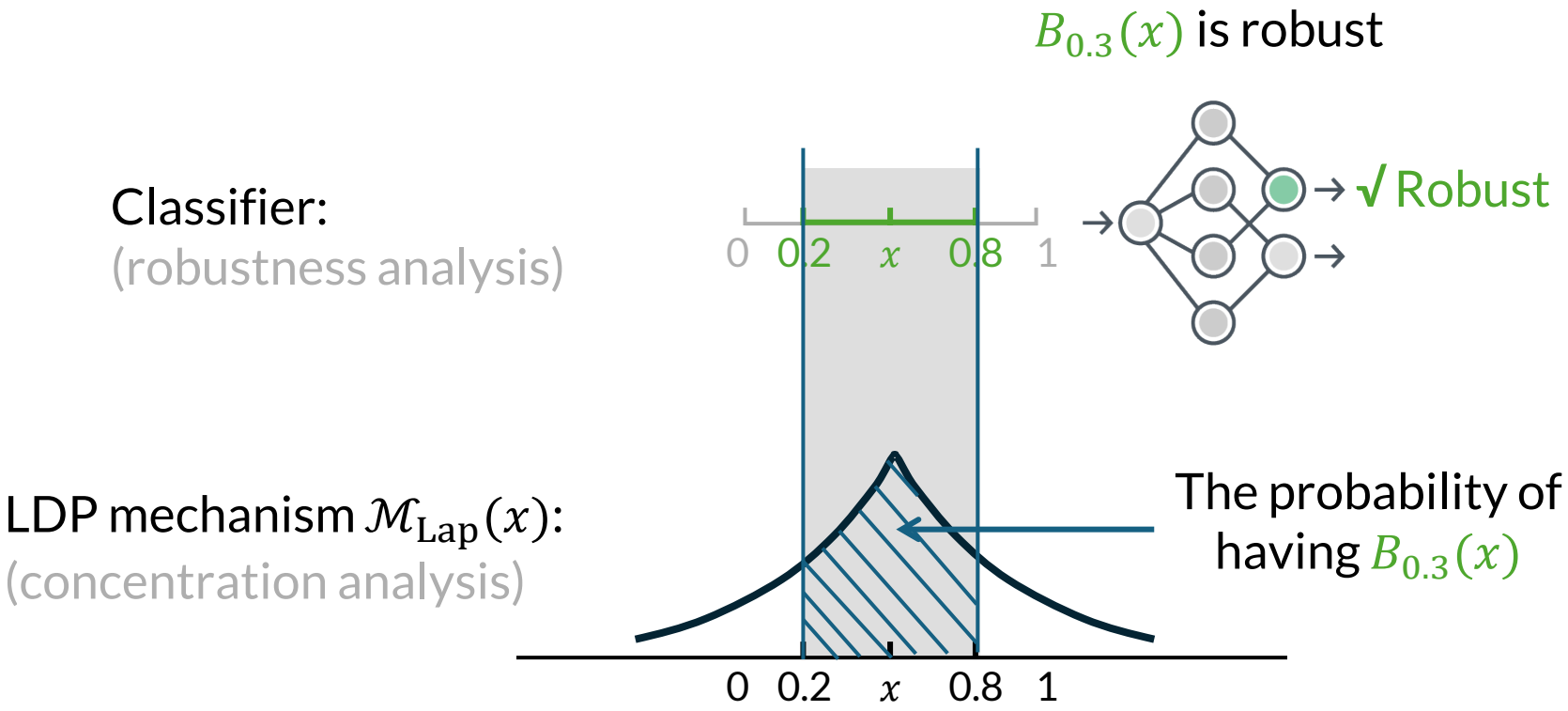

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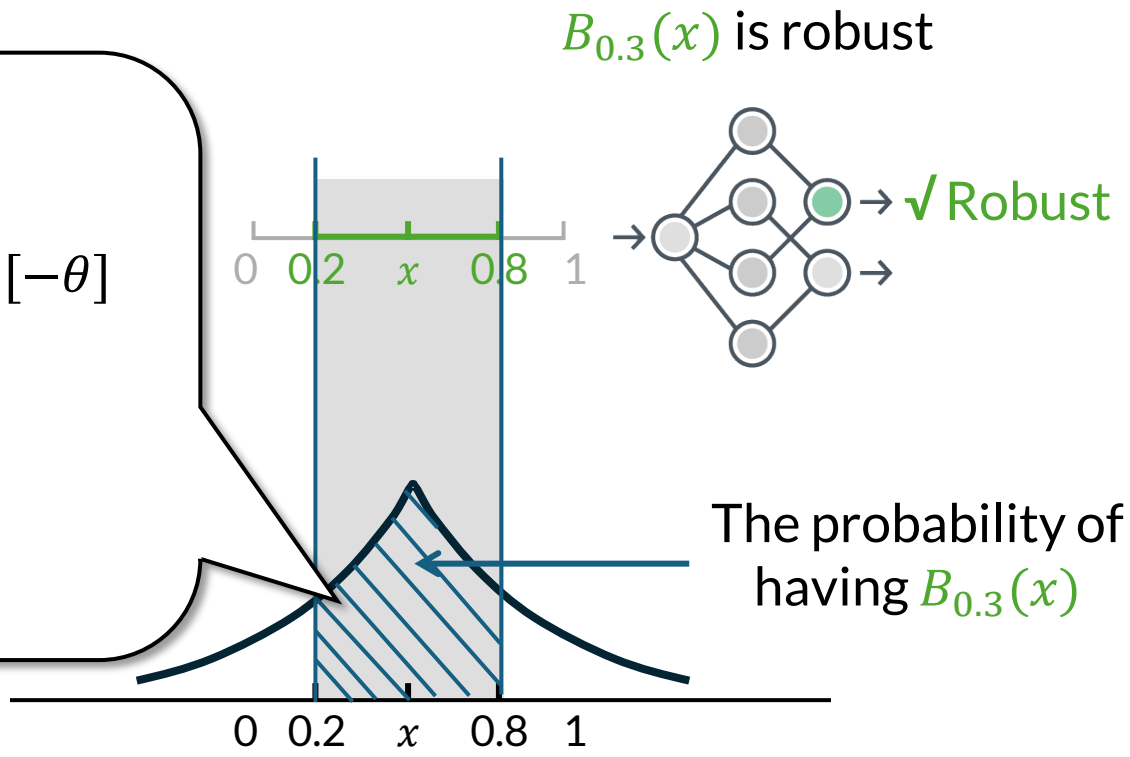
Concentration analysis

$$p(\varepsilon, \theta) = \text{cdf}_{\text{Lap}}[\theta] - \text{cdf}_{\text{Lap}}[-\theta]$$
$$= 1 - e^{-\theta\varepsilon}$$



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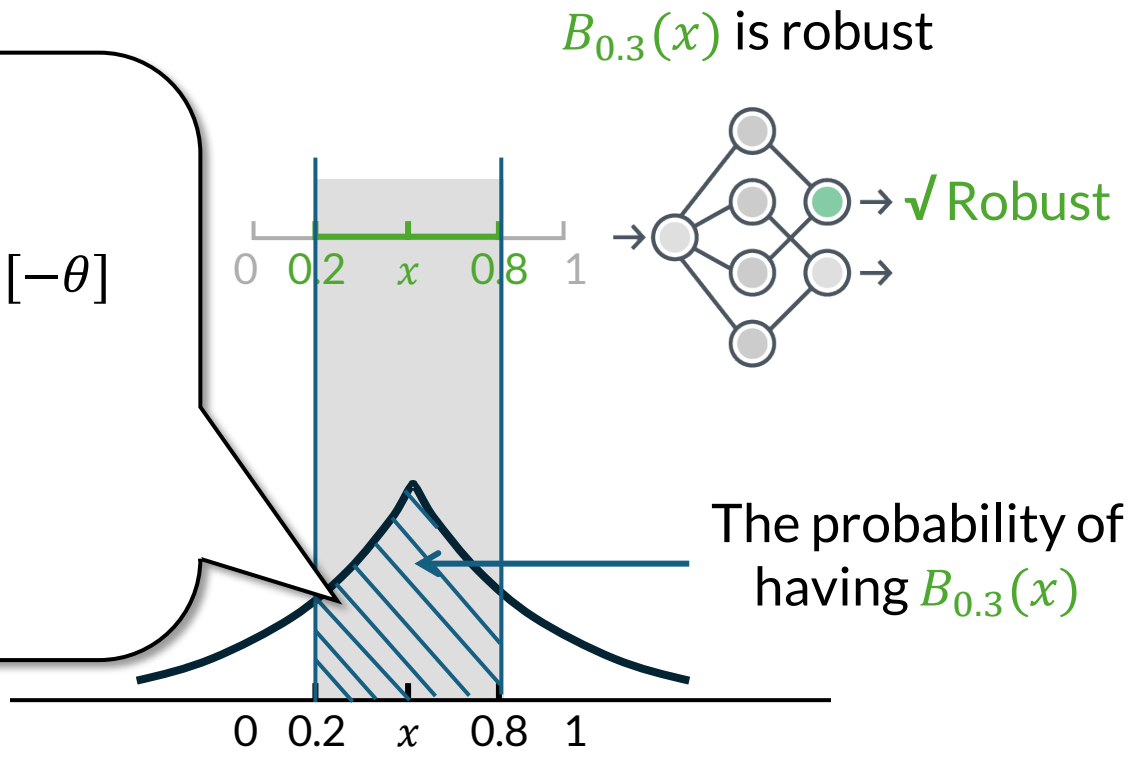
$$p(\varepsilon, \theta) = \text{cdf}_{\text{Lap}}[\theta] - \text{cdf}_{\text{Lap}}[-\theta]$$
$$= 1 - e^{-\theta\varepsilon}$$

$$p(\varepsilon = 2, \theta = 0.3) = 0.46$$



One-Dimension Example

RIT

- Classifier h: $[0,1] \rightarrow \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

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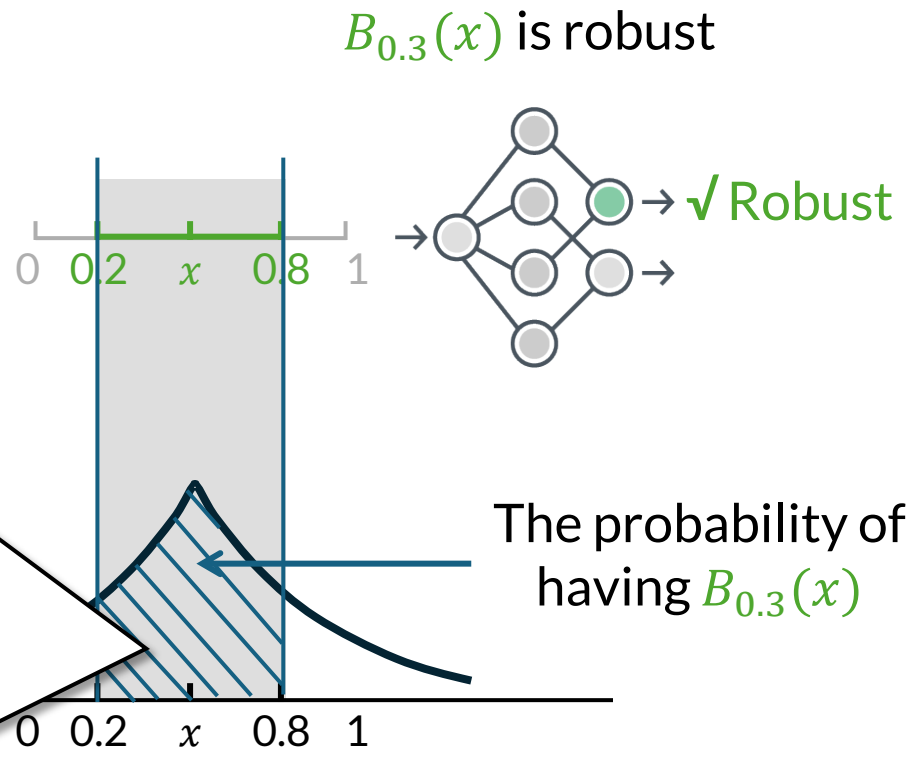
$$p(\varepsilon, \theta) = \text{cdf}_{\text{Lap}}[\theta] - \text{cdf}_{\text{Lap}}[-\theta]$$
$$= 1 - e^{-\theta\varepsilon}$$

$p(\varepsilon = 2, \theta = 0.3) = 0.46$

With probability at least $p(2,0.3) = 0.46$, the classifier preserves its correct classification under input $\mathcal{M}_{\varepsilon=2}(0.5)$.



One-Dimension Example

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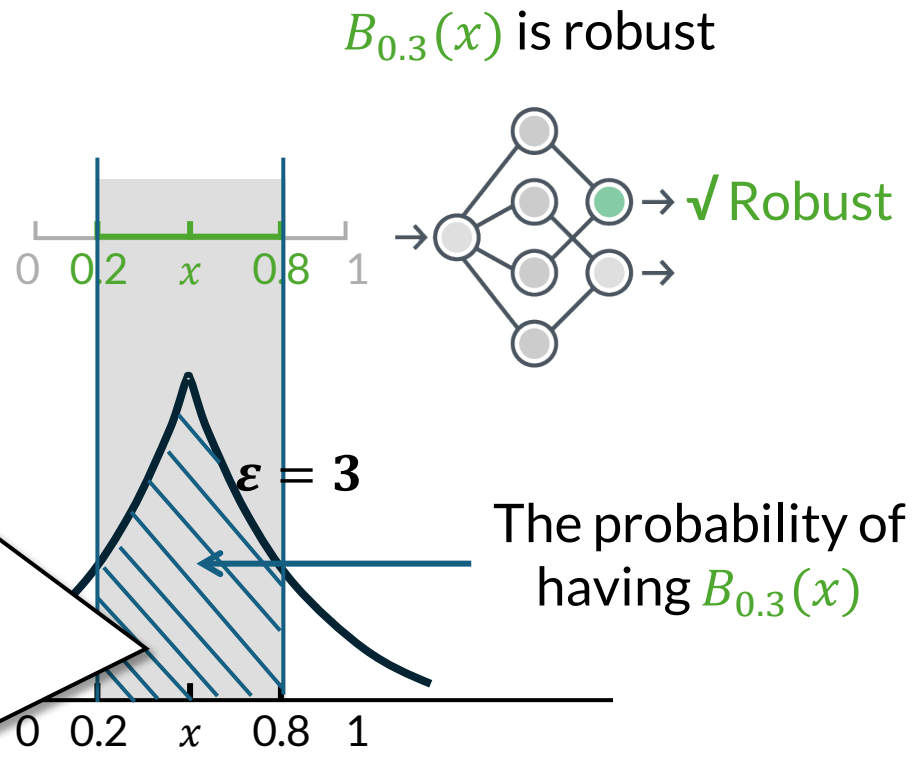
$$= 1 - e^{-\theta\varepsilon}$$

For any ε:

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_\varepsilon(x)$.

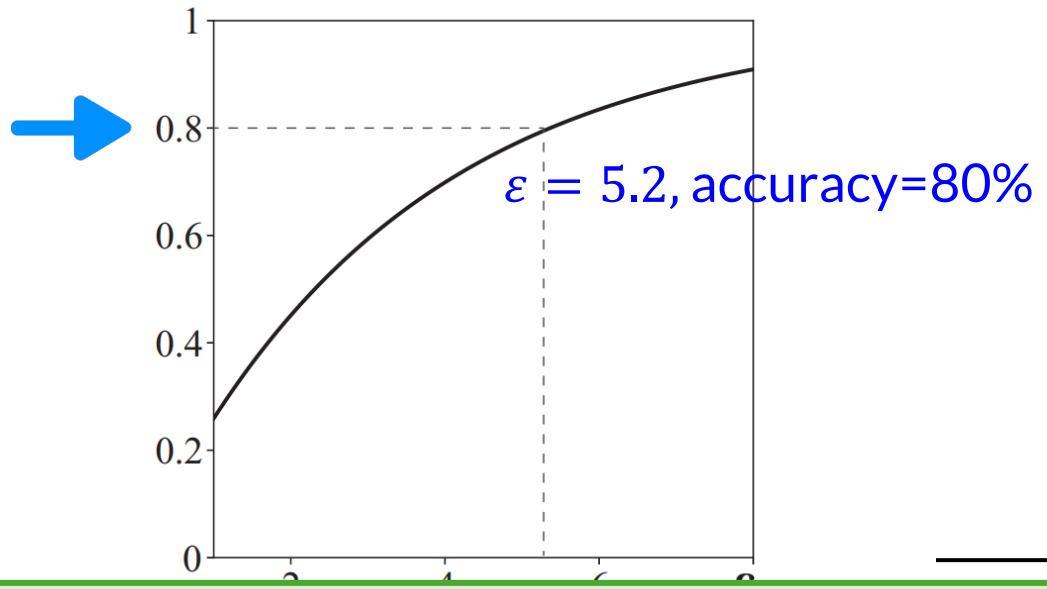


Fixed Classifier (θ)

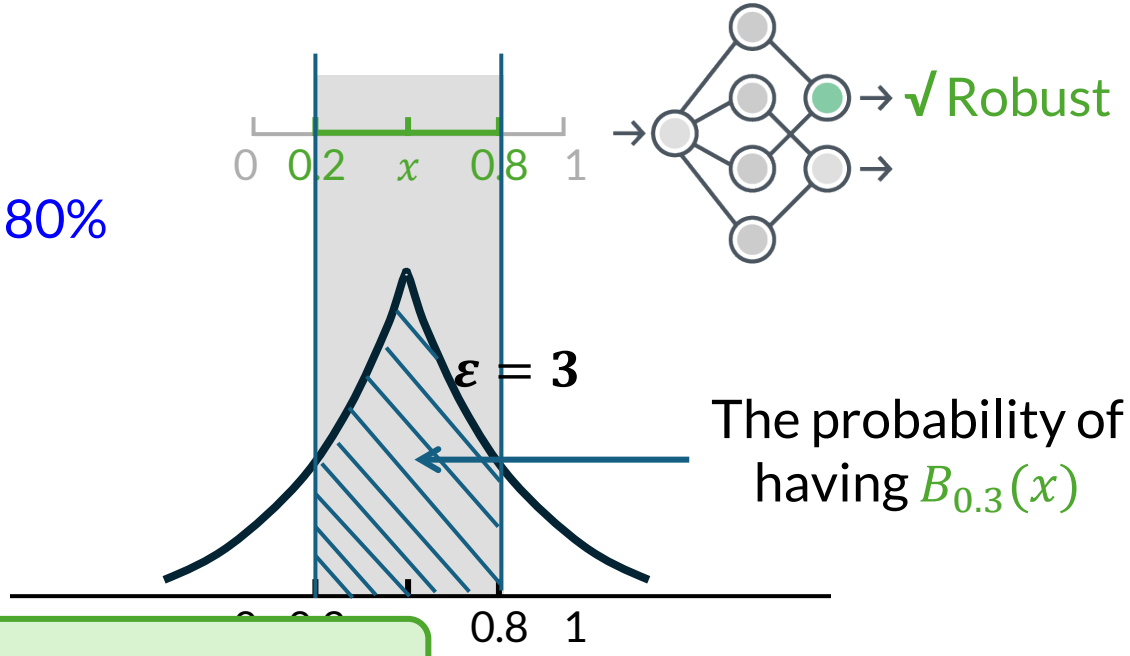
RIT

- Classifier h: $[0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

This classifier's utility $p(\varepsilon, 0.3)$



$B_{0.3}(x)$ is robust

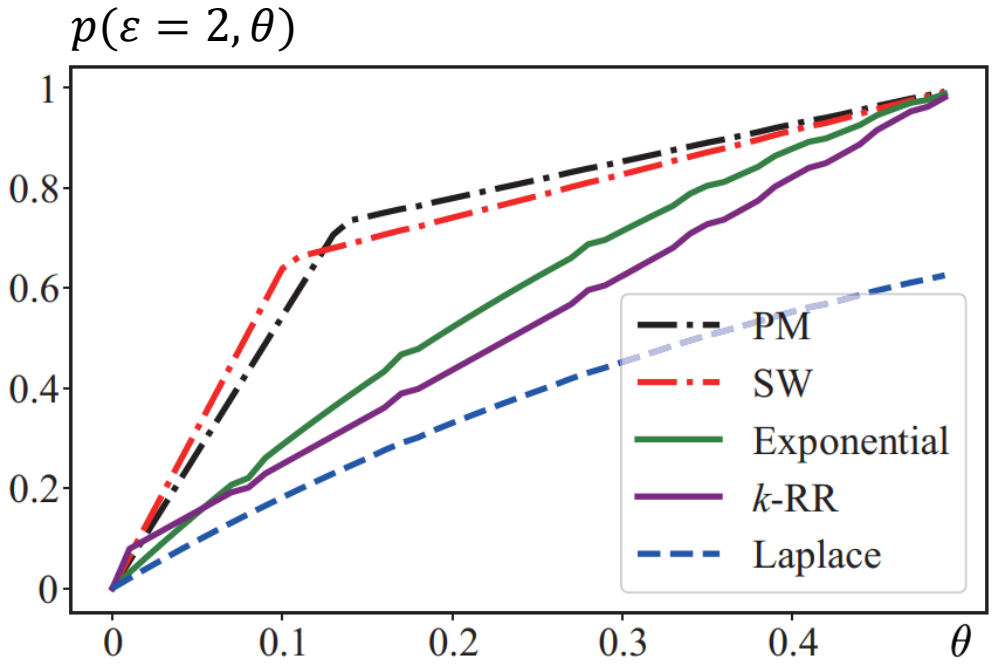


Benefit 1: Choose the **best ε** for a desired classifier utility

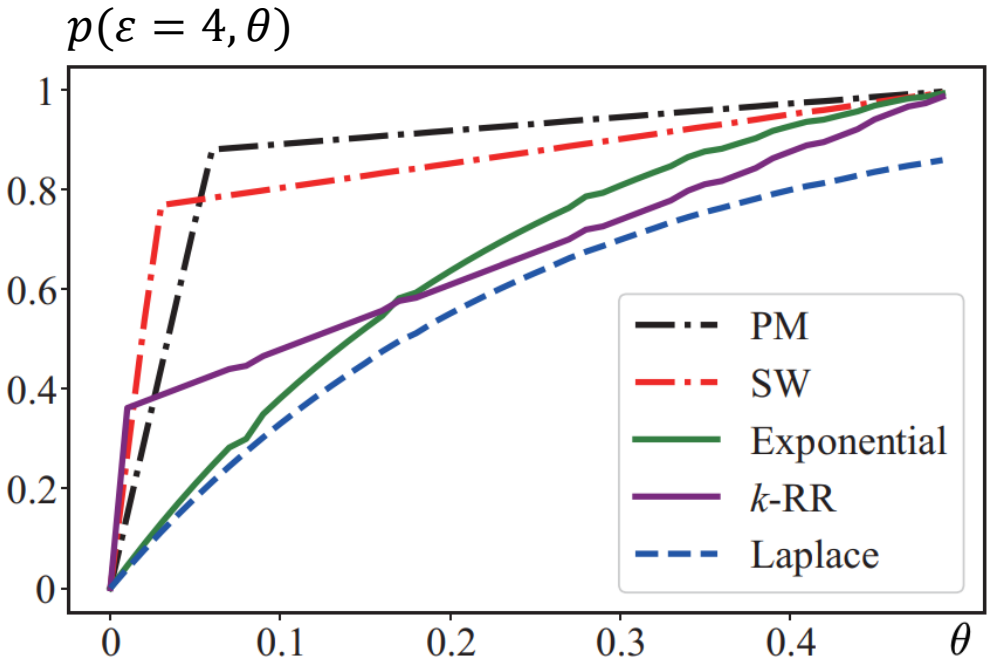
Different $\mathcal{M}$ & Different Classifiers

RIT

- No universally optimal LDP mechanism for all ε and θ



(a) $\varepsilon = 2$



(b) $\varepsilon = 4$

Different $\mathcal{M}$ & Different Classifiers

RIT

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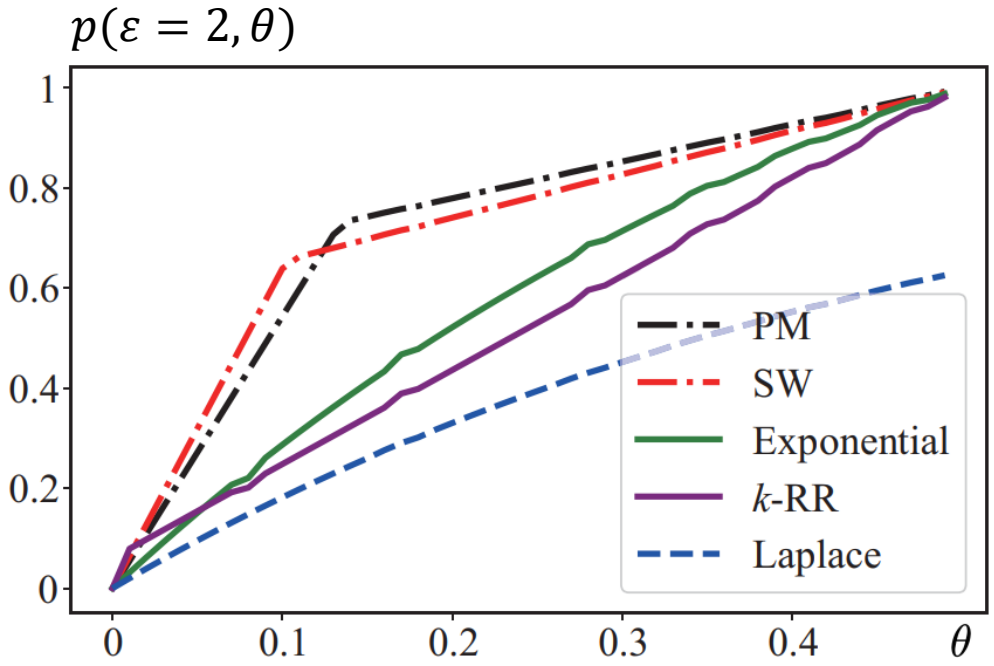
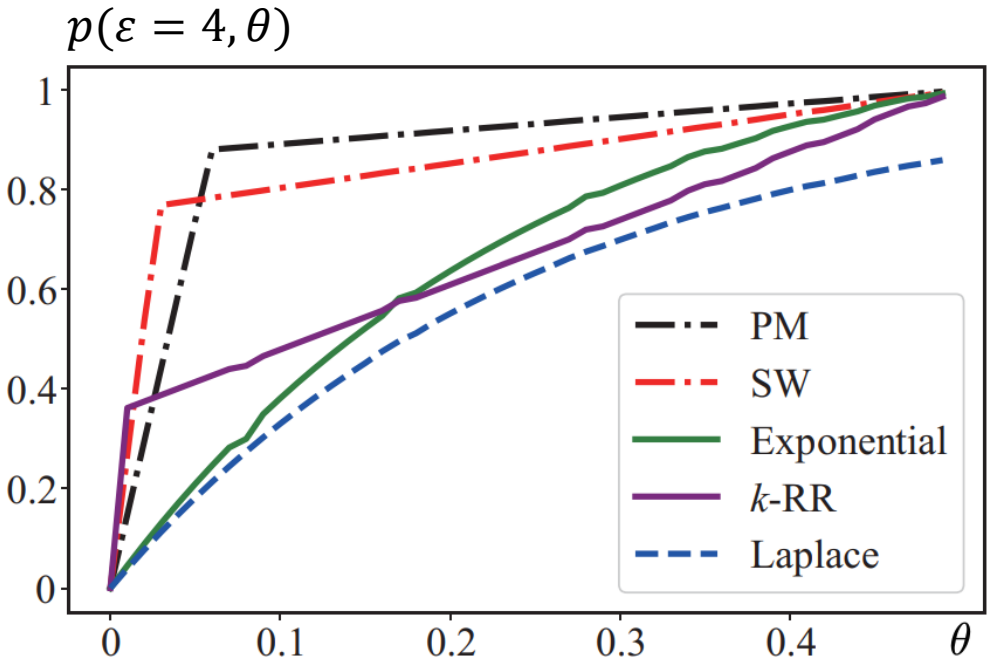




(b) $\varepsilon = 4$

Benefit 2: **Systematic comparison** of different LDP mechanisms

High-Dimensional Cases

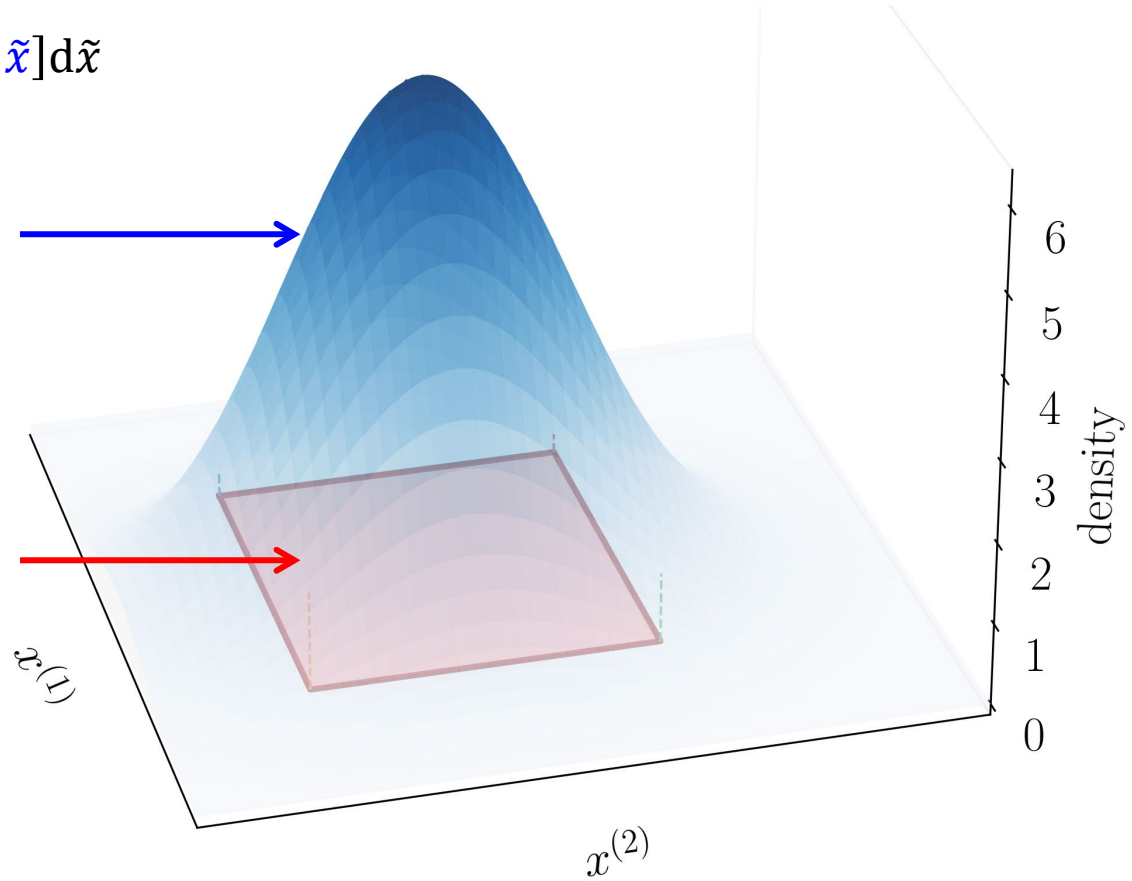
RIT

- Concentration analysis on the robustness region S

$$p(\varepsilon, S) = \int_S \mathrm{pdf}[\mathcal{M}_\varepsilon(x) = \tilde{x}]\mathrm{d}\tilde{x}$$

LDP mechanism $\mathcal{M}_\varepsilon(x)$

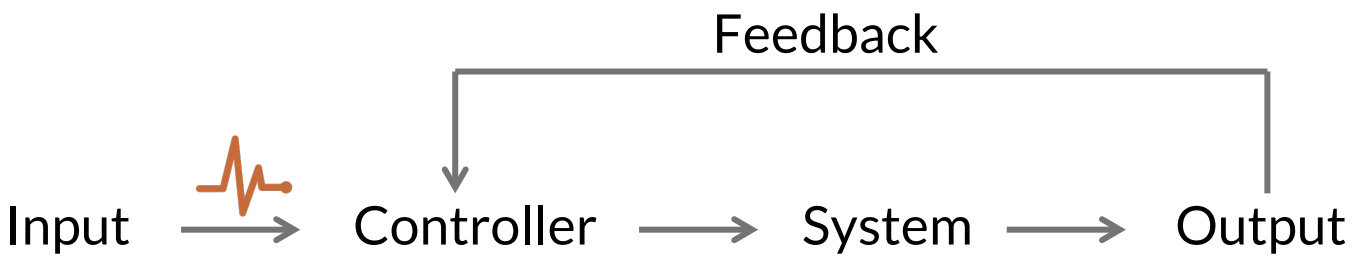
Robustness region S



Beyond Classifier – Robustness in Other Systems

- **Communication systems**

- **Control systems**



Case Studies – Quality of The Analytical Utility

- Claim: Accurate (compared with the empirical utility) & efficient
- Classifiers:
  - Logistic Regression, Random Forest
- Datasets: Stroke Prediction (medical data), Bank Customer Attrition (financial data),

2 sensitive dims

Case Studies – Quality of The Analytical Utility

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2 sensitive dims

- Datasets: Stroke Prediction (medical data), Bank Customer Attrition (financial data),

- Classifier utility:

  - analytical utility $p(\varepsilon, S)$: approximated S for black-box classifiers

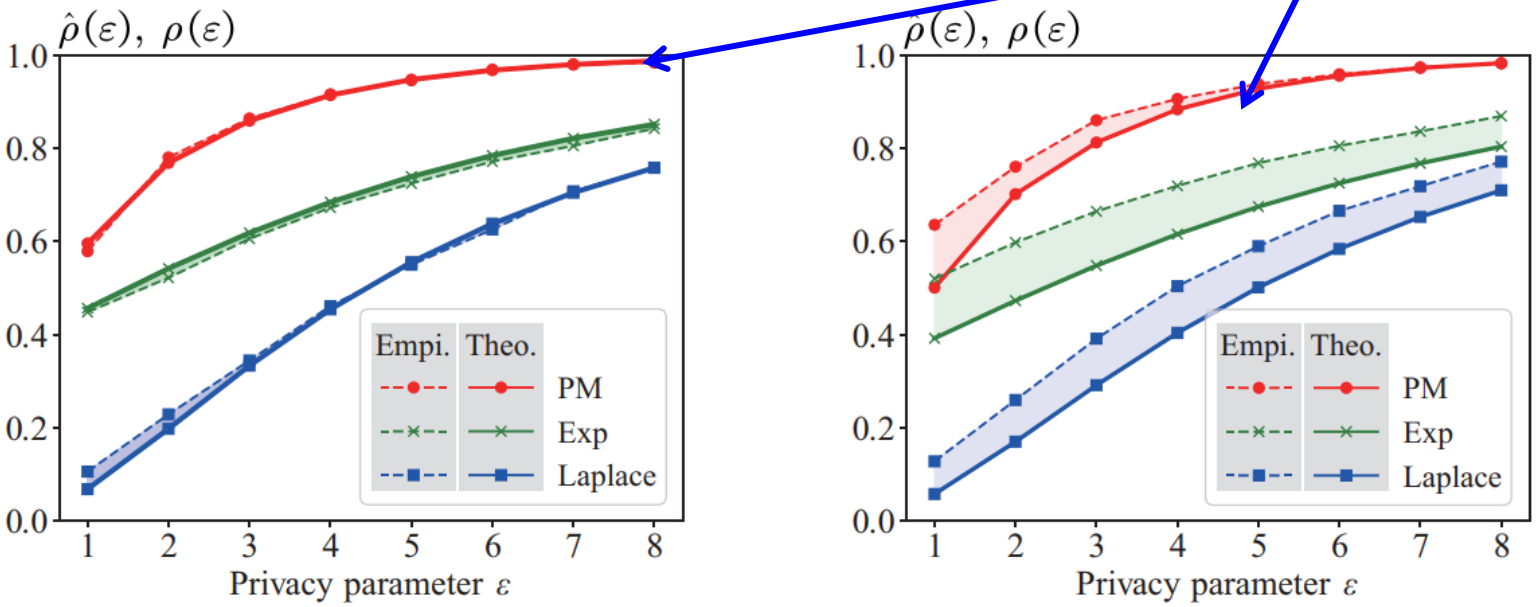
  - empirical utility $\hat{p}(\varepsilon)$: 2000 samples from $\mathcal{M}_\varepsilon$ and then used for testing

Case Studies – Stoke Predication

RIT

- x = the first record, with noisy "Age" and "BMI"

Accurate analytical utility



(a) Logistic Regression. **(b) Random Forest.**

Case Studies – Stoke Predication

RIT

- x = the first record, with noisy "Age" and "BMI"

Time cost comparison (ms)

	PM	Exponential	Laplace
Empirical[a]	$6.56 + 1.38$	$859.83 + 1.53$	$11.29 + 1.53$
Theoretical[b]	0.24	0.94	0.30

← Negligible time cost

[a] Time of 2000 samples + inference.

[b] Time to compute $\rho(\varepsilon, S)$ only; computing S takes 5.80 ms but is a one-time cost amortized across all ε values.

Future Research Plan

RIT

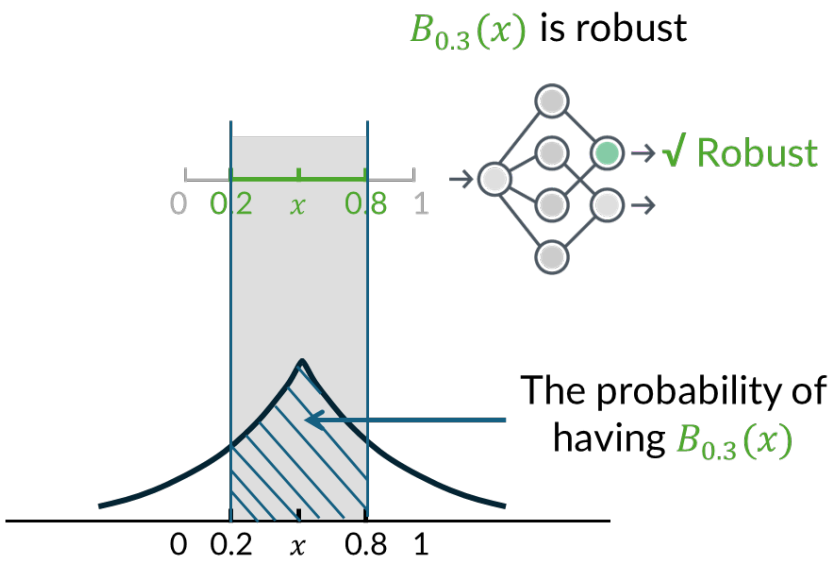
- High-dimensional classifiers, e.g. Neural Networks
  - their analytical utility under LDP-data
  - if the analytical bound is loose, how to refine?
- Average-case and worst-case classifier utility
  - can we have an average-case classifier utility? (for classifier designers)
- What if the noisy data is used for training?

Summary & Takeaway

RIT

- **RQ: Classifier utility under LDP-perturbed inputs**

- Our approach:*
  - **connects LDP with robustness**
  - **provides analytical classifier utility**
  - the analytical utility is accurate for low-dim classifiers



* Quantifying Classifier Utility under Local Differential Privacy, PETS'26

LDP: Refined Mechanism Design and Utility Analysis

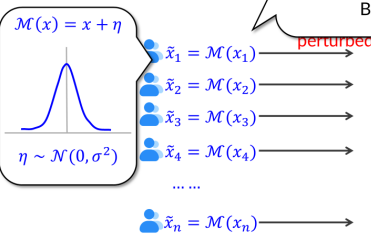
Privacy-Preserving Computation - **LDP**

RIT

- Local differential privacy (L
  - hard to differentiate the
  - each user locally perturb

Advantages:
Negligible computational complexity
No communication between users

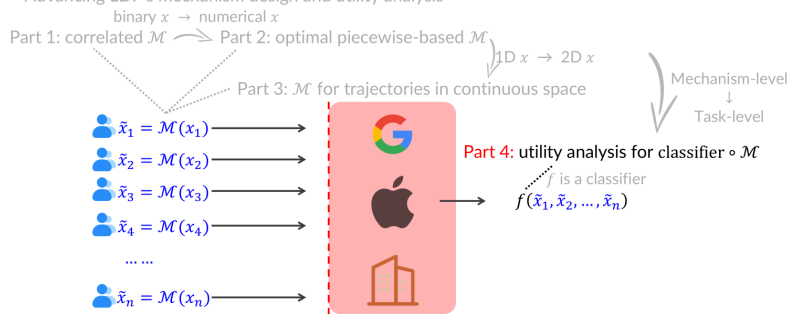
But approximated f



This Proposal: LDP Theory

RIT

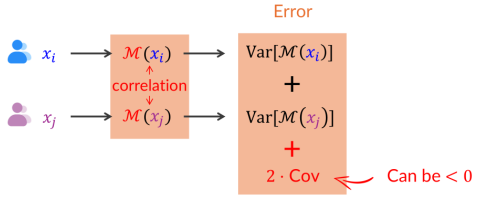
- Advancing LDP's mechanism design and utility analysis



This Paper: Joint/Correlated RR (JRR)

RIT

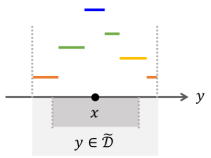
- Existing LDP mechanisms: Each user perturbs their data independently
- Correlated LDP mechanisms: Users' data are perturbed by correlated $\mathcal{M}$



Optimal Piecewise-based Mechanism

RIT

- **Most generalized version:** m-piecewise distribution



$$pdf[\mathcal{M}(x) = y] = \begin{cases} p_{1,\varepsilon} & \text{if } y \in [l_{1,x,\varepsilon}, r_{1,x,\varepsilon}] \\ p_{2,\varepsilon} & \text{if } y \in [l_{2,x,\varepsilon}, r_{2,x,\varepsilon}] \\ \dots & \\ p_{m,\varepsilon} & \text{if } y \in [l_{m,x,\varepsilon}, r_{m,x,\varepsilon}] \end{cases}$$

$\frac{p_{i,\varepsilon}}{p_{j,\varepsilon}} \le e^{\varepsilon}$ (LDP constraint)

$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \max_{x \in \mathcal{D}} \int_{\tilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

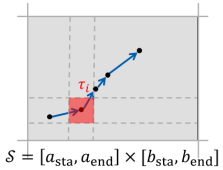
Find $\mathcal{M}$ to minimize the worst-case error

Solved $\mathcal{M}$ is
the optimal piecewise-based mechanism

Mathematically ≡ to find the **optimal** piecewise distribution under the LDP constraint

Continuous Spaces: Better & Universal

- Operate on a continuous space (covering the discrete space)



Simple sampling: (satisfying LDP for $\mathcal{S}$)

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_{\text{high}}$

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \square] = p_{\text{low}}$

- Benefits:
  1. **Efficiency:** $\mathcal{O}(1)$ sampling complexity
  2. **Trajectory utility:** "n-independent"
  3. **Applicability:** Both continuous spaces & discrete spaces

One-Dimension Example

RIT

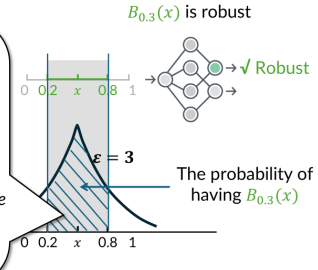
- Classifier $h: [0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

Concentration analysis

$$p(\varepsilon, \theta) = \mathrm{cdf}_{\text{Lap}}[\theta] - \mathrm{cdf}_{\text{Lap}}[-\theta] = 1 - e^{-\theta\varepsilon}$$

For any ε:

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_\varepsilon(x)$.



Thank you!

& Q A